SSC JE

FLUID MECHANICS & HYDRAULIC MACHINES

Topper-Style Concept Notes

SSC JE Mechanical / Civil • Paper-I & Paper-II • Concept + Formula + PYQ-Trend + Solved Numericals

Fluid Properties • Statics • Buoyancy • Kinematics • Dynamics • Flow Measurement • Laminar & Turbulent Flow • Pipe Flow • Boundary Layer • Dimensional Analysis • Compressible Flow • Open Channels • Impact of Jets • Turbines • Pumps

Concept-based notes – no MCQ practice. "PYQ-Trend" boxes list concepts SSC JE has repeatedly tested (based on recurring topics, not copied questions). Cross-check values with a standard textbook.

Contents

How to Use These Notes 8

Why Fluid Mechanics Decides Your Rank . . . 8

How Each Chapter Is Built . . . 8

1. Properties of Fluids 9

1.1 What Is a Fluid? . . . 9

1.2 Mass and Weight Properties . . . 9

1.3 Viscosity . . . 9

1.4 Compressibility and Bulk Modulus . . . 11

1.5 Surface Tension and Capillarity . . . 11

1.6 Vapour Pressure and Cavitation . . . 12

1.7 Other Properties and Equation of State . . . 12

2. Fluid Statics and Pressure Measurement 13

2.1 Pressure at a Point . . . 13

2.2 Hydrostatic Law . . . 13

2.3 Absolute, Gauge and Vacuum Pressure . . . 13

2.4 Manometers . . . 14

2.5 Mechanical and Electrical Pressure Gauges . . . 15

2.6 Fluids in Rigid-Body Motion (Relative Equilibrium) . . . 15

3. Hydrostatic Forces on Surfaces 17

3.1 Total Pressure and Centre of Pressure . . . 17

3.2 Curved Surfaces . . . 17

3.3 Applications – Dams, Gates, Lock Gates . . . 18

4. Buoyancy, Floatation and Stability 19

4.1 Archimedes' Principle and Buoyancy . . . 19

4.2 Metacentre and Metacentric Height . . . 19

4.3 Stability of Floating Bodies – Practical Points . . . 20

5. Fluid Kinematics 21

5.1 Description of Flow . . . 21

5.2 Flow Lines . . . 21

5.3 Continuity Equation . . . 22

5.4 Acceleration of a Fluid Particle . . . 22

5.5 Rotation, Vorticity and Circulation . . . 22

5.6 Velocity Potential and Stream Function . . . 22

5.7 Basic Potential Flows and Vortex Flows . . . 23

6. Fluid Dynamics – Energy and Momentum 25

6.1 Forces and Equations of Motion . . . 25

6.2 Bernoulli's Equation . . . 25

6.3 HGL and TEL . . . 25

6.4 Correction Factors . . . 26

6.5 Impulse–Momentum Equation and Its Applications . . . 26
6.6 Free Liquid Jets . . . 26

7. Flow Measurement – Meters, Orifices, Mouthpieces, Notches 28

7.1 Venturimeter . . . 28
7.2 Orificemeter, Flow Nozzle and Others . . . 28
7.3 Orifices . . . 29
7.4 Mouthpieces . . . 29
7.5 Notches and Weirs . . . 30

8. Viscous (Laminar) Flow 32

8.1 Reynolds' Experiment . . . 32
8.2 Flow Through Circular Pipes – Hagen–Poiseuille . . . 32
8.3 Flow Between Parallel Plates . . . 33
8.4 Lubrication, Dashpot and Stokes' Law . . . 33
8.5 Measurement of Viscosity . . . 34

9. Turbulent Flow and Flow Through Pipes 35

9.1 Turbulent Flow Basics . . . 35
9.2 Major Losses – Darcy–Weisbach and Others . . . 35
9.3 Minor Losses . . . 36
9.4 Pipes in Series, Parallel, Branching, Siphons . . . 37
9.5 Power Transmission Through Pipes and Nozzles . . . 37
9.6 Water Hammer and Surge Tanks . . . 37
9.7 Pipe Networks and Other Topics . . . 38

10. Boundary Layer, Drag and Lift 39

10.1 Boundary Layer Concept . . . 39
10.2 Boundary Layer Separation . . . 39
10.3 Drag and Lift . . . 40

11. Dimensional Analysis and Model Similitude 42

11.1 Dimensions of Common Quantities . . . 42
11.2 Methods of Dimensional Analysis . . . 42
11.3 Dimensionless Numbers . . . 43
11.4 Similitude and Model Laws . . . 43

12. Compressible Flow (Basics) 45

12.1 Speed of Sound and Mach Number . . . 45
12.2 Stagnation Properties and Isentropic Flow . . . 45
12.3 Normal Shock Waves . . . 46

13. Open Channel Flow 47

13.1 Basics . . . 47
13.2 Uniform Flow Formulae . . . 47
13.3 Most Economical (Efficient) Sections . . . 47
13.4 Specific Energy and Critical Flow . . . 48

13.5 Hydraulic Jump . . . 48

14. Impact of Jets **50**

14.1 Force of a Jet on Fixed Plates and Vanes . . . 50
14.2 Moving Plates and Vanes . . . 50
14.3 Hinged Plates . . . 50

15. Hydraulic Turbines **52**

15.1 Hydroelectric Plant and Heads . . . 52
15.2 Classification of Turbines . . . 52
15.3 Pelton Wheel . . . 53
15.4 Francis Turbine . . . 53
15.5 Kaplan and Propeller Turbines . . . 54
15.6 Draft Tube . . . 54
15.7 Cavitation in Turbines . . . 54
15.8 Performance, Unit Quantities and Governing . . . 55

16. Pumps and Hydraulic Devices **57**

16.1 Centrifugal Pumps – Construction and Working . . . 57
16.2 Heads, Work and Efficiencies . . . 57
16.3 Similarity Laws, Specific Speed and Characteristics . . . 57
16.4 Cavitation and NPSH in Pumps . . . 58
16.5 Reciprocating Pumps . . . 59
16.6 Other Pumps and Hydraulic Devices . . . 59

17. Solved Numericals – Set 1 (Statics to Pipe Flow) **61**

17.1 Properties . . . 61
17.2 Statics and Hydrostatic Forces . . . 61
17.3 Buoyancy and Stability . . . 63
17.4 Kinematics and Dynamics . . . 63
17.5 Flow Measurement . . . 64
17.6 Laminar and Pipe Flow . . . 64

17B. Solved Numericals – Set 2 (BL, Models, Channels, Machines) and Ratio Twisters **66**

17B.1 Boundary Layer, Drag, Similitude, Compressible . . . 66
17B.2 Open Channels . . . 67
17B.3 Jets and Turbines . . . 67
17B.4 Pumps . . . 68
17B.5 Ratio Twisters (Most Time-Saving) . . . 68

17C. Concept Recall Bank **70**

R1. Properties and Statics . . . 70
R2. Forces, Buoyancy, Kinematics . . . 71
R3. Dynamics and Measurement . . . 71
R4. Laminar, Turbulent, Pipe Flow . . . 72
R5. BL, Models, Channels, Machines . . . 73

17D. Paper-II Model Answers 75

P1. Statics and Kinematics . . . 75

P2. Dynamics and Pipe Flow . . . 75

P3. Machines . . . 76

17E. Solved Numericals – Set 3 (Velocity Triangles & Mixed Practice) 78

17E.1 Velocity Triangles . . . 78

17E.2 Mixed Practice . . . 78

18. Last-Minute Revision Sheet 81

18.1 Must-Know Values . . . 81

18.2 Commonly Confused Pairs . . . 81

18.3 Paper-II Quick Answers (5-mark style) . . . 82

Appendix A – One-Place Formula Sheet 84

Properties of Fluids . . . 84

Fluid Statics and Pressure Measurement . . . 84

Hydrostatic Forces on Surfaces . . . 85

Buoyancy, Floatation and Stability . . . 85

Fluid Kinematics . . . 86

Fluid Dynamics – Energy and Momentum . . . 86

Flow Measurement – Meters, Orifices, Mouthpieces, Notches . . . 87

Viscous (Laminar) Flow . . . 87

Turbulent Flow and Flow Through Pipes . . . 88

Boundary Layer, Drag and Lift . . . 89

Dimensional Analysis and Model Similitude . . . 89

Compressible Flow (Basics) . . . 89

Open Channel Flow . . . 90

Impact of Jets . . . 90

Hydraulic Turbines . . . 90

Pumps and Hydraulic Devices . . . 91

Appendix B – PYQ-Trend Concept Map 93

Properties of Fluids . . . 93

Fluid Statics and Pressure Measurement . . . 93

Hydrostatic Forces on Surfaces . . . 93

Buoyancy, Floatation and Stability . . . 94

Fluid Kinematics . . . 94

Fluid Dynamics – Energy and Momentum . . . 94

Flow Measurement – Meters, Orifices, Mouthpieces, Notches . . . 94

Viscous (Laminar) Flow . . . 95

Turbulent Flow and Flow Through Pipes . . . 95

Boundary Layer, Drag and Lift . . . 95

Dimensional Analysis and Model Similitude . . . 95

Compressible Flow (Basics) . . . 96

Open Channel Flow 96
Impact of Jets 96
Hydraulic Turbines 96
Pumps and Hydraulic Devices 97

INTRODUCTION

How to Use These Notes

Why Fluid Mechanics Decides Your Rank

In SSC JE Mechanical, Fluid Mechanics and Hydraulic Machines together give roughly 15–20 of the 100 technical questions in Paper-I, and a full block in Paper-II. Civil candidates get an even larger share (hydraulics, open channels, pipe flow). The subject repeats concepts heavily: viscosity behaviour, manometers, centre of pressure, metacentric height, Bernoulli applications, venturimeter, laminar flow (f = 64/Re), Darcy–Weisbach, minor losses, model laws, specific speed and turbine selection appear almost every year.

Area	Typical questions (Mech, Paper-I)	Priority
Fluid properties & statics	3–4	Very high
Buoyancy & kinematics	2–3	High
Bernoulli, flow measurement, momentum	3–4	Very high
Laminar, turbulent & pipe flow	3–4	Very high
Boundary layer, dimensional analysis	1–2	Medium
Impact of jets & turbines	2–3	Very high
Pumps & hydraulic devices	2–3	High

How Each Chapter Is Built

- Concept points first, then a yellow formula box, then a blue PYQ-TREND box listing what SSC asks repeatedly, a green TOPPER TRICK, a red COMMON MISTAKE, and a purple descriptive question for Paper-II.
- Chapter 17 has solved numericals in SSC style (step by step, no options). Chapter 18 is the last-minute sheet. Appendix A collects every formula; Appendix B collects every PYQ-trend box.
- Use SI units throughout: ρ_{water} = 1000 kg/m³, g = 9.81 m/s², γ_{water} = 9810 N/m³, $\rho_{mercury}$ = 13,600 kg/m³ (S = 13.6).

✔ **TOPPER TRICK:** Fluid mechanics numericals in SSC are short (30–60 seconds). Memorise the ratio results (how a quantity changes when D, V or H doubles) – they save the most time.

✖ **COMMON MISTAKE:** Don't mix up gauge and absolute pressure, dynamic (μ) and kinematic (ν) viscosity, or the turbine and pump specific-speed formulas. These three slips cost the most marks.

CHAPTER 1

Properties of Fluids

1.1 What Is a Fluid?

- A fluid is a substance that deforms continuously under the action of a shear stress, however small. A solid resists shear by a static deformation; a fluid cannot.
- Liquids have a definite volume and form a free surface; gases fill the whole container. Both are fluids.
- Continuum hypothesis: a fluid is treated as continuous matter, ignoring molecular gaps. Valid when the mean free path is much smaller than the characteristic length (Knudsen number $Kn = \lambda/L \ll 0.01$). Fails in rarefied gas (very high altitude, vacuum systems).
- Fluid mechanics divides into fluid statics (fluid at rest), kinematics (motion without forces) and dynamics (motion with forces).

★ **PYQ-TREND:** Definition of fluid based on continuous deformation under shear. Continuum assumption and Knudsen number.

1.2 Mass and Weight Properties

Mass density $\rho = m/V$ (kg/m³); water 1000 at 4 °C, mercury 13,600, air ≈ 1.225 at 15 °C sea level

Specific weight $w = \gamma = \rho g$ (N/m³); water 9810 N/m³ = 9.81 kN/m³

Specific volume $v = 1/\rho$ (m³/kg)

Specific gravity (relative density) $S = \rho_{fluid}/\rho_{water}$ (gases: relative to air or hydrogen)

- Specific gravity of some fluids: mercury 13.6, sea water 1.025, petrol ≈ 0.7, kerosene ≈ 0.8, oil (typical) 0.8–0.9, glycerine 1.26.
- Density of liquids changes little with pressure; it decreases slightly with temperature (except water between 0 and 4 °C – anomalous behaviour).
- Weight density depends on g and therefore on location; mass density does not.

★ **PYQ-TREND:** Specific gravity of mercury (13.6). Unit of specific weight (N/m³). Specific volume is the reciprocal of density. Water has maximum density at 4 °C.

1.3 Viscosity

- Viscosity is the property by which a fluid resists the relative motion (shear) between adjacent layers. It is the 'internal friction' of a fluid.

Newton's law of viscosity: $\tau = \mu \, (du/dy)$

Dynamic viscosity μ – unit N·s/m² = Pa·s; CGS: poise (1 P = 1 dyne·s/cm² = 0.1 Pa·s); 1 centipoise = 10^{-3} Pa·s

Kinematic viscosity $\nu = \mu/\rho$ – unit m²/s; CGS: stoke (1 St = 1 cm²/s = 10^{-4} m²/s)

Dimensions: $\mu = [ML^{-1}T^{-1}]$; $\nu = [L^2T^{-1}]$

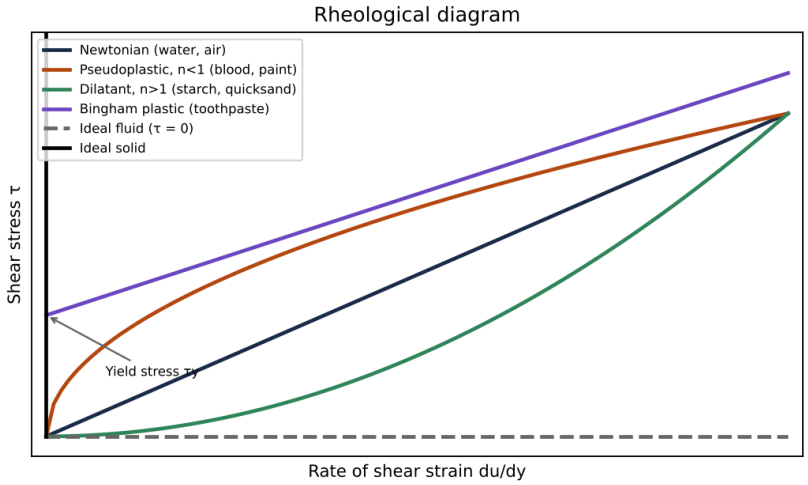
- Values at 20 °C: water $\mu \approx 1.0 \times 10^{-3}$ Pa·s (1 cP), $\nu \approx 1.0 \times 10^{-6}$ m²/s; air $\mu \approx 1.8 \times 10^{-5}$ Pa·s, $\nu \approx 1.5 \times 10^{-5}$ m²/s. Air has lower μ but higher ν than water (because its density is very low).
- Effect of temperature: in liquids viscosity is due to cohesion between molecules → viscosity decreases as temperature rises. In gases it is due to molecular momentum exchange → viscosity increases with temperature.
- Effect of pressure: practically independent for liquids and gases at moderate pressures.
- Viscosity index (lubricants): measure of how little the viscosity changes with temperature; higher VI is better.
- Velocity gradient du/dy is also called rate of shear strain (angular deformation rate).

Rheological classification

Fluid	τ – du/dy relation	Examples
Ideal fluid	No viscosity (τ = 0), incompressible	Imaginary; used for analysis
Newtonian	Straight line through origin (constant μ)	Water, air, petrol, kerosene, thin oils, alcohol
Pseudoplastic (shear-thinning)	Curve, slope decreases; n < 1	Blood, paint, polymer solutions, milk, paper pulp
Dilatant (shear-thickening)	Curve, slope increases; n > 1	Starch in water, quicksand, sugar in water, rice starch
Bingham plastic	Yield stress, then linear	Toothpaste, drilling mud, sewage sludge, chalk–water
Thixotropic	Viscosity falls with time of shearing	Printer's ink, lipstick, some paints, enamels
Rheopectic	Viscosity rises with time of shearing	Gypsum paste, bentonite clay
Ideal plastic	Yield stress, then constant slope	–
Real fluid	Has viscosity	All actual fluids

Power law (Ostwald–de Waele): $\tau = K (du/dy)^n$ (n = 1 Newtonian; n < 1 pseudoplastic; n > 1 dilatant)

Bingham plastic: $\tau = \tau_y + \mu_p (du/dy)$



★ **PYQ-TREND:** Viscosity variation with temperature (liquids ↓, gases ↑) is asked every year. Units/dimensions of μ and ν; poise and stoke conversions. Examples of non-Newtonian fluids (blood – pseudoplastic, toothpaste – Bingham, printer's ink – thixotropic, quicksand – dilatant). Newton's law of viscosity.

✔ **TOPPER TRICK:** Kinematic viscosity has no mass in it – it is purely kinematic (L^2/T), like a diffusivity. 'Stoke' sounds like 'stroke of area per second' = cm^2/s.

✖ **COMMON MISTAKE:** 1 poise = 0.1 Pa·s, not 10 Pa·s. And 1 stoke = 10^{-4} m^2/s, not 10^{-6} (10^{-6} m^2/s is one centistoke).

1.4 Compressibility and Bulk Modulus

Bulk modulus $K = -dp/(dV/V) = dp/(d\rho/\rho)$ (N/m^2); compressibility $\beta = 1/K$

For water $K \approx 2.1–2.2 \times 10^9$ Pa (2.1 GPa); steel ≈ 160 GPa; air at 1 atm (isothermal) ≈ 1.01×10^5 Pa

Gas: isothermal $K = p$; adiabatic (isentropic) $K = \gamma p$

Speed of sound $c = \sqrt{(K/\rho)}$; in ideal gas $c = \sqrt{(\gamma RT)}$

- Liquids are treated as incompressible in most problems; compressibility matters in water hammer and high-speed gas flow (Mach > 0.3).
- A higher bulk modulus means a less compressible fluid.

★ **PYQ-TREND:** K of water; K = p (isothermal) and γp (adiabatic) for gases; compressibility as the reciprocal of bulk modulus; speed of sound √(K/ρ).

1.5 Surface Tension and Capillarity

- Surface tension σ: tensile force per unit length acting along the free surface of a liquid, due to unbalanced cohesive forces on surface molecules. Unit N/m (= J/m^2). Water–air at 20 °C ≈ 0.073 N/m; mercury–air ≈ 0.48 N/m.
- Surface tension decreases with increasing temperature (zero at the critical point) and with soaps/detergents (surfactants); dissolved salts slightly increase it.

Pressure excess inside a liquid droplet: $\Delta p = 4\sigma/d = 2\sigma/R$

Inside a soap bubble (two free surfaces): $\Delta p = 8\sigma/d = 4\sigma/R$

Inside a liquid jet (cylinder): $\Delta p = 2\sigma/d = \sigma/R$

Capillary rise/fall: $h = 4\sigma \cos\theta / (\rho g d)$ (θ = contact angle)

Rise between two parallel plates (gap t): $h = 2\sigma \cos\theta / (\rho g t)$

- Contact angle: water–clean glass θ ≈ 0° (wets, concave meniscus, capillary rise); mercury–glass θ ≈ 130°–140° (does not wet, convex meniscus, capillary depression).
- Adhesion > cohesion → liquid wets the surface, rises. Cohesion > adhesion → falls.
- Capillary rise is inversely proportional to tube diameter ($h \propto 1/d$). In a tube of insufficient height, the liquid does not overflow; the meniscus radius simply increases.
- Applications: rise of water in soil and plant tissues (partly), wick action of lamps, blotting paper, measurement errors in manometers (use tubes > 6 mm diameter to make capillary error negligible).

★ **PYQ-TREND:** Pressure inside a soap bubble 8σ/d vs droplet 4σ/d (very frequent). Capillary rise formula and $h \propto 1/d$. Mercury shows capillary depression. Unit of surface tension (N/m). Effect of temperature on surface tension.

✔ **TOPPER TRICK:** Count the number of surfaces: droplet – 1 surface → 4σ/d; soap bubble – 2 surfaces → 8σ/d; jet – cylinder with 1 surface → 2σ/d.

1.6 Vapour Pressure and Cavitation

- Vapour pressure: the partial pressure exerted by vapour molecules in equilibrium with the liquid at a given temperature. It rises with temperature; boiling occurs when vapour pressure equals the surrounding pressure (so water boils below 100 °C on mountains).
- Water vapour pressure ≈ 2.34 kPa at 20 °C, 101.3 kPa at 100 °C. Mercury has extremely low vapour pressure (≈ 0.16 Pa at 20 °C) – one reason it is used in barometers.
- Cavitation: formation of vapour bubbles in a flowing liquid where local pressure drops below vapour pressure, followed by their violent collapse in higher-pressure regions. Effects: pitting/erosion of metal, noise, vibration, loss of efficiency. Occurs at pump inlets, turbine runner outlets, propellers, valves, venturi throats.

★ **PYQ-TREND:** Cavitation occurs when pressure falls below vapour pressure. Why mercury is used in barometers (high density, low vapour pressure, does not wet glass). Boiling point lower at higher altitude.

1.7 Other Properties and Equation of State

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Perfect gas: p = ρRT; R_air = 287 J/kg·K
Isothermal: p/ρ = constant; adiabatic: p/ρ^γ = constant
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- Cohesion: attraction between molecules of the same liquid. Adhesion: attraction between liquid molecules and solid surface molecules.
- Real fluid: has viscosity, surface tension and compressibility. Ideal fluid: none of these – a mathematical model used for potential flow and Bernoulli derivations.
- Newtonian fluid with very low viscosity can still be 'real'; ideal ≠ inviscid real fluid in practice.

```
Shear force between plates: F = τA = μ A V / h (linear velocity profile)
Torque on a rotating shaft in a journal bearing (thin oil film t): T = μ (πDN/60)/t × (πDL) × D/2
Power lost P = T ω = 2πNT/60
```

★ **PYQ-TREND:** Ideal vs real fluid. Shear stress between two plates with oil film (τ = μV/h). Power lost in a journal bearing due to viscous friction.

✎ **DESCRIPTIVE Q:** State Newton's law of viscosity and explain why the viscosity of liquids decreases and that of gases increases with temperature.

Newton's law states that the shear stress between adjacent fluid layers is proportional to the velocity gradient normal to the layers: $\tau = \mu \, du/dy$, where μ is the coefficient of dynamic viscosity. In liquids, viscosity arises mainly from cohesive forces between closely packed molecules. As temperature rises, molecular kinetic energy increases and cohesion weakens, so the resistance to shear falls – viscosity decreases (by about 2% per °C for water near room temperature). In gases the molecules are far apart, and viscosity arises from the transfer of momentum by molecules moving randomly between layers. A higher temperature increases molecular speed and this momentum exchange, so gas viscosity increases with temperature (roughly $\propto \sqrt{T}$ by kinetic theory).

CHAPTER 2

Fluid Statics and Pressure Measurement

2.1 Pressure at a Point

- Pressure (intensity of pressure) p = dF/dA, normal compressive force per unit area. Unit Pa (N/m²); 1 bar = 10^5 Pa; 1 atm = 101.325 kPa = 760 mm Hg = 10.33 m of water; 1 kgf/cm² = 98.1 kPa; 1 psi = 6.895 kPa.
- Pascal's law: in a fluid at rest, pressure at a point is the same in all directions (px = py = pz). It holds for fluids at rest and in rigid-body motion (no shear).
- Applications of Pascal's law: hydraulic press, hydraulic jack, hydraulic brakes, hydraulic lift.

Hydraulic press: F/a = W/A → mechanical advantage W/F = A/a = $(D/d)^2$

★ **PYQ-TREND:** Statement and applications of Pascal's law; mechanical advantage of hydraulic press; unit conversions (1 atm = 10.33 m water = 760 mm Hg).

2.2 Hydrostatic Law

$dp/dz = -\rho g$ (z upward) → $p = \rho g h$ (h measured downward from the free surface)

Pressure head $h = p/(\rho g) = p/w$; converting heads: $h_1 S_1 = h_2 S_2$

- Pressure depends only on depth, density and g, not on the shape of the container (hydrostatic paradox). Dams are thicker at the base because pressure increases linearly with depth.
- Free surfaces of a liquid at rest are horizontal; points at the same level in a continuous body of the same liquid have equal pressure.
- In compressible fluids (atmosphere): isothermal atmosphere $p = p_0$ e^(−gz/RT); troposphere temperature lapse rate ≈ 6.5 K/km (ISA).

Atmospheric pressure variation (linear lapse): $p = p_0 (1 - Lz/T_0)^{(g/(LR))}$

★ **PYQ-TREND:** p = ρgh and its independence from container shape; head conversions between fluids (e.g., 10 m of water = 0.735 m of mercury); thick dam base.

✔ **TOPPER TRICK:** Head of fluid 2 = head of fluid 1 × S_1/S_2. For water to mercury divide by 13.6; 10.33 m water ÷ 13.6 = 0.76 m Hg.

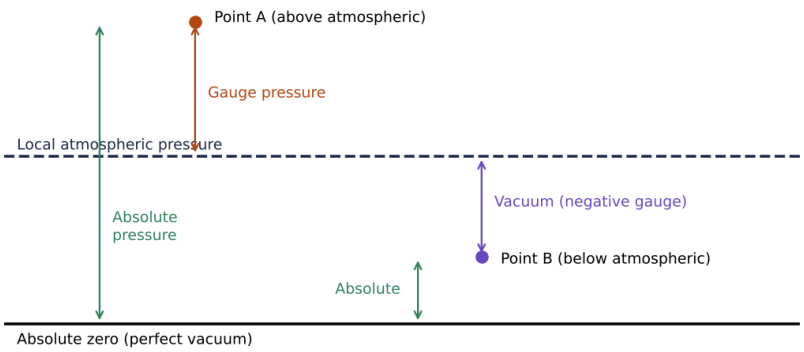
2.3 Absolute, Gauge and Vacuum Pressure

Absolute pressure = atmospheric pressure + gauge pressure

Vacuum (negative gauge) pressure = atmospheric − absolute

- Absolute zero pressure = perfect vacuum. Gauges (Bourdon, manometers open to air) read gauge pressure. Barometers read atmospheric (absolute) pressure.
- Local atmospheric pressure varies with altitude and weather; standard value 101.325 kPa.

Absolute, gauge and vacuum pressure



★ **PYQ-TREND:** Relation among absolute, gauge, vacuum pressure; which instrument measures which (barometer – atmospheric; Bourdon – gauge).

✖ **COMMON MISTAKE:** In gas law and cavitation problems use absolute pressure. In manometer balances you may use gauge pressure consistently.

2.4 Manometers

- Manometers measure pressure by balancing a column of liquid. They can measure positive and negative (vacuum) gauge pressures and pressure differences.

Manometer	Use / feature
Piezometer	Simplest; vertical tube; only liquid gauge pressure; cannot measure negative pressure, gas pressure or high pressure (tube too long)
Simple U-tube	Gauge pressure (positive or negative) with a heavier manometric liquid (mercury)
Single-column (micromanometer)	One limb a large reservoir; read only one limb; area ratio makes it sensitive
Inclined single-column	More sensitive; reading magnified by 1/sinθ; used for low pressures (air ducts)
Differential U-tube	Pressure difference between two points (e.g., venturimeter) using heavier liquid
Inverted U-tube differential	Small pressure differences between two points in a liquid; lighter manometric fluid (air/oil) at the top
Micromanometer (two liquids)	Very small differences

```
U-tube (mercury), pipe centre h₁ above mercury in left limb, deflection h₂: p_A + ρ₁gh₁ = ρ₂gh₂ → p_A = ρ₂gh₂ − ρ₁gh₁
Vacuum with U-tube: p_A + ρ₁gh₁ + ρ₂gh₂ = 0 (gauge) → p_A = −(ρ₁gh₁ + ρ₂gh₂)
Differential U-tube, same fluid in both pipes at same level: p_A − p_B = gx(ρ_m − ρ)
Inverted U-tube: p_A − p_B = gx(ρ − ρ_light)
Single column: p_A = ρ₂gh₂ (1 + a/A) − ρ₁gh₁ ≈ ρ₂gh₂ − ρ₁gh₁ when A □ a
Inclined manometer: h₂ = L sinθ
```

- Method: start from one end, add ρgh going down, subtract going up, equate at the other end.
- Requirements of manometric liquid: should not mix with the fluid, low vapour pressure, stable density, good visibility, low surface tension effects; mercury is common for large

differences, water/alcohol/red oil for small.

★ **PYQ-TREND:** Limitations of piezometer; differential manometer reading $p_A - p_B = gx(\rho_m - \rho)$ (very frequent for venturimeter); inclined manometer for small pressures; inverted U-tube for small differences between liquids.

✔ **TOPPER TRICK:** For a water pipe with a mercury differential manometer, 1 cm of mercury deflection = 12.6 cm of water head difference (13.6 − 1 = 12.6).

2.5 Mechanical and Electrical Pressure Gauges

- Bourdon tube gauge: elliptical-section curved tube straightens under pressure; most common; measures gauge pressure (positive or vacuum types); C-type, spiral, helical.
- Diaphragm gauge: low pressures, corrosive fluids. Bellows gauge: low pressure and differential pressure.
- Dead-weight piston gauge: primary standard to calibrate Bourdon gauges.
- Electrical: strain-gauge transducers, piezoelectric (rapidly fluctuating pressure – engine cylinders), capacitive, LVDT-based.
- Vacuum gauges: McLeod (reference/calibration), Pirani (thermal conductivity), ionisation (very high vacuum), Knudsen.
- Barometers: mercury barometer (Torricelli, 1643) and aneroid barometer (no liquid; altimeter in aircraft).

★ **PYQ-TREND:** Bourdon gauge measures gauge pressure; dead-weight tester for calibration; McLeod for vacuum; aneroid barometer has no liquid.

2.6 Fluids in Rigid-Body Motion (Relative Equilibrium)

Linear acceleration

Tank accelerating horizontally with a: free surface tilts, $\tan\theta = a/g$

Vertical acceleration a (upwards): $p = \rho h(g + a)$; downwards: $p = \rho h(g - a)$; free fall: $p = 0$ (gauge)

General: $\tan\theta = a_x/(g + a_y)$

Rotation about a vertical axis (forced vortex)

Free surface is a paraboloid: $z = \omega^2 r^2/(2g)$; height rise at rim above vertex $h = \omega^2 R^2/(2g)$

Volume of paraboloid = $\frac{1}{2}$ × volume of circumscribing cylinder → rise at wall = fall at centre = $h/2$ (from original level)

Pressure: $p = \rho\omega^2 r^2/2 - \rho gz + C$

- Rotating tank applications: centrifugal pumps (forced vortex in impeller), cream separators.

★ **PYQ-TREND:** Inclination of free surface $\tan\theta = a/g$; pressure in an accelerating lift; paraboloid free surface in rotating vessel and $h = \omega^2 R^2/2g$; equal rise and fall about the original level.

✎ **DESCRIPTIVE Q:** Explain the working of a differential U-tube manometer and derive the expression for pressure difference between two pipes carrying the same liquid.

A differential U-tube manometer has a U-shaped glass tube containing a heavy manometric liquid (often mercury, density ρ_m) with each limb connected to one pipe. When pressures differ, the mercury surfaces stand at different levels; let x be the difference in mercury levels,

and let pipe A centre be y above the higher mercury surface level reference. Writing pressures at the lower mercury surface (datum) in both limbs: left limb $p_A + \rho g(y + x)$; right limb $p_B + \rho g y + \rho_m g x$. Equating: $p_A - p_B = gx(\rho_m - \rho)$. If the pipes are at different levels or carry different liquids, the corresponding $\rho g h$ terms are added. It is used with venturimeters and orifice meters to measure flow.

CHAPTER 3

Hydrostatic Forces on Surfaces

3.1 Total Pressure and Centre of Pressure

- Total pressure: the resultant force exerted by a static fluid on a surface, always normal to the surface.
- Centre of pressure: the point through which the total pressure acts. For any non-horizontal submerged surface it lies below the centroid, because pressure increases with depth.

Total pressure on any plane surface: $F = \rho g A \bar{h}$ ($\bar{h}$ = vertical depth of centroid below free surface)

Vertical plane surface: $h^* = \bar{h} + I_G/(A \bar{h})$

Inclined plane surface (angle θ with free surface): $h^* = \bar{h} + I_G \sin^2\theta / (A \bar{h})$

Horizontal surface: $F = \rho g A h$, acts at the centroid ($h^* = \bar{h}$)

- As depth of immersion increases, the centre of pressure moves closer to the centroid ($I_G/(A \bar{h}) \to 0$).
- The total force does not depend on the inclination; only the centre-of-pressure location does.

Vertical surface (free surface at top edge unless stated)	Depth of centre of pressure h*
Rectangle of depth d, top edge at surface	2d/3
Triangle, apex at surface, base horizontal at depth h	3h/4
Triangle, base at surface, apex at depth h	h/2
Circle of diameter d, touching surface	5d/8
Semicircle, diameter at surface (radius R)	3πR/16
Rectangle with top at depth a and bottom at depth b	$(2/3)(b^3 - a^3)/(b^2 - a^2)$

★ **PYQ-TREND:** Centre of pressure is below the centroid; $h^* = 2d/3$ for a rectangle with top at the surface; $F = \rho g A \bar{h}$ independent of inclination; formula $h^* = \bar{h} + I_G/(A\bar{h})$; CP approaches centroid with depth.

✔ **TOPPER TRICK:** Pressure prism: for a rectangle reaching the surface the pressure diagram is a triangle, and a triangle's centroid is at 2/3 of its height from the apex – hence 2d/3.

3.2 Curved Surfaces

Horizontal component F_H = force on the vertical projection of the curved surface ($\rho g A_{proj} \bar{h}_{proj}$), acting at its centre of pressure

Vertical component F_V = weight of the liquid (real or imaginary) directly above the curved surface up to the free surface, acting through its centroid

Resultant $F = \sqrt{F_H^2 + F_V^2}$; inclination $\tan\theta = F_V/F_H$

- If the liquid is below the curved surface, F_V equals the weight of the imaginary liquid above and acts upward.

- For a circular surface, the resultant passes through the centre of the circle (all pressure forces are radial).

★ **PYQ-TREND:** Horizontal component = force on vertical projection; vertical component = weight of liquid above; resultant on a circular arc passes through the centre.

3.3 Applications – Dams, Gates, Lock Gates

```
Force on vertical gate (width b, water depth H): F = ρgbH²/2 acting at H/3 from the bottom
Lock gates: reaction between gates R = force on each gate P, with P = R when gates make 120° (angle with the normal 30°): P = F/(2 sinα)
Gravity dam stability: no tension at heel if the resultant falls within the middle third of the base (e ≤ b/6)
```

- Gravity dam checks: overturning (factor of safety against overturning = resisting moment/overturning moment, ≥ 1.5), sliding (μΣV/ΣH ≥ 1–1.5), compressive stress, no tension (middle third).
- Uplift pressure under the dam reduces stability; drainage galleries reduce it.
- Sluice gates and Tainter (radial) gates on spillways – radial gates have the pressure resultant passing through the hinge.

★ **PYQ-TREND:** Force on a vertical gate ρgbH²/2 at H/3 above base; middle-third rule for dams; resultant on Tainter gate through the hinge.

✖ **COMMON MISTAKE:** The centre of pressure is H/3 from the bottom, i.e., 2H/3 from the free surface – check which reference the question asks.

✎ **DESCRIPTIVE Q:** Derive the expression for the depth of centre of pressure of a vertically immersed plane surface.

Consider a plane area A immersed vertically, with its centroid at depth $\bar{h}$. A strip of width b and thickness dh at depth h carries force dF = ρgh b dh. Total force F = ρg ∫h dA = ρgA$\bar{h}$. Taking moments about the free surface line: F·h* = ∫ρgh² dA = ρg I_o, where I_o is the second moment of area about the free surface line. By the parallel-axis theorem $I_o = I_G + A\bar{h}^2$. So h* = ρg(I_G + A$\bar{h}^2$)/(ρgA$\bar{h}$) = $\bar{h}$ + I_G/(A$\bar{h}$). Since I_G/(A$\bar{h}$) is always positive, the centre of pressure lies below the centroid; the gap shrinks as $\bar{h}$ increases. For an inclined surface, distances along the plane give h* = $\bar{h}$ + I_G sin²θ/(A$\bar{h}$).

CHAPTER 4

Buoyancy, Floatation and Stability

4.1 Archimedes' Principle and Buoyancy

- Buoyant force: the net upward force exerted by a fluid on a fully or partly immersed body, equal to the weight of fluid displaced (Archimedes' principle).
- Centre of buoyancy (B): the centroid of the displaced volume, through which the buoyant force acts.

```
F_B = ρ_fluid g V_displaced
Floating body: weight W = F_B → fraction submerged = ρ_body/ρ_fluid
Apparent weight of a submerged body = W − ρ_fluid g V
```

- Ice (ρ ≈ 917 kg/m³) floats in fresh water with about 92% of its volume submerged; in sea water about 89%.
- A ship rises slightly when it moves from a river into the sea (sea water is denser) – Plimsoll (load) line marks safe draughts.
- Hydrometer and lactometer work on floatation; a hydrometer sinks deeper in a lighter liquid.

★ **PYQ-TREND:** Archimedes' principle; centre of buoyancy = centroid of displaced volume; fraction submerged = ρ_body/ρ_liquid; ship rises when entering sea; hydrometer principle.

4.2 Metacentre and Metacentric Height

- Metacentre (M): the point where the line of action of the buoyant force in the slightly tilted position meets the original vertical axis of the body.
- Metacentric height GM: distance between the centre of gravity G and the metacentre M.

```
BM = I/V (I = least second moment of area of the water-plane section about the tilt axis; V = displaced volume)
GM = BM − BG = I/V − BG (BG positive when G above B)
Experimental: GM = w x / (W tanθ) (moving a load w through distance x across the deck tilts by θ)
Time period of rolling (oscillation): T = 2π √(k² / (g·GM)) (k = radius of gyration about the roll axis)
```

Stability conditions

Body	Stable	Neutral	Unstable
Floating	M above G (GM > 0)	M coincides with G	M below G
Fully submerged	B above G	B coincides with G	B below G

- For submerged bodies (submarines, balloons) the water-plane area is zero, so BM = 0 and stability depends only on the relative positions of B and G.
- Typical GM: merchant ships 0.3–1.2 m (comfortable, slow roll); warships 1–1.5 m; river craft up to 3.5 m.

- Large GM → greater stability but a shorter rolling period (uncomfortable, 'stiff' ship). Small GM → longer, gentler roll ('tender' ship).
- Weight added high above G lowers GM; ballast added low raises it.

★ **PYQ-TREND:** Stability conditions (M above G for floating; B above G for submerged) – very frequent. Metacentric height formula I/V − BG. Effect of GM on time period of rolling. Experimental determination of GM.

✔ **TOPPER TRICK:** For a floating rectangular block of width b (rolling about its long axis), $I = Lb^3/12$ and $V = L\,b\,d$, so $BM = b^2/(12d)$. This shortcut solves most block problems in seconds.

✖ **COMMON MISTAKE:** Use the least moment of inertia (about the longitudinal axis) for BM; rolling about the short axis gives a larger I and is not critical.

4.3 Stability of Floating Bodies – Practical Points

- A body floating with its long side vertical may become unstable; a cylinder of small diameter-to-length ratio floats with its axis horizontal.
- Liquid cargo with a free surface reduces effective GM (free-surface effect) – tankers use baffles and divided tanks.
- Hydrostatic stability is also important for floating platforms, pontoons and buoys.

✎ **DESCRIPTIVE Q:** Define metacentre and metacentric height and state the conditions of equilibrium for floating and submerged bodies.

When a floating body is tilted through a small angle, the shape of the displaced volume changes and the centre of buoyancy shifts from B to B'. The vertical through B' intersects the original axis of symmetry at the metacentre M. The distance GM between the centre of gravity G and M is the metacentric height. If M lies above G, the weight and buoyancy form a restoring couple and the body is in stable equilibrium; if M coincides with G, neutral; if M is below G, an overturning couple makes it unstable. For a fully submerged body, B does not shift on tilting, so the body is stable if B is above G, neutral if they coincide, and unstable if B is below G. GM is found analytically from GM = I/V − BG or experimentally by shifting a known weight across the deck and measuring the tilt.

CHAPTER 5

Fluid Kinematics

5.1 Description of Flow

- Lagrangian approach: follow individual fluid particles (like tracking a boat). Eulerian approach: observe properties at fixed points in space (like a fixed gauge on a river bank) – used in most fluid mechanics.
- Velocity field V = u i + v j + w k, each a function of (x, y, z, t).

Types of flow

Classification	Meaning
Steady / unsteady	∂(properties)/∂t = 0 at a point / not zero
Uniform / non-uniform	∂V/∂s = 0 along a streamline at an instant / not zero
Laminar / turbulent	Smooth layers / random eddying mixing (Re-based)
Compressible / incompressible	Density varies / constant (ρ = const)
Rotational / irrotational	Particles rotate about their own axes / do not (ω = 0)
1-D, 2-D, 3-D	Number of space coordinates the velocity depends on
Internal / external	Inside pipes, ducts / around bodies

- Combinations: steady uniform flow – constant-diameter pipe with constant discharge; steady non-uniform – tapering pipe with constant discharge; unsteady uniform – constant-diameter pipe with changing discharge (valve being closed); unsteady non-uniform – tapering pipe with changing discharge, waves.

★ **PYQ-TREND:** Examples of steady/unsteady and uniform/non-uniform flows (very frequent). Lagrangian vs Eulerian.

5.2 Flow Lines

- Streamline: an imaginary line whose tangent at every point gives the velocity direction at that instant. No flow crosses a streamline; two streamlines never intersect (except at stagnation points/singularities).
- Pathline: actual path traced by a single fluid particle over time.
- Streakline: locus of all particles that have passed through a fixed point (seen with continuous dye or smoke injection).
- Timeline: line formed by a set of adjacent particles marked at the same instant.
- Stream tube: a tube formed by streamlines through a closed curve; no flow through its walls.
- In steady flow, streamlines, pathlines and streaklines coincide.

```
Streamline equation: dx/u = dy/v = dz/w
```

★ **PYQ-TREND:** Definitions and coincidence of streamline/pathline/streakline in steady flow; smoke from a chimney shows a streakline; no flow across a streamline.

5.3 Continuity Equation

General (3-D, compressible): $\partial\rho/\partial t + \partial(\rho u)/\partial x + \partial(\rho v)/\partial y + \partial(\rho w)/\partial z = 0$

Steady compressible: $\nabla\cdot(\rho V) = 0$; 1-D: $\rho_1 A_1 V_1 = \rho_2 A_2 V_2$ (mass flow constant)

Incompressible (steady or unsteady): $\partial u/\partial x + \partial v/\partial y + \partial w/\partial z = 0$; 1-D: $A_1V_1 = A_2V_2 = Q$

Cylindrical: $\partial(r\, v_r)/(r\partial r) + \partial v_\theta/(r\partial\theta) + \partial v_z/\partial z = 0$

- The continuity equation expresses conservation of mass. A flow field 'exists' (is possible) only if it satisfies continuity.
- Branch pipes: $Q_1 = Q_2 + Q_3$.

★ **PYQ-TREND:** Checking whether a given velocity field satisfies continuity; $A_1V_1 = A_2V_2$; velocity $\propto 1/D^2$ in pipes.

✔ **TOPPER TRICK:** If the diameter halves, velocity becomes 4 times ($V \propto 1/D^2$).

5.4 Acceleration of a Fluid Particle

$a_x = \partial u/\partial t + u\,\partial u/\partial x + v\,\partial u/\partial y + w\,\partial u/\partial z$ (local + convective)

Along a streamline: $a_s = \partial V/\partial t + V\,\partial V/\partial s$; normal (centripetal): $a_n = V^2/R + \partial V_n/\partial t$

- Local (temporal) acceleration: due to change with time at a point (zero in steady flow).
- Convective acceleration: due to change of position (zero in uniform flow). A tapering pipe with steady flow has only convective acceleration.

★ **PYQ-TREND:** Local vs convective acceleration; which is zero in steady/uniform flow; computing acceleration from a velocity field.

5.5 Rotation, Vorticity and Circulation

Rotation about z: $\omega_z = \frac{1}{2}(\partial v/\partial x - \partial u/\partial y)$; similarly ω_x, ω_y

Vorticity $\Omega = 2\omega$ (curl of velocity)

Circulation $\Gamma = \oint V\cdot ds$ = vorticity × enclosed area (Stokes' theorem)

Angular deformation (shear strain rate) $\dot{\gamma}_{xy} = \partial v/\partial x + \partial u/\partial y$

- Irrotational flow: $\omega = 0$ everywhere – a velocity potential exists. Flow outside the boundary layer is approximately irrotational. Flow inside the boundary layer and in forced vortices is rotational.

★ **PYQ-TREND:** Condition of irrotational flow ($\partial v/\partial x = \partial u/\partial y$); vorticity = twice the rotation; circulation definition.

5.6 Velocity Potential and Stream Function

Velocity potential ϕ: $u = -\partial\phi/\partial x$, $v = -\partial\phi/\partial y$ (some texts use + sign)

Stream function ψ: $u = -\partial\psi/\partial y$, $v = \partial\psi/\partial x$ (or $u = \partial\psi/\partial y$, $v = -\partial\psi/\partial x$ – sign convention varies)

Existence: ϕ exists only for irrotational flow; ϕ satisfies Laplace's equation $\nabla^2\phi = 0$ if the flow is also incompressible

ψ exists for any 2-D incompressible flow (rotational or irrotational); $\nabla^2\psi = 0$ only if irrotational

Discharge per unit width between two streamlines $q = \psi_2 - \psi_1$

Lines of constant φ (equipotentials) and constant ψ (streamlines) intersect at right angles: slope product = −1

- Flow net: grid of streamlines and equipotential lines forming curvilinear squares; used for seepage under dams, flow around bodies. Cannot be drawn near boundaries where viscous effects dominate.
- Cauchy–Riemann relations: $\partial\phi/\partial x = \partial\psi/\partial y$ and $\partial\phi/\partial y = -\partial\psi/\partial x$.

★ **PYQ-TREND:** Existence of φ (irrotational only) vs ψ (2-D incompressible); orthogonality of φ and ψ lines; discharge = $\psi_2 - \psi_1$; flow nets and their limitations.

5.7 Basic Potential Flows and Vortex Flows

Flow	φ	ψ
Uniform flow U along x	−Ux	−Uy
Source (strength q)	−(q/2π) ln r	−(q/2π) θ
Sink	opposite signs	–
Free vortex (circulation Γ)	−(Γ/2π) θ	(Γ/2π) ln r
Doublet	(μ cosθ)/r	−(μ sinθ)/r

- Combinations: uniform flow + doublet → flow past a circular cylinder; + free vortex → lift on a rotating cylinder (Magnus effect); source + uniform flow → Rankine half-body.

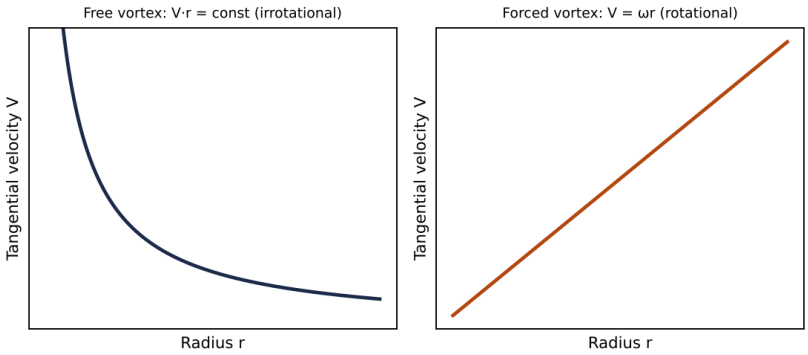
Vortex flows

Free vortex (irrotational): $V \cdot r$ = constant; pressure rises outward: $p + \rho V^2/2$ = const across streamlines

Forced vortex (rotational, rigid-body rotation): $V = \omega r$; free surface paraboloid, $h = \omega^2 R^2/(2g)$

Rankine combined vortex: forced core ($r < R$) + free vortex outside

- Free vortex examples: whirlpool in a river, flow in a sink/bath drain, flow in the casing of a centrifugal pump after the impeller, flow around a bend. Forced vortex examples: liquid in a rotating cylinder, flow inside a centrifugal pump impeller, flow in turbine runners.
- Bernoulli applies across streamlines in a free vortex (irrotational) but only along a streamline in a forced vortex.



★ **PYQ-TREND:** Free vortex Vr = const vs forced vortex V = ωr; examples of each; which is rotational (forced) and irrotational (free); Rankine vortex; potential-flow combinations (Magnus effect).

✎ **DESCRIPTIVE Q:** Distinguish between stream function and velocity potential, and prove that streamlines and equipotential lines are orthogonal.

The stream function ψ is defined so that its derivatives give the velocity components and ψ is constant along a streamline; it exists for every two-dimensional incompressible flow because it automatically satisfies continuity. The velocity potential φ is a scalar whose negative gradient gives velocity; it exists only when the flow is irrotational. Along a streamline $d\psi = (\partial\psi/\partial x)dx + (\partial\psi/\partial y)dy = v\,dx - u\,dy = 0$, so its slope is $dy/dx = v/u$. Along an equipotential line $d\varphi = -u\,dx - v\,dy = 0$, so its slope is $dy/dx = -u/v$. The product of the two slopes is -1, so the lines cross at right angles, forming the flow net.

CHAPTER 6

Fluid Dynamics – Energy and Momentum

6.1 Forces and Equations of Motion

- Forces on a fluid element: gravity (Fg), pressure (Fp), viscous (Fv), turbulence (Ft), compressibility/elastic (Fe), surface tension (Fs).

Equation	Forces considered
Euler's equation	Gravity + pressure (ideal fluid)
Navier–Stokes	Gravity + pressure + viscous
Reynolds equations	Gravity + pressure + viscous + turbulent
Newton's law (full)	All forces

Euler's equation along a streamline: $dp/\rho + g\,dz + V\,dV = 0$

★ **PYQ-TREND:** Which forces are included in Euler's vs Navier–Stokes vs Reynolds equations.

6.2 Bernoulli's Equation

$p/(\rho g) + V^2/(2g) + z$ = constant (pressure head + velocity (kinetic) head + datum (potential) head = total head)

Energy form per unit mass: $p/\rho + V^2/2 + gz$ = const; per unit volume: $p + \rho V^2/2 + \rho gz$ = const

Real fluid between sections 1 and 2: $p_1/\rho g + V_1^2/2g + z_1 = p_2/\rho g + V_2^2/2g + z_2 + h_L$

With pump (h_p) and turbine (h_t): $H_1 + h_p = H_2 + h_t + h_L$

- Assumptions: steady flow; incompressible; inviscid (frictionless); along a streamline (or irrotational flow for any two points); no shaft work or heat transfer.
- It is a statement of conservation of energy (derived by integrating Euler's equation).
- Static pressure p, dynamic pressure $\rho V^2/2$, stagnation (total) pressure $p_0 = p + \rho V^2/2$.
- Applications: venturimeter, orificemeter, pitot tube, flow nozzle, siphon, aerofoil lift, atomiser/sprayer, carburettor (low pressure at venturi throat draws fuel), chimney draught, roofs lifted in storms, two boats moving side by side drawn together.

★ **PYQ-TREND:** Assumptions of Bernoulli's equation (very frequent). Bernoulli is energy conservation. Terms as heads (m) and their names. Everyday applications (carburettor, aerofoil, atomiser).

✖ **COMMON MISTAKE:** Bernoulli's equation cannot be applied across a hydraulic jump, through a pump or turbine without adding/subtracting head, or in a region of strong viscous losses without a loss term.

6.3 HGL and TEL

- Hydraulic gradient line (HGL): line joining piezometric heads ($p/\rho g + z$) along the flow.

- Total energy line (TEL): line joining total heads ($p/\rho g + V^2/2g + z$); lies above the HGL by $V^2/2g$.
- For real fluids, the TEL always falls in the direction of flow (losses). The HGL may rise where velocity decreases (sudden enlargement) – but TEL never rises without a pump.
- Where the HGL falls below the pipe centreline, pressure is negative (below atmospheric) – risk of air release/cavitation (siphons).
- For a pipe discharging into a reservoir, HGL meets the reservoir surface at the outlet; TEL is $V^2/2g$ above it.

★ **PYQ-TREND:** TEL always slopes down in flow direction; HGL can rise; vertical distance between TEL and HGL = velocity head; HGL below pipe → negative pressure.

6.4 Correction Factors

Kinetic energy correction factor $\alpha = (1/(AV^3)) \int u^3 \, dA$

Momentum correction factor $\beta = (1/(AV^2)) \int u^2 \, dA$

Flow	α	β
Uniform (ideal)	1	1
Laminar pipe flow	2	4/3
Laminar flow between parallel plates	1.543	1.2
Turbulent pipe flow	1.03–1.10	1.01–1.04

★ **PYQ-TREND:** $\alpha = 2$ and $\beta = 4/3$ for laminar pipe flow (frequent). $\alpha > \beta > 1$ for non-uniform profiles.

6.5 Impulse–Momentum Equation and Its Applications

Momentum equation (steady): $\Sigma F = \rho Q (V_2 - V_1)$ (vector, per direction)

Force on a pipe bend (x-direction): $F_x = \rho Q(V_1 - V_2 \cos\theta) + p_1A_1 - p_2A_2 \cos\theta$

y-direction: $F_y = \rho Q(-V_2 \sin\theta) - p_2A_2 \sin\theta$ (sign depends on the coordinate choice); resultant $F = \sqrt{(F_x^2 + F_y^2)}$

Force on a nozzle (water exerts on nozzle): $F = p_1A_1 - \rho Q(V_2 - V_1)$

Moment of momentum (torque): $T = \rho Q(V_2r_2 - V_1r_1)$

- The momentum equation is used where internal losses are unknown or energy is not conserved: forces on bends, reducers, nozzles, jets on vanes, hydraulic jumps, sudden enlargement loss, jet propulsion, lawn sprinklers (moment of momentum).
- Jet propulsion: ships (water jet), rockets (no atmospheric air needed), jet aircraft.

Propulsive (Froude) efficiency of a jet: $\eta = 2u/(u + V_j)$ (u = vehicle speed, V_j = jet velocity relative to the vehicle) — maximum as $V_j \to u$

★ **PYQ-TREND:** Impulse-momentum equation; force on pipe bends; momentum equation used for sudden enlargement loss and hydraulic jump; lawn sprinkler uses moment of momentum.

6.6 Free Liquid Jets

Jet issuing at angle θ with velocity V: max height $= V^2 \sin^2\theta/(2g)$; horizontal range $= V^2 \sin2\theta/g$ (max at 45°)

Equation of jet path: $y = x \tan\theta - g x^2/(2V^2 \cos^2\theta)$

`Horizontal jet from an orifice at height y above ground: horizontal distance x = V √(2y/g); Cv = x/(2√(yH))`

★ **PYQ-TREND:** Range of a liquid jet max at 45°; Cv from jet trajectory method.

✎ **DESCRIPTIVE Q:** State Bernoulli's theorem, derive it from Euler's equation and list its limitations.

Euler's equation along a streamline for steady flow of an ideal fluid is $dp/\rho + g\,dz + V\,dV = 0$. For an incompressible fluid (ρ constant), integration gives $p/\rho + gz + V^2/2 = \text{constant}$; dividing by g, $p/\rho g + z + V^2/2g = \text{constant}$ – Bernoulli's equation, stating that the sum of pressure, potential and kinetic energy per unit weight is constant along a streamline. Limitations: (1) real fluids are viscous, so energy is lost to friction – a loss term h_L must be added; (2) the velocity is not uniform across a section, so kinetic-energy correction factor α is needed; (3) it applies only to steady flow; (4) it does not hold for compressible flow without modification; (5) it applies along a streamline unless the flow is irrotational; (6) it ignores shaft work and heat transfer (pumps, turbines) unless those heads are added explicitly.

CHAPTER 7

Flow Measurement – Meters, Orifices, Mouthpieces, Notches

7.1 Venturimeter

- Parts: short converging cone (angle ≈ 20–22°), throat, long diverging cone (angle ≈ 5–7°, to avoid flow separation and recover pressure). Pressure difference between inlet and throat is measured with a differential manometer.

Theoretical discharge $Q_{th} = A_1A_2\sqrt{(2gh)} / \sqrt{(A_1^2 - A_2^2)}$

Actual discharge $Q = C_d Q_{th}$; $C_d \approx 0.95–0.98$

Differential head (mercury–water): $h = x (S_m/S - 1) = 12.6 x$ (for water)

Inclined venturimeter: $h = (p_1/\rho g + z_1) - (p_2/\rho g + z_2)$ – same formula; the manometer reading is unaffected by inclination

- Throat diameter usually D/3 to D/2. Overall head loss is small (≈ 10–15% of differential head).
- Advantages: low energy loss, high C_d, accurate. Disadvantages: long, costly, not easy to install.

★ **PYQ-TREND:** Divergent cone angle 5–7° to avoid separation; C_d 0.95–0.98; differential manometer reading independent of inclination; why the divergent cone is longer than the convergent cone.

7.2 Orificemeter, Flow Nozzle and Others

Orificemeter: $Q = C_d a_0a_1\sqrt{(2gh)}/\sqrt{(a_1^2 - a_0^2)}$; $C_d \approx 0.6–0.65$

- Orificemeter: a thin plate with a sharp-edged hole; cheap and compact but high permanent head loss. Vena contracta forms downstream (≈ D/2).
- Flow nozzle: between venturi and orifice in cost, loss and C_d (≈ 0.93–0.98).
- Rotameter: variable-area meter; a float rises in a tapered vertical tube; reading proportional to flow; works for liquids and gases.
- Elbow (bend) meter: uses pressure difference between inner and outer walls of a bend.
- Pitot tube: measures stagnation pressure; velocity $V = C_v\sqrt{(2gh)}$, $C_v \approx 0.98–1.0$. Pitot-static (Prandtl) tube measures both static and total pressure – velocity at a point.
- Current meter: river velocity. Hot-wire anemometer: rapidly fluctuating (turbulent) velocities in gases. Laser Doppler anemometer: non-intrusive.
- Other volumetric meters: turbine meter, positive displacement (domestic water meter), electromagnetic (Faraday's law; conductive liquids only, no obstruction), ultrasonic, vortex-shedding, Coriolis (mass flow).

Meter	C_d (approx.)	Head loss
Venturimeter	0.95–0.98	Lowest
Flow nozzle	0.93–0.98	Medium

Meter	C_d (approx.)	Head loss
Orificemeter	0.60–0.65	Highest

Pitot tube: V = √(2gh); with differential manometer: h = x(S_m/S − 1)

★ **PYQ-TREND:** Order of C_d (venturi > nozzle > orifice) and head loss (reverse). Pitot tube measures point velocity/stagnation pressure. Rotameter is a variable-area meter. Electromagnetic flowmeter needs conductive liquid.

✔ **TOPPER TRICK:** Venturi, nozzle and orificemeters all use the same formula; only C_d changes.

7.3 Orifices

- Orifice: an opening in the wall or base of a tank, with a closed perimeter. Classified by size (small/large), shape (circular, rectangular, triangular), edge (sharp/bell-mouthed), discharge condition (free, drowned, partially submerged).
- Vena contracta: section of the jet where the area is minimum and streamlines are parallel; located about d/2 from the orifice plane; velocity is maximum and pressure is atmospheric there.

Coefficient of contraction C_c = a_c/a ≈ 0.61–0.69 (≈ 0.64)

Coefficient of velocity C_v = V_actual/√(2gH) ≈ 0.95–0.99

Coefficient of discharge C_d = C_c × C_v ≈ 0.62

Coefficient of resistance C_r = (1/C_v²) − 1 (loss of head = C_r × actual velocity head)

Discharge through a small orifice: Q = C_d a √(2gH)

Large rectangular orifice (breadth b, top H_1, bottom H_2): Q = (2/3) C_d b √(2g) (H_2^(3/2) − H_1^(3/2))

Fully drowned orifice: Q = C_d a √(2gH), H = difference of water levels

Time to empty a tank (area A) from H_1 to H_2 through orifice a: T = 2A(√H_1 − √H_2)/(C_d a √(2g))

Time to empty completely: T = 2A√H/(C_d a √(2g))

Hemispherical tank of radius R emptying from full: T = (14π R^(5/2))/(15 C_d a √(2g))

Time to empty a circular horizontal cylinder (length L, radius R) from full: T = (4L/(3C_d a√(2g))) (2R)^(3/2)

- Hydraulic coefficients are found experimentally – C_v by the jet trajectory method (x, y coordinates), C_c by measuring jet diameter at vena contracta, C_d by measuring actual discharge.

★ **PYQ-TREND:** C_d = C_c × C_v; typical values (0.64, 0.98, 0.62); vena contracta location; time to empty a tank ∝ √H; drowned orifice uses difference of levels.

7.4 Mouthpieces

- Mouthpiece: short tube (length 2–3 times its diameter) fitted to an orifice; increases discharge because the vena contracta forms inside and creates a partial vacuum.

Mouthpiece	C_d
External cylindrical	0.855
Convergent	≈ 0.975
Convergent–divergent	≈ 1.0 (based on throat area)
Internal (Borda's re-entrant) – running free	0.5

Mouthpiece	C_d
Internal (Borda's re-entrant) – running full	0.707

Loss at entrance of external mouthpiece ≈ 0.375 $V^2/2g$; pressure at vena contracta ≈ 2.5 m below atmospheric (absolute 7.8 m of water)

Maximum head for a convergent–divergent mouthpiece to avoid separation: H ≈ 2.5–8 m depending on area ratio

★ **PYQ-TREND:** C_d values of mouthpieces (external 0.855; Borda's running free 0.5, running full 0.707). Why a mouthpiece discharges more than an orifice.

7.5 Notches and Weirs

- Notch: an opening in the side of a small channel/tank where the liquid surface is below the top edge (usually metal plate). Weir: a notch on a large scale, usually of masonry across a river or channel. Nappe: the sheet of water flowing over; crest/sill: the bottom edge.

Rectangular notch/weir: Q = (2/3) C_d L √(2g) H^(3/2)

Triangular (V) notch: Q = (8/15) C_d √(2g) tan(θ/2) H^(5/2); for 90° V-notch with C_d = 0.6: Q ≈ 1.417 H^(5/2) (m^3/s)

Trapezoidal notch = rectangular + triangular parts

Cipolletti weir (trapezoidal, side slope 1 horizontal : 4 vertical, i.e., ≈ 14°): Q = 1.86 L H^(3/2) (m^3/s) — compensates end contractions

Stepped notch = sum of rectangular notches

Francis' formula with end contractions: Q = 1.84 (L − 0.1 n H) H^(3/2) (n = number of end contractions)

Bazin's formula (velocity of approach included): Q = m √(2g) L H^(3/2), m = 0.405 + 0.003/H

Error in discharge due to error in head: rectangular dQ/Q = (3/2) dH/H; triangular dQ/Q = (5/2) dH/H

Time to lower water level from H_1 to H_2 over a rectangular weir: T = (3A/(C_d L √(2g))) $(1/\sqrt{H_2} - 1/\sqrt{H_1})$

- Advantages of a V-notch: more accurate for small discharges (larger head for a small flow), only one parameter (H) to measure, C_d nearly constant, ventilation not needed.
- Ventilation of weirs: the space under the nappe must be at atmospheric pressure; otherwise the nappe clings (depressed or clinging nappe), increasing discharge and causing error.
- Broad-crested weir: Q_max = 1.705 C_d L H^(3/2) (critical depth on the crest = 2H/3).
- Ogee (spillway) weir follows the profile of the lower nappe of a sharp-crested weir.
- Submerged (drowned) weir: downstream level above the crest – treated as free + drowned parts.

★ **PYQ-TREND:** Q ∝ H^(3/2) (rectangular) and H^(5/2) (triangular) – very frequent. V-notch preferred for low flows. Cipolletti slope 1:4. Error ratios 3/2 and 5/2. Francis' end-contraction correction. Ventilation of weirs.

✔ **TOPPER TRICK:** Exponent rule: rectangle → 3/2, triangle → 5/2 (one more power from the widening sides).

✖ **COMMON MISTAKE:** In the V-notch formula use tan(θ/2), not tanθ. For a 90° notch, tan 45° = 1.

✎ **DESCRIPTIVE Q:** Explain the construction and working of a venturimeter, and why its divergent cone is longer than its convergent cone.

A venturimeter has a short convergent section (about 20° included angle), a throat of diameter D/3–D/2, and a long divergent section (5–7°). As fluid accelerates in the convergent cone, its pressure falls (Bernoulli); a differential manometer between the inlet and throat measures the drop h. Continuity and Bernoulli give $Q_{th} = A_1A_2\sqrt{(2gh)}/\sqrt{(A_1^2 - A_2^2)}$, and the actual discharge is $C_d Q_{th}$ with $C_d \approx 0.97$. The divergent cone is long and gentle because the flow there decelerates against an adverse pressure gradient; a rapid expansion would cause boundary-layer separation, eddies and a large energy loss. A gradual expansion recovers most of the pressure, keeping the overall loss small. The convergent cone can be short because accelerating flow (favourable pressure gradient) does not separate.

CHAPTER 8

Viscous (Laminar) Flow

8.1 Reynolds' Experiment

- Osborne Reynolds (1883) injected dye into water flowing through a glass tube. At low velocity the dye formed a straight thread (laminar); at higher velocity it wavered (transition); at still higher velocity it mixed across the tube (turbulent).

```
Reynolds number Re = ρVD/μ = VD/ν (ratio of inertia force to viscous force)
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Pipe flow regime	Re range
Laminar	Re < 2000 (some texts 2300)
Transitional	2000 – 4000
Turbulent	Re > 4000

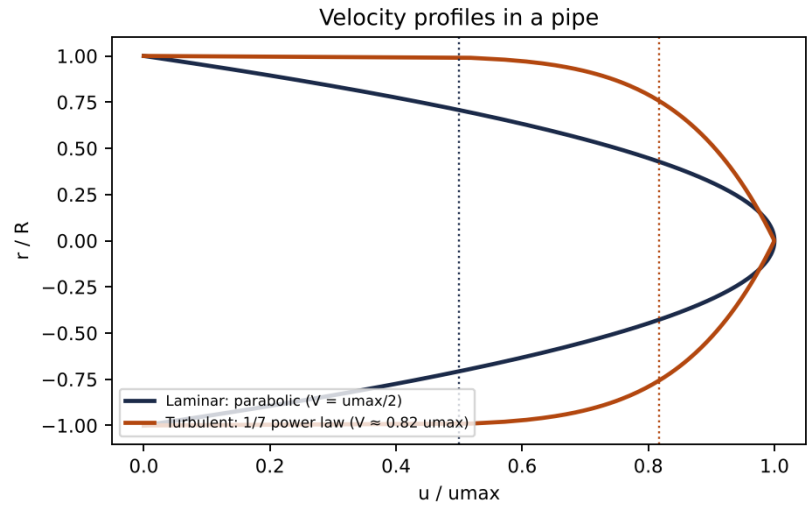
- Lower critical Re (≈ 2000) is fairly definite; the upper critical value depends on disturbances (can be very high in a carefully controlled experiment).
- For open channels (using hydraulic radius R): laminar Re < 500, turbulent Re > 2000 (approx.). For flow over a flat plate: transition at $Re_x \approx 5 \times 10^5$.

★ **PYQ-TREND:** Reynolds number definition and critical values (2000/4000); Re = inertia/viscous forces; dye experiment.

8.2 Flow Through Circular Pipes – Hagen–Poiseuille

```
Shear stress distribution: τ = (−∂p/∂x)(r/2) → zero at centre, maximum at wall τ₀ = (−∂p/∂x)(R/2)
Velocity profile: u = (1/4μ)(−∂p/∂x)(R² − r²) = u_max (1 − r²/R²) (parabolic)
Maximum velocity u_max = (1/4μ)(−∂p/∂x) R²; average velocity V = u_max/2
Discharge Q = (π/8μ)(−∂p/∂x) R⁴ = (π D⁴ Δp)/(128 μ L)
Pressure drop (Hagen–Poiseuille): Δp = 32 μ V L / D²; head loss h_f = 32 μ V L/(ρ g D²)
Darcy friction factor (laminar): f = 64/Re; Fanning friction factor f' = 16/Re
Wall shear stress τ₀ = 8μV/D; power required P = Δp × Q
Radius where local velocity equals the average velocity: r = R/√2 = 0.707R
```

- Assumptions: steady, laminar, fully developed, incompressible, Newtonian, horizontal circular pipe of constant diameter.
- Entrance (development) length for laminar flow: L_e ≈ 0.05 Re D (up to ≈ 120D at Re = 2300); for turbulent flow L_e ≈ 10–60D (≈ 4.4 Re^(1/6) D).



★ **PYQ-TREND:** Parabolic velocity profile; V_avg = u_max/2; shear stress linear (zero at centre, max at wall); $h_f = 32\mu VL/\rho gD^2$; $f = 64/Re$ (most frequent); $Q \propto D^4$ at constant pressure gradient; $\alpha = 2$, $\beta = 4/3$.

✔ **TOPPER TRICK:** At the same flow rate in laminar flow, $\Delta p \propto 1/D^4$; doubling D reduces Δp to 1/16.

8.3 Flow Between Parallel Plates

Both plates fixed (gap t): $u = (1/2\mu)(-\partial p/\partial x)(t\,y - y^2)$; $u_{max} = (1/8\mu)(-\partial p/\partial x)\,t^2$; $V_{avg} = (2/3)\,u_{max}$

Pressure drop: $\Delta p = 12\,\mu\,V\,L\,/\,t^2$

Shear stress $\tau = (-\partial p/\partial x)(t/2 - y)$ (zero at mid-plane)

One plate moving with velocity U, no pressure gradient (Couette flow): $u = U\,y/t$ (linear); $\tau = \mu U/t$

Couette flow with pressure gradient: $u = U\,y/t - (1/2\mu)(\partial p/\partial x)(t\,y - y^2)$

- Applications: leakage past pistons, flow in thin lubricating films (journal bearings), dashpots, hydraulic shock absorbers.

★ **PYQ-TREND:** $V_{avg} = (2/3)u_{max}$ between fixed plates; $\Delta p = 12\mu VL/t^2$; Couette flow linear profile; comparison with pipe flow (1/2 vs 2/3).

8.4 Lubrication, Dashpot and Stokes' Law

Journal bearing viscous torque: $T = \mu\,\pi^2\,N\,D^3\,L\,/\,(120\,t)$; power $P = 2\pi NT/60 = \mu\,\pi^3\,N^2\,D^3\,L/(3600\,t)$

Footstep bearing: $T = \mu\,\pi^2\,N\,R^4/(60\,t)$; collar bearing: $T = \mu\,\pi^2\,N\,(R_2^4 - R_1^4)/(60\,t)$

Dashpot (piston of diameter D, length L, clearance t, velocity V): $\mu = 4\,W\,t^3/(3\pi\,L\,D^3\,V)$

Stokes' law (drag on a sphere in creeping flow, $Re < 1$): $F_D = 3\pi\mu DV$; $C_D = 24/Re$

Terminal velocity of a falling sphere: $V_t = D^2(\rho_s - \rho_f)\,g\,/\,(18\mu)$

- Stokes' law is used in falling-sphere viscometers, settling of dust/sediment, Millikan's oil-drop experiment.

★ **PYQ-TREND:** Stokes' drag $3\pi\mu DV$ and $C_D = 24/Re$; terminal velocity $\propto D^2$; viscous torque on journal bearing $\propto \mu ND^3L/t$.

8.5 Measurement of Viscosity

Viscometer	Principle
Capillary tube	Hagen–Poiseuille (Δp, Q, D, L)
Falling sphere	Stokes' law (terminal velocity)
Rotating cylinder (Couette)	Torque on inner cylinder
Orifice type – Redwood (UK), Saybolt (USA), Engler (Europe)	Time for a fixed volume to flow through an orifice (commercial)
Cone-and-plate	Uniform shear rate (non-Newtonian fluids)
Ostwald	Capillary, relative to a standard liquid

★ **PYQ-TREND:** Falling-sphere viscometer uses Stokes' law; Redwood/Saybolt are orifice-type; rotating cylinder method.

✎ **DESCRIPTIVE Q:** Derive the Hagen–Poiseuille equation for laminar flow in a circular pipe.

Consider a coaxial cylindrical fluid element of radius r and length dx in steady, fully developed laminar flow. Pressure force on the ends = $(-\partial p/\partial x)\,dx\,\pi r^2$; shear force on the curved surface = $\tau\,2\pi r\,dx$. For equilibrium $\tau = (-\partial p/\partial x)(r/2)$: shear varies linearly from zero at the axis to a maximum at the wall. Using Newton's law $\tau = -\mu\,du/dr$ (u decreases with r): $du/dr = -(1/2\mu)(-\partial p/\partial x)\,r$. Integrating with $u = 0$ at $r = R$: $u = (1/4\mu)(-\partial p/\partial x)(R^2 - r^2)$, a parabola. Discharge $Q = \int_0^R u\,2\pi r\,dr = (\pi/8\mu)(-\partial p/\partial x)R^4$, so $V = Q/\pi R^2 = (1/8\mu)(-\partial p/\partial x)R^2 = u_{max}/2$. Over length L with $D = 2R$: $\Delta p = 32\mu VL/D^2$, i.e., $h_f = 32\mu VL/(\rho g D^2)$. Writing h_f in Darcy form $fLV^2/(2gD)$ gives $f = 64/Re$.

CHAPTER 9

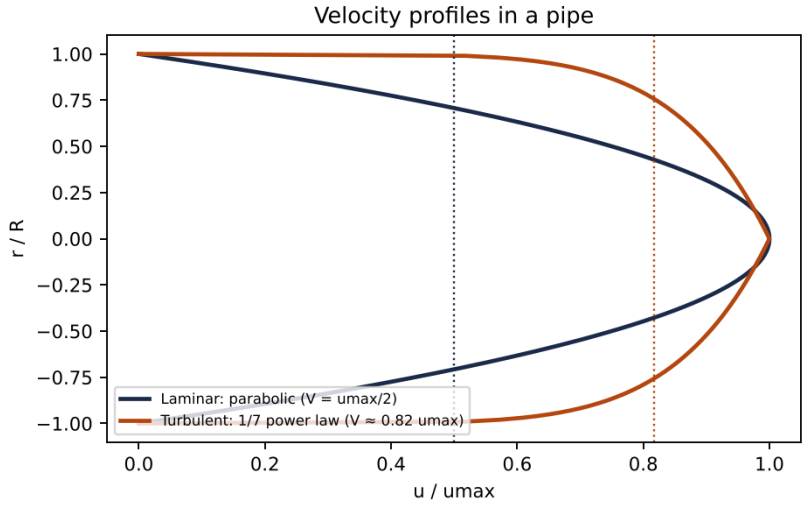
Turbulent Flow and Flow Through Pipes

9.1 Turbulent Flow Basics

- Turbulence: random, three-dimensional fluctuations of velocity superimposed on the mean flow. u = ū + u'. Mixing is far stronger than in laminar flow, so the velocity profile is flatter.

```
Turbulent shear stress: τ = τ_laminar + τ_turbulent = μ dū/dy − ρ u'v' (Reynolds stress)
Boussinesq eddy viscosity: τ_t = η dū/dy; Prandtl mixing length: τ_t = ρ l² (dū/dy)²
Power-law profile (smooth pipe, Re ≈ 10⁵): u/u_max = (y/R)^(1/7); V ≈ 0.817 u_max (often taken ≈ 0.8–0.85 u_max)
Shear (friction) velocity u* = √(τ₀/ρ)
Laminar sub-layer thickness δ' = 11.6 ν/u*
```

- Hydrodynamically smooth: roughness height k < δ' (k/δ' < 0.25); rough: k > 6δ'. In between – transition.
- In rough pipes at high Re, friction factor depends only on relative roughness (ε/D), not on Re (fully rough zone).



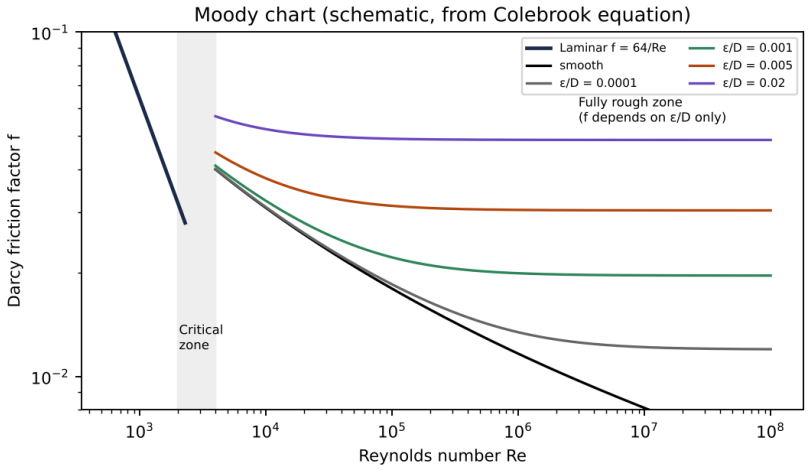
★ **PYQ-TREND:** Velocity profile flatter in turbulent flow; 1/7th power law; laminar sub-layer and smooth/rough classification; Reynolds stress; Prandtl mixing length.

9.2 Major Losses – Darcy–Weisbach and Others

```
Darcy–Weisbach: h_f = f L V² / (2gD) (f = Darcy friction factor)
In terms of discharge: h_f = f L Q² / (12.1 D⁵) = 8fLQ²/(π² g D⁵)
Fanning friction factor f' = f/4 → h_f = 4f' L V²/(2gD)
Wall shear stress τ₀ = (f/8) ρ V²
Blasius (smooth pipes, 4000 < Re < 10⁵): f = 0.316/Re^0.25
```

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Chezy's formula: V = C √(m i); m = hydraulic mean depth = A/P (= D/4 for full circular pipe); i = h_f/L; C = √(8g/f)
Manning's formula: V = (1/n) R^(2/3) S^(1/2)
Hazen–Williams (water supply): V = 0.849 C_HW R^0.63 S^0.54
Colebrook–White: 1/√f = −2 log₁₀(ε/(3.7D) + 2.51/(Re√f)) (basis of the Moody diagram)
```

- Moody diagram: f vs Re for various ε/D. Laminar zone (f = 64/Re, independent of roughness), critical zone, transition zone (f depends on both), fully rough zone (f depends on ε/D only).
- For a given Q and f, $h_f \propto 1/D^5$ – a small increase in diameter greatly reduces friction loss.



★ **PYQ-TREND:** Darcy–Weisbach formula; $h_f \propto Q^2/D^5$; hydraulic mean depth D/4; Chezy and Manning formulas; Moody diagram zones; Blasius equation.

✔ **TOPPER TRICK:** If diameter is doubled at constant Q (and f), head loss becomes 1/32. If Q is doubled, loss becomes 4 times.

✖ **COMMON MISTAKE:** Some books use 'f' for the Fanning factor ($h_f = 4fLV^2/2gD$). Darcy f = 4 × Fanning f. Check which one the question uses.

9.3 Minor Losses

Loss	Head loss
Sudden enlargement	$(V_1 - V_2)^2/2g$
Sudden contraction	$k\,V_2^2/2g$, with $k = (1/C_c - 1)^2 \approx 0.5$ (often taken 0.375–0.5)
Entrance (sharp-edged)	$0.5\,V^2/2g$
Entrance (bell-mouthed)	≈ 0.04–$0.05\,V^2/2g$
Exit (into reservoir)	$V^2/2g$
Bend / elbow / fittings	$k\,V^2/2g$ (90° elbow $k \approx 0.9$; gate valve open $k \approx 0.2$; globe valve $k \approx 10$)
Obstruction in pipe	$(A/(C_c(A - a)) - 1)^2\,V^2/2g$
Gradual enlargement	$k\,(V_1 - V_2)^2/2g$, k depends on cone angle (minimum near 7–8°)

- Equivalent length of a fitting: $L_e = k\,D/f$ – minor loss expressed as an extra length of straight pipe.

- Minor losses are 'minor' only in long pipes; in short pipes with many fittings they can dominate.
- Loss at sudden enlargement is derived using the momentum equation (Borda–Carnot loss).

Sudden enlargement loss as fraction of initial KE: $h_L = (1 - A_1/A_2)^2 V_1^2/2g$

★ **PYQ-TREND:** Sudden enlargement loss $(V_1-V_2)^2/2g$ (very frequent); exit loss = $V^2/2g$; entrance $0.5V^2/2g$; equivalent length; momentum used to derive enlargement loss.

9.4 Pipes in Series, Parallel, Branching, Siphons

Series: Q same; H = Σh_f + minor losses; equivalent pipe (Dupuit): $L/D^5 = L_1/D_1^5 + L_2/D_2^5 + \ldots$

Parallel: h_f same in each branch; $Q = Q_1 + Q_2 + \ldots$

Three-reservoir problem: continuity at the junction + equal piezometric head at the junction for all branches

- Siphon: a long bent pipe carrying liquid from a higher reservoir over a ridge (summit) to a lower one. The summit is above the HGL, so pressure there is below atmospheric. Practical limit of the summit above the upstream water level ≈ 7.6 m of water (to avoid vapour/air release; theoretical 10.3 m).
- Applications: emptying tanks, irrigation canals over embankments, water supply over hills.

★ **PYQ-TREND:** Series vs parallel rules; Dupuit's equation; siphon summit pressure negative and practical height limit ≈ 7.6–8 m.

9.5 Power Transmission Through Pipes and Nozzles

Power available at outlet: $P = \rho g Q (H - h_f)$

Efficiency of transmission $\eta = (H - h_f)/H$

Condition for maximum power transmitted: $h_f = H/3 \rightarrow \eta_{max} = 66.67\%$

Nozzle at end of pipe: maximum power when the velocity head of the jet = 2H/3, i.e., $h_f = H/3$

Diameter of nozzle for maximum power: $d = (D^5/(8fL))^{(1/4)}$ (Darcy f)

★ **PYQ-TREND:** Maximum power transmission when $h_f = H/3$ and $\eta = 66.7\%$ (very frequent); nozzle diameter for maximum power.

9.6 Water Hammer and Surge Tanks

- Water hammer: pressure surge caused by sudden closure of a valve in a pipe carrying liquid; the kinetic energy of the moving column is converted into elastic strain energy of the liquid and pipe.

Pressure wave velocity: $c = \sqrt{(K/\rho)}$ (rigid pipe); elastic pipe: $c = \sqrt{((K/\rho)/(1 + KD/(Et)))}$

Critical time of closure $t_c = 2L/c$

Sudden (instantaneous, $t < 2L/c$) closure: $\Delta p = \rho V c$

Gradual closure ($t > 2L/c$): $\Delta p = \rho L V / t$

- c for water in a rigid pipe ≈ 1,450 m/s; in elastic pipes 600–1,200 m/s.
- Protection: slow valve closure, surge tanks (between penstock and turbine in hydro plants – absorb surges and supply water during sudden load increase), relief valves, air vessels.

★ **PYQ-TREND:** Water hammer pressure ρVc for sudden closure; critical time 2L/c; surge tank function; c = √(K/ρ).

9.7 Pipe Networks and Other Topics

- Hardy Cross method: iterative method for pipe networks (loops); Σh_f around a loop = 0 and continuity at nodes.
- Hydraulic gradient for economic pipe design; commercial pipe roughness: drawn tubing ≈ 0.0015 mm, commercial steel ≈ 0.045 mm, cast iron ≈ 0.26 mm, concrete 0.3–3 mm.
- Ageing of pipes increases roughness (tuberculation) and reduces carrying capacity.

★ **PYQ-TREND:** Hardy Cross method for networks; effect of ageing on roughness.

✎ **DESCRIPTIVE Q:** Derive the expression for loss of head due to sudden enlargement of a pipe.

Let a pipe of area A_1 (velocity V_1, pressure p_1) suddenly enlarge to area A_2 (V_2, p_2). Just after the enlargement, eddies form in the corners, and experiments show the pressure on the annular face is approximately p_1 (call it p'). Applying the momentum equation to the control volume between section 1 (just after enlargement) and section 2 (where flow is re-established): $(p_1 - p_2)A_2 = \rho Q(V_2 - V_1) = \rho A_2 V_2(V_2 - V_1)$. So $(p_1 - p_2)/\rho g = V_2(V_2 - V_1)/g$. Applying the energy equation: $p_1/\rho g + V_1^2/2g = p_2/\rho g + V_2^2/2g + h_L$, so $h_L = (p_1 - p_2)/\rho g + (V_1^2 - V_2^2)/2g = [2V_2^2 - 2V_1V_2 + V_1^2 - V_2^2]/2g = (V_1 - V_2)^2/2g$. This is the Borda–Carnot result.

CHAPTER 10

Boundary Layer, Drag and Lift

10.1 Boundary Layer Concept

- Prandtl (1904): when a real fluid flows past a solid surface, the no-slip condition makes the fluid velocity zero at the wall. Velocity rises to the free-stream value U within a thin layer – the boundary layer – where viscous effects are significant. Outside it the flow is nearly inviscid (potential flow).
- Along a flat plate the boundary layer starts laminar, passes through a transition region and becomes turbulent beyond a critical Reynolds number $Re_x = Ux/\nu \approx 5 \times 10^5$. Below the turbulent boundary layer a very thin laminar (viscous) sub-layer remains.
- Boundary-layer thickness grows downstream: laminar $\delta \propto \sqrt{x}$; turbulent $\delta \propto x^{(4/5)}$.

Thickness definitions

Boundary layer thickness δ: distance where $u = 0.99\,U$

Displacement thickness $\delta^* = \int_0^{\delta} (1 - u/U)\,dy$ (distance by which the outer flow is displaced)

Momentum thickness $\theta = \int_0^{\delta} (u/U)(1 - u/U)\,dy$ (loss of momentum flux)

Energy thickness $\delta^{**} = \int_0^{\delta} (u/U)(1 - u^2/U^2)\,dy$

Shape factor $H = \delta^*/\theta$ (Blasius laminar ≈ 2.59; turbulent ≈ 1.3–1.4)

Order (Blasius profile): $\delta > \delta^* > \delta^{**} > \theta$ ($\delta^* = 1.72$, $\delta^{**} \approx 1.04$, $\theta = 0.664$ in units of $x/\sqrt{Re_x}$)

Linear profile $u/U = y/\delta$: $\delta^* = \delta/2$, $\theta = \delta/6$, $H = 3$

Parabolic profile $u/U = 2(y/\delta) - (y/\delta)^2$: $\delta^* = \delta/3$, $\theta = 2\delta/15$, $H = 2.5$

Flat plate results

Laminar (Blasius): $\delta = 5x/\sqrt{Re_x}$ (exact ≈ 4.91); $\delta^* = 1.72x/\sqrt{Re_x}$; $\theta = 0.664x/\sqrt{Re_x}$

Local skin-friction coefficient (laminar): $C_f = 0.664/\sqrt{Re_x}$; average $C_D = 1.328/\sqrt{Re_L}$

Turbulent (1/7 power law): $\delta = 0.37x/Re_x^{(1/5)}$; $C_D = 0.074/Re_L^{(1/5)}$ ($5\times10^5 < Re < 10^7$)

Von Kármán momentum integral equation: $\tau_0/(\rho U^2) = d\theta/dx$

★ **PYQ-TREND:** Definitions of δ, δ^*, θ; $\delta \propto \sqrt{x}$ for laminar; Blasius result $5x/\sqrt{Re_x}$; displacement thickness meaning; critical Re for flat plate 5×10^5; shape factor; momentum integral equation.

10.2 Boundary Layer Separation

- When the pressure increases in the flow direction (adverse pressure gradient, $dp/dx > 0$), fluid near the wall, already slowed by friction, is further decelerated. At some point $(\partial u/\partial y)_{wall} = 0$ – the separation point – and beyond it reverse flow occurs. The main flow detaches, forming a wake of eddies.
- Separation occurs behind bluff bodies (cylinders, spheres), in diffusers with large angles, on aerofoils at high angles of attack (stall), at sudden enlargements.

- Conditions: favourable gradient (dp/dx < 0) – no separation; zero gradient (flat plate) – no separation; adverse (dp/dx > 0) – possible separation.
- Methods to control/delay separation: streamlining the body, suction of the slow boundary layer, injection of high-energy fluid (blowing), slotted wings/flaps, rotating cylinders at the boundary (Magnus), vortex generators, guide vanes in bends, roughening to trigger turbulence (golf ball dimples).
- A turbulent boundary layer has more momentum near the wall and resists separation longer than a laminar one.

★ **PYQ-TREND:** Separation occurs with adverse pressure gradient and at (∂u/∂y)_wall = 0 (very frequent). Methods to prevent separation. Why golf balls have dimples. Stall of aerofoils.

10.3 Drag and Lift

Drag $F_D = C_D (\frac{1}{2} \rho U^2) A$; Lift $F_L = C_L (\frac{1}{2} \rho U^2) A$ (A = projected/frontal area for drag, plan area for aerofoils)

- Total drag = pressure (form) drag + skin-friction (viscous) drag.
- Streamlined body (aerofoil, thin plate parallel to flow): mostly skin-friction drag; small wake. Bluff body (sphere, cylinder, plate normal to flow): mostly pressure drag; large wake.

Body	Approx. C_D (high Re)
Flat plate normal to flow	2.0 (2-D), 1.1–1.2 (disc)
Circular cylinder (laminar BL, $10^4 < Re < 2\times10^5$)	1.2
Sphere ($10^3 < Re < 2\times10^5$)	0.4–0.5
Streamlined body	0.05–0.1
Sphere in Stokes regime (Re < 1)	24/Re

- Drag crisis: at Re ≈ 2–3 × 10^5 for spheres/cylinders the boundary layer turns turbulent before separation, the wake narrows, and C_D drops sharply.
- Karman vortex street: alternate shedding of vortices behind a cylinder ($40 < Re < 10^5$); shedding frequency given by the Strouhal number St = fD/U ≈ 0.2 – causes vibration of chimneys, power lines ('singing wires') and was a factor in the Tacoma Narrows bridge collapse.

Kutta–Joukowski: Lift per unit span $L = \rho U \Gamma$

Magnus effect – rotating cylinder/ball in a stream experiences lift: swing of a cricket ball, spin of a tennis ball, Flettner rotor ships

- Aerofoil terms: chord, camber, angle of attack (lift rises with it until stall at about 12–16°), lift-to-drag ratio (glide ratio).
- Terminal velocity: weight = buoyancy + drag.

★ **PYQ-TREND:** Form drag dominates bluff bodies and skin friction dominates streamlined bodies; Magnus effect and Kutta–Joukowski theorem; drag crisis; Karman vortex street and Strouhal number; drag formula with ½ρU²A.

✎ **DESCRIPTIVE Q:** What is boundary layer separation? Explain its causes and methods of control.

In a boundary layer the velocity rises from zero at the wall to the free-stream value. If the pressure outside increases along the flow (adverse pressure gradient), the slow fluid near the wall cannot overcome both friction and the rising pressure. It decelerates until the wall velocity gradient becomes zero; this is the separation point. Downstream the near-wall flow reverses

and the main stream detaches, leaving a low-pressure wake of eddies that greatly increases pressure drag and energy loss. Causes: adverse pressure gradient from body curvature (rear of cylinders and spheres), diverging passages (diffusers, sudden enlargements), high angles of attack on aerofoils. Control: streamlining to reduce the adverse gradient; suction of the retarded fluid through slots; blowing high-velocity fluid into the boundary layer; slotted wings and flaps; vortex generators and surface roughness to make the layer turbulent; moving-wall devices like rotating cylinders.

CHAPTER 11

Dimensional Analysis and Model Similitude

11.1 Dimensions of Common Quantities

Quantity	Dimensions (MLT)
Velocity	LT^{-1}
Acceleration	LT^{-2}
Discharge	L^3T^{-1}
Force	MLT^{-2}
Pressure, stress, bulk modulus	$ML^{-1}T^{-2}$
Energy, work, torque	ML^2T^{-2}
Power	ML^2T^{-3}
Density	ML^{-3}
Specific weight	$ML^{-2}T^{-2}$
Dynamic viscosity	$ML^{-1}T^{-1}$
Kinematic viscosity	L^2T^{-1}
Surface tension	MT^{-2}
Momentum	MLT^{-1}
Angular velocity	T^{-1}
Strain, angle, specific gravity	Dimensionless

- Dimensional homogeneity: every term in a physically meaningful equation has the same dimensions (Fourier's principle).

★ **PYQ-TREND:** Dimensions of viscosity, surface tension, power, kinematic viscosity (very frequent); principle of dimensional homogeneity.

11.2 Methods of Dimensional Analysis

- Rayleigh's method: express the dependent variable as a product of powers of independent variables; suitable when the number of variables is small (≤ 4).
- Buckingham π-theorem: if a phenomenon involves n variables with m fundamental dimensions, it can be expressed in terms of $(n - m)$ independent dimensionless π-terms. Repeating variables (m of them) should together contain all fundamental dimensions, should not form a dimensionless group, and should not be the dependent variable; typically choose one geometric (D or L), one kinematic (V) and one dynamic (ρ) property.

```
Number of π-terms = n − m
```

★ **PYQ-TREND:** Buckingham π-theorem statement and $n - m$; choice of repeating variables; Rayleigh method.

11.3 Dimensionless Numbers

Number	Definition	Ratio	Significant in
Reynolds (Re)	$\rho VL/\mu$	Inertia / viscous	Pipe flow, closed conduits, fully submerged bodies (submarines, aircraft at low speed), viscous flows
Froude (Fr)	$V/\sqrt{(gL)}$	√(Inertia / gravity)	Free-surface flows: open channels, spillways, weirs, ship resistance (waves)
Euler (Eu)	$V/\sqrt{(p/\rho)}$ (or $\Delta p/\rho V^2$)	Inertia / pressure (or reverse)	Cavitation, pressure-dominated flows, pipe flow with pressure drop
Weber (We)	$V/\sqrt{(\sigma/\rho L)}$	√(Inertia / surface tension)	Capillary flows, droplets, thin films, small waves, atomisation
Mach (M)	$V/c = V/\sqrt{(K/\rho)}$	√(Inertia / elastic)	Compressible flow, high-speed aircraft, projectiles, water hammer
Cauchy (Ca)	$\rho V^2/K$	Inertia / elastic	Compressible flow ($Ca = M^2$)
Strouhal (St)	fL/V	Unsteady/oscillation	Vortex shedding

★ **PYQ-TREND:** Meaning and application of Re, Fr, We, M, Eu (very frequent). Froude for open channels and ships; Reynolds for pipes and submarines; Mach for compressible flow; Weber for surface tension.

11.4 Similitude and Model Laws

- Geometric similarity: ratios of all corresponding lengths are equal (L_r).
- Kinematic similarity: ratios of velocities and accelerations at corresponding points are equal.
- Dynamic similarity: ratios of corresponding forces are equal. Complete similarity requires all three.

Model law	Condition	Velocity ratio V_r	Discharge ratio Q_r	Time ratio T_r	Force ratio F_r
Reynolds (same fluid)	Re_m = Re_p	1/L_r	L_r	L_r^2	1
Froude	Fr_m = Fr_p	√L_r	L_r^(5/2)	√L_r	L_r^3
Mach	M_m = M_p	c_r	–	–	–
Weber	We_m = We_p	1/√L_r (same fluid)	–	–	–

```
Froude law (same g): power ratio P_r = L_r^(7/2); pressure ratio p_r = L_r; acceleration ratio = 1
Reynolds law: V_r L_r / ν_r = 1
```

- Distorted models: different horizontal and vertical scales (rivers, harbours, estuaries) – to keep depths measurable and flow turbulent. Undistorted models have the same scale in all

directions.

- Scale effect: differences between model and prototype behaviour because all force ratios cannot be matched simultaneously (e.g., ship models satisfy Froude but not Reynolds; skin friction corrected separately – Froude's hypothesis).

★ **PYQ-TREND:** Froude model ratios (V ∝ √L, Q ∝ L^2.5, T ∝ √L, F ∝ L^3, P ∝ L^3.5) – very frequent. Reynolds model V ∝ 1/L for the same fluid. Distorted models for rivers. Types of similarity.

✔ **TOPPER TRICK:** For Froude models remember the exponents in steps of ½ on length: velocity ½, time ½, discharge 5/2, force 3, power 7/2.

✖ **COMMON MISTAKE:** Under the Reynolds law with the same fluid, a smaller model needs a HIGHER velocity ($V_m/V_p = L_p/L_m$). Under the Froude law the model velocity is LOWER.

✎ **DESCRIPTIVE Q:** State Buckingham's π-theorem and explain the selection of repeating variables.

If a physical phenomenon involves n dimensional variables and these contain m fundamental dimensions (M, L, T), the relationship $f(X_1, X_2, \ldots, X_n) = 0$ can be rewritten as $F(\pi_1, \pi_2, \ldots, \pi_{n-m}) = 0$, where each π is an independent dimensionless product. To form the π-terms, m repeating variables are chosen and each remaining variable is combined with them in turn. Rules for repeating variables: (1) the dependent variable should not be chosen; (2) together they must contain all m fundamental dimensions; (3) they must not form a dimensionless group among themselves; (4) they should not have the same dimensions; (5) preferably one geometric property (diameter, length), one flow property (velocity) and one fluid property (density) are chosen. For example, pipe-friction pressure drop depends on D, V, ρ, μ, L, ε; with D, V, ρ repeating, the π-terms are $\Delta p/\rho V^2$, $\rho VD/\mu$, L/D and ε/D.

CHAPTER 12

Compressible Flow (Basics)

12.1 Speed of Sound and Mach Number

Speed of sound: $c = \sqrt{(dp/d\rho)} = \sqrt{(K/\rho)}$; ideal gas $c = \sqrt{(\gamma RT)}$; air at 15 °C ≈ 340 m/s, 20 °C ≈ 343 m/s

Mach number $M = V/c$

Regime	Mach number
Incompressible (approx.)	$M < 0.3$
Subsonic	$M < 1$
Transonic	$0.8 < M < 1.2$
Sonic	$M = 1$
Supersonic	$1 < M < 5$
Hypersonic	$M > 5$

Mach angle: $\sin \alpha = 1/M$ (Mach cone exists only for $M \geq 1$)

- The region inside the Mach cone is the 'zone of action'; outside is the 'zone of silence' – an observer hears a supersonic aircraft only after the cone passes.
- Density changes below about 5% ($M < 0.3$) are negligible – the flow can be treated as incompressible.

★ **PYQ-TREND:** Speed of sound $\sqrt{(\gamma RT)}$; Mach angle $\sin \alpha = 1/M$; flow regimes by Mach number; zone of silence.

12.2 Stagnation Properties and Isentropic Flow

Energy equation (adiabatic): $h + V^2/2 = h_0 \rightarrow c_pT + V^2/2 = c_pT_0$

Stagnation temperature: $T_0/T = 1 + ((\gamma - 1)/2)\, M^2$

Stagnation pressure: $p_0/p = [1 + ((\gamma - 1)/2)\, M^2]^{(\gamma/(\gamma-1))}$

Stagnation density: $\rho_0/\rho = [1 + ((\gamma - 1)/2)\, M^2]^{(1/(\gamma-1))}$

Critical (sonic, $M = 1$) ratios for air ($\gamma = 1.4$): $T^*/T_0 = 0.833$, $p^*/p_0 = 0.528$, $\rho^*/\rho_0 = 0.634$

Area–velocity relation: $dA/A = (M^2 - 1)\, dV/V$

- Subsonic flow ($M < 1$): velocity increases in a converging duct (nozzle) and decreases in a diverging duct (diffuser). Supersonic flow ($M > 1$): the opposite – velocity increases in a diverging duct. To accelerate from subsonic to supersonic, a convergent–divergent (de Laval) nozzle is needed, with $M = 1$ at the throat.
- Choked flow: when the throat reaches $M = 1$, mass flow is maximum; reducing the back pressure further does not increase the flow.

Maximum mass flow (choked): $\dot{m}_{max} = 0.0404\, A^* p_0/\sqrt{T_0}$ (air, SI)

★ **PYQ-TREND:** Stagnation temperature relation; critical pressure ratio 0.528 for air; area–velocity relation and behaviour of nozzles/diffusers in subsonic vs supersonic flow; choking at M = 1; de Laval nozzle.

12.3 Normal Shock Waves

- A normal shock is an abrupt, irreversible compression from supersonic ($M_1 > 1$) to subsonic ($M_2 < 1$) flow, across a very thin region.

Property across normal shock	Change
Mach number	Decreases (supersonic → subsonic)
Static pressure	Increases
Static temperature	Increases
Density	Increases
Velocity	Decreases
Stagnation temperature	Constant (adiabatic)
Stagnation pressure	Decreases (loss)
Entropy	Increases

$$M_2^2 = (M_1^2 + 2/(\gamma - 1)) / ((2\gamma/(\gamma - 1)) M_1^2 - 1)$$

- Oblique shocks form at wedges and in supersonic inlets; expansion fans (Prandtl–Meyer) occur at convex corners.
- Fanno flow: adiabatic flow with friction in a constant-area duct. Rayleigh flow: frictionless flow with heat transfer in a constant-area duct. Both drive the Mach number towards 1.

★ **PYQ-TREND:** Across a normal shock stagnation temperature is constant and stagnation pressure falls; shock converts supersonic to subsonic; entropy increases; Fanno vs Rayleigh lines.

✖ **COMMON MISTAKE:** A normal shock cannot occur in subsonic flow (it would decrease entropy).

CHAPTER 13

Open Channel Flow

13.1 Basics

- Open-channel flow has a free surface at atmospheric pressure; it is driven by gravity (bed slope). Examples: rivers, canals, partly full sewers, culverts, flumes.

Classification	Meaning
Steady / unsteady	Depth at a section constant / varying with time
Uniform / non-uniform (varied)	Depth constant along the channel / changing
Gradually varied flow (GVF)	Depth changes slowly (backwater curves)
Rapidly varied flow (RVF)	Depth changes abruptly (hydraulic jump, flow over weirs)
Subcritical / critical / supercritical	Fr < 1 / = 1 / > 1

```
Hydraulic radius (hydraulic mean depth) R = A/P; hydraulic depth D_h = A/T (T = top width)
Froude number Fr = V/√(g D_h) (rectangular: V/√(gy))
```

- In uniform flow the bed slope, water-surface slope and energy-line slope are all equal.

★ **PYQ-TREND:** Hydraulic radius; Froude number classification; uniform flow – all slopes equal; examples of GVF and RVF.

13.2 Uniform Flow Formulae

```
Chezy: V = C √(R S); Q = A C √(R S)
Manning: V = (1/n) R^(2/3) S^(1/2); Chezy C related: C = R^(1/6)/n
Bazin: C = 87/(1 + K/√R); Kutter's formula (older)
```

- Manning's n: smooth concrete ≈ 0.012–0.013, brickwork ≈ 0.015, earth channels ≈ 0.02–0.03, natural rivers 0.03–0.05.

★ **PYQ-TREND:** Chezy and Manning formulas; relationship C = R^(1/6)/n; typical n values.

13.3 Most Economical (Efficient) Sections

- The most economical section carries the maximum discharge for a given area (or needs minimum area for a given discharge) – i.e., minimum wetted perimeter.

Section	Condition	Hydraulic radius
Rectangular	b = 2y (width = 2 × depth)	R = y/2
Trapezoidal	Half of a regular hexagon; side slopes 60° with horizontal (1/√3 H : 1 V); top width = 2 × sloping side length; a semicircle of radius y can be inscribed	R = y/2
Triangular	Vertex angle 90° (sides at 45°)	R = y/(2√2)

Section	Condition	Hydraulic radius
Circular (max discharge)	y ≈ 0.938 D (Chezy) / 0.95 D (Manning)	–
Circular (max velocity)	y ≈ 0.81 D	R ≈ 0.304 D

★ **PYQ-TREND:** Most economical rectangular section b = 2y and R = y/2 (very frequent); trapezoidal as half hexagon; triangular 90° vertex; circular channel depths for max velocity (0.81D) and max discharge (0.95D).

13.4 Specific Energy and Critical Flow

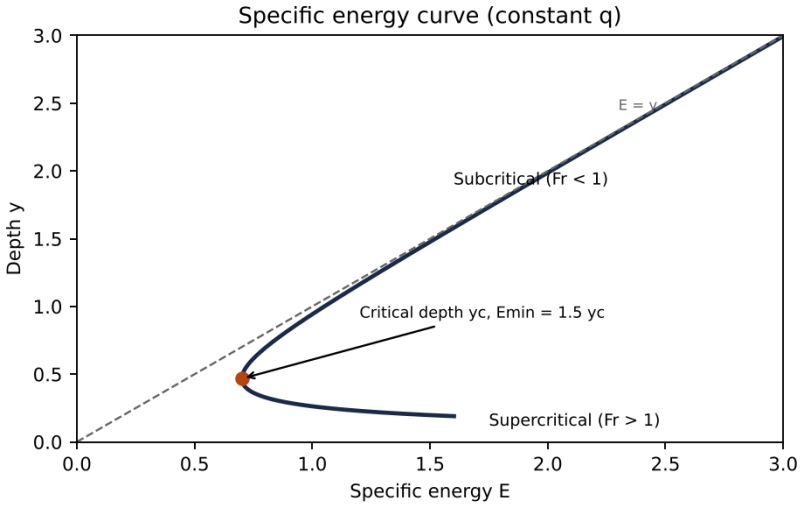
Specific energy $E = y + V^2/(2g) = y + q^2/(2gy^2)$ (q = discharge per unit width)

Critical depth (rectangular): $y_c = (q^2/g)^{1/3}$; $E_{min} = (3/2)\, y_c$; $V_c = \sqrt{g\, y_c}$

At critical flow: $Fr = 1$, specific energy minimum for given q, discharge maximum for given E

General critical condition: $Q^2\, T / (g\, A^3) = 1$

- For a given E there are two alternate depths (subcritical deep/slow and supercritical shallow/fast) except at critical depth.
- Critical slope: slope that maintains uniform flow at critical depth. Mild slope ($y_n > y_c$), steep slope ($y_n < y_c$).
- Uses: critical-depth flumes (Parshall flume, venturi flume) measure discharge; broad-crested weir.



★ **PYQ-TREND:** Critical depth $(q^2/g)^{1/3}$ and $E_{min} = 1.5\, y_c$ (very frequent); minimum specific energy at critical depth; alternate depths; Parshall flume.

13.5 Hydraulic Jump

- A hydraulic jump is the abrupt rise of the water surface when supercritical flow changes to subcritical flow. It involves intense turbulence and rollers and dissipates large amounts of energy.

Sequent (conjugate) depths: $y_2/y_1 = \frac{1}{2}\,[-1 + \sqrt{(1 + 8\, Fr_1^2)}]$

Energy loss: $\Delta E = (y_2 - y_1)^3 / (4\, y_1\, y_2)$

Length of jump ≈ 5–7 $(y_2 - y_1)$ (≈ 6.1 y_2 for many designs)

- Derived from the momentum equation (energy is not conserved across the jump).

- Uses: energy dissipation below spillways and sluice gates (stilling basins), protecting the channel bed from scour, raising the downstream level for irrigation, mixing chemicals in water treatment, aeration, desalting.
- Classification by Fr_1: undular (1–1.7), weak (1.7–2.5), oscillating (2.5–4.5), steady (4.5–9, best for design), strong (> 9).
- Surges (moving hydraulic jumps), tidal bores.

★ **PYQ-TREND:** Jump converts supercritical to subcritical; sequent depth formula; energy loss $(y_2 - y_1)^3/4y_1y_2$; momentum (not energy) is conserved; uses in stilling basins.

✎ **DESCRIPTIVE Q:** Explain the most economical rectangular channel section and prove that the width is twice the depth.

For a given area A = by and given slope and roughness, discharge $Q = AC\sqrt{(RS)}$ is maximum when the hydraulic radius R = A/P is maximum, i.e., when the wetted perimeter P is minimum. For a rectangle $P = b + 2y = A/y + 2y$. Setting $dP/dy = -A/y^2 + 2 = 0$ gives $A = 2y^2$, so $by = 2y^2$, i.e., $b = 2y$. Then $R = A/P = 2y^2/(4y) = y/2$. Hence the most economical rectangular section has a width twice the depth and a hydraulic radius equal to half the depth.

CHAPTER 14

Impact of Jets

14.1 Force of a Jet on Fixed Plates and Vanes

Fixed flat plate normal to jet: $F = \rho a V^2$

Fixed flat plate inclined at θ to the jet (normal force): $F_n = \rho a V^2 \sin\theta$; component along jet $F_x = \rho a V^2 \sin^2\theta$; across $F_y = \rho a V^2 \sin\theta \cos\theta$

Fixed curved vane, jet striking at centre and deflected through angle $(180° - \varphi)$: $F_x = \rho a V^2 (1 + \cos\varphi)$

Semicircular vane (jet reversed, $\varphi = 0$): $F = 2\rho a V^2$

Curved plate, jet at one tip (unsymmetrical): $F_x = \rho a V(V_1\cos\theta + V_2\cos\varphi)$

- Force = rate of change of momentum (impulse–momentum principle). Friction on vanes reduces the outlet relative velocity: $V_{r2} = k V_{r1}$.

★ **PYQ-TREND:** $F = \rho a V^2$ for fixed flat plate; semicircular vane doubles the force ($2\rho a V^2$); inclined plate normal force $\rho a V^2 \sin\theta$.

14.2 Moving Plates and Vanes

Single flat plate moving in the jet direction with u: $F = \rho a (V - u)^2$; work done per second = $\rho a (V - u)^2 u$

Efficiency (single moving plate): $\eta = 2u(V - u)^2/V^3$; maximum at $u = V/3$, $\eta_{max} = 8/27 = 29.6\%$

Series of flat plates (wheel): $F = \rho a V (V - u)$; $\eta = 2u(V - u)/V^2$; maximum at $u = V/2$, $\eta_{max} = 50\%$

Series of radial curved vanes, jet deflected by $(180° - \varphi)$: $\eta_{max} = (1 + \cos\varphi)/2$ at $u = V/2$

Semicircular buckets ($\varphi = 0$): $\eta_{max} = 100\%$ (theoretical)

Curved moving vane, velocity triangles: work done per second = $\rho a V_1 (V_{w1} \pm V_{w2}) u$; per unit weight = $(V_{w1} \pm V_{w2}) u / g$

- Take '+' when the outlet whirl component V_{w2} is opposite in direction to V_{w1} (the vane turns the jet back), and '−' when it is in the same direction.
- For a single moving vane, the mass striking per second is $\rho a(V - u)$; for a series of vanes (wheel) the whole jet $\rho a V$ is used because successive vanes catch it.
- Velocity triangles: absolute velocity V = vector sum of blade velocity u and relative velocity V_r. Inlet blade angle chosen so that the jet enters without shock (tangential to blade).

★ **PYQ-TREND:** $\eta_{max} = 50\%$ at $u = V/2$ for a series of flat plates (very frequent); single moving plate $u = V/3$, 8/27; curved vanes give higher efficiency; series vs single plate mass flow.

✔ **TOPPER TRICK:** Single plate → V/3; series of plates → V/2. Remember 'series = Pelton style = V/2'.

14.3 Hinged Plates

Plate hinged at the top, jet striking at the centre of gravity: $\sin\theta = \rho a V^2/W$ (W = weight of plate)

★ **PYQ-TREND:** Angle of deflection of a hinged plate struck by a jet.

CHAPTER 15

Hydraulic Turbines

15.1 Hydroelectric Plant and Heads

- Layout: reservoir/dam → intake with trash rack → penstock (with surge tank) → turbine (scroll casing, guide vanes, runner) → draft tube → tail race. The turbine drives a generator.

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Gross head H_g = head race level − tail race level
Net (effective) head H = H_g − h_f (penstock losses); Pelton: H measured at the nozzle base; reaction: H = (p₁/ρg + V₁²/2g + z₁) − (V₂²/2g at draft tube exit)
Water power (input) = ρ g Q H
```

Efficiencies

```
Hydraulic efficiency η_h = power developed by runner / water power = (V_w1 u₁ ± V_w2 u₂)/(gH)
Mechanical efficiency η_m = shaft power / runner power
Volumetric efficiency η_v = Q actually striking runner / Q supplied
Overall efficiency η_o = shaft power / (ρgQH) = η_h × η_m (× η_v)
Euler's turbine equation: work per unit mass = V_w1 u₁ ± V_w2 u₂
```

- Maximum hydraulic efficiency in reaction turbines occurs with radial discharge (V_w2 = 0): η_h = V_w1 u₁/(gH).

★ **PYQ-TREND:** Gross vs net head; definitions of hydraulic, mechanical, overall efficiency; Euler's equation; radial discharge for max efficiency.

15.2 Classification of Turbines

Basis	Types
Energy at inlet	Impulse (only kinetic energy; atmospheric pressure throughout) – Pelton, Turgo, cross-flow (Banki). Reaction (kinetic + pressure energy) – Francis, Kaplan, Propeller, Deriaz
Direction of flow through runner	Tangential (Pelton); radial inward (old Francis, Thomson); radial outward (Fourneyron); mixed (modern Francis); axial (Kaplan, Propeller)
Head	High (> 250 m): Pelton; medium (60–250 m): Francis; low (< 60 m): Kaplan/Propeller
Specific speed	Low: Pelton; medium: Francis; high: Kaplan
Disposition of shaft	Horizontal (usually Pelton) / vertical (Francis, Kaplan)

Turbine	Head (m)	Ns (SI, kW)	Flow direction	Type
Pelton, single jet	> 250 (up to ~1800)	8.5–30	Tangential	Impulse
Pelton, multi-jet	> 250	30–51	Tangential	Impulse

Turbine	Head (m)	Ns (SI, kW)	Flow direction	Type
Francis	60–250 (up to ~300)	50–340	Radial inward / mixed	Reaction
Kaplan / Propeller	< 60 (2–70)	255–860	Axial	Reaction

Specific speed of turbine: $N_s = N \sqrt{P} / H^{(5/4)}$ (P in kW, H in m, N in rpm)

★ **PYQ-TREND:** Turbine selection by head and specific speed (most frequent hydraulic-machines question); Pelton – tangential impulse; Kaplan – axial reaction; Francis – mixed/radial reaction; Ns formula with H^(5/4).

✔ **TOPPER TRICK:** Head and specific speed move in opposite directions: High head → low Ns → Pelton; low head → high Ns → Kaplan.

15.3 Pelton Wheel

- Components: nozzle with spear (needle) valve to regulate flow; runner with buckets (double hemispherical cups with a central splitter that divides the jet into two halves and avoids axial thrust); casing (only prevents splashing – no hydraulic function); breaking jet (stops the runner when the main jet is closed); deflector (diverts the jet during sudden load reduction, avoiding water hammer from rapid spear closure).
- Jet deflection by buckets ≈ 160°–165° (so the leaving water does not hit the next bucket).
- Pelton wheel works at atmospheric pressure; it must be placed above the tail race; no draft tube.

Jet velocity $V_1 = C_v \sqrt{(2gH)}$, $C_v \approx 0.97–0.99$

Bucket speed $u = \varphi \sqrt{(2gH)}$, speed ratio $\varphi \approx 0.43–0.48$

Work per second = $\rho a V_1 (V_1 - u)(1 + k \cos\varphi) u$ (k = bucket friction factor, φ = outlet angle)

Hydraulic efficiency $\eta_h = 2(V_1 - u)(1 + k \cos\varphi) u / V_1^2$; maximum at $u = V_1/2$: $\eta_{h,max} = (1 + k \cos\varphi)/2$

Jet ratio $m = D/d$ (pitch circle dia / jet dia) ≈ 12 (range 11–14; minimum ≈ 6)

Number of buckets $z = 15 + m/2 = 15 + D/(2d)$ (Taygun's formula)

Bucket width ≈ 5d; depth ≈ 1.2d

Number of jets n: $Q = n \times (\pi/4)d^2 V_1$

★ **PYQ-TREND:** Spear valve regulates discharge; splitter divides the jet; casing has no hydraulic function; deflector avoids water hammer; jet ratio ≈ 12; number of buckets 15 + m/2; max efficiency at u = V/2; no draft tube for Pelton.

15.4 Francis Turbine

- Components: spiral (scroll) casing – decreasing area to keep velocity uniform around the runner; stay vanes – structural, direct water to guide vanes; guide vanes (wicket gates) – adjustable, regulate flow and direct it at the correct angle; runner – fixed blades; draft tube.
- Inward radial/mixed-flow reaction turbine; the runner is completely full of water under pressure.

Flow ratio $\psi = V_{f1}/\sqrt{(2gH)} \approx 0.15–0.30$

Speed ratio $K_u = u_1/\sqrt{(2gH)} \approx 0.6–0.9$

Discharge $Q = \pi D_1 B_1 V_{f1}$ (× blade thickness factor); width ratio $n = B_1/D_1 \approx 0.10–0.40$

Hydraulic efficiency (radial discharge) $\eta_h = V_{w1} u_1/(gH)$

- Advantages of inward flow over outward flow: centrifugal head opposes the flow, making it self-regulating (runaway speed limited).

★ **PYQ-TREND:** Guide vanes (wicket gates) regulate flow; scroll casing gives uniform velocity; flow ratio range; Francis is inward/mixed-flow reaction turbine; runner full of water.

15.5 Kaplan and Propeller Turbines

- Axial-flow reaction turbines for low heads and large discharges. Runner looks like a ship's propeller with few blades (3–8). Hub (boss) diameter $D_b \approx 0.35–0.6\ D_o$.
- Kaplan: runner blades adjustable (double regulation – both guide vanes and runner blades) → high efficiency over a wide load range. Propeller: fixed blades → poor part-load efficiency.

Discharge $Q = (\pi/4)(D_o^2 - D_b^2)\ V_{f1}$

Peripheral velocity at outer diameter $u = \pi D_o N/60$

- Bulb turbine: Kaplan with the generator in a bulb in the flow – tidal and very-low-head plants. Deriaz: diagonal (mixed) flow with adjustable blades; can work as pump-turbine.

★ **PYQ-TREND:** Kaplan has adjustable runner blades (double regulation); propeller has fixed blades; axial flow; flat part-load efficiency curve.

15.6 Draft Tube

- A gradually diverging pipe connecting the runner exit to the tail race, used only with reaction turbines.
- Functions: (1) creates a negative (suction) head at the runner outlet, so the turbine can be installed above the tail race without loss of head (convenient for inspection); (2) converts a large part of the kinetic energy at the runner exit into pressure energy (regains head), increasing the efficiency.

Draft tube	Efficiency (approx.)
Conical (straight divergent), cone angle < 8°	≈ 90%
Moody spreading tube (bell-mouthed)	≈ 85%
Simple elbow	≈ 60%
Elbow with varying cross-section (rectangular outlet)	Used in large Kaplan/Francis plants

Draft tube efficiency $\eta_d = [(V_2^2 - V_3^2)/2g - h_f] / (V_2^2/2g)$

Pressure head at draft-tube inlet: $p_2/\rho g = H_{atm} - H_s - (V_2^2 - V_3^2)/2g + h_f$ (H_s = height of runner exit above tail race)

- The draft tube outlet must be submerged below the tail race level.

★ **PYQ-TREND:** Draft tube used only in reaction turbines (not Pelton); functions (suction head + KE recovery); conical type most efficient; angle < 8° to avoid separation.

15.7 Cavitation in Turbines

- Occurs where local pressure falls below vapour pressure – typically at the runner exit/draft tube inlet of reaction turbines (lowest pressure point).

Thoma's cavitation factor $\sigma = (H_{atm} - H_v - H_s)/H$; cavitation avoided if $\sigma > \sigma_{critical}$

```
Critical σ increases with specific speed (Kaplan needs higher σ → must be set lower relative to tail race)
```

- Prevention: install the turbine lower (reduce H_s), use cavitation-resistant materials (stainless steel, bronze), polished surfaces, avoid operation at part load or overload far from design.

★ **PYQ-TREND:** Location of cavitation in reaction turbines; Thoma's factor; σ_critical increases with Ns.

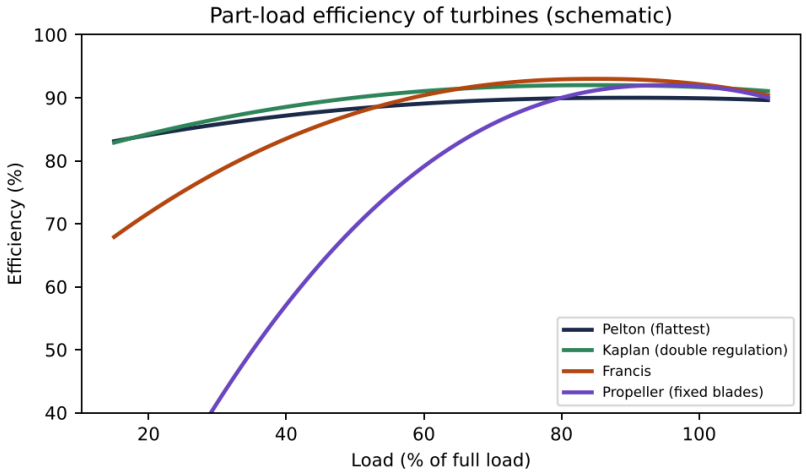
15.8 Performance, Unit Quantities and Governing

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Unit speed N_u = N/√H; unit discharge Q_u = Q/√H; unit power P_u = P/H^(3/2)
Same turbine at different heads: N ∝ √H, Q ∝ √H, P ∝ H^(3/2)
Model–prototype (geometrically similar): N D/√H = const, Q/(D²√H) = const, P/(D²H^(3/2)) = const
```

- Characteristic curves: main (constant head) curves, operating (constant speed) curves, constant-efficiency (Muschel/iso-efficiency) curves. Pelton and Kaplan have flat efficiency curves at part load; Francis is steeper; propeller is the worst.
- Runaway speed: maximum speed of a turbine at full gate opening under no load (Pelton ≈ 1.8–1.9 N; Francis ≈ 2–2.2 N; Kaplan ≈ 2.5–3 N). Generators are designed to withstand it.

```
Synchronous speed of generator: N = 120 f / p (p = number of poles; f = 50 Hz in India)
```

- Governing: keeping speed constant (synchronous) under varying load. Oil-pressure governor: centrifugal sensor (or electronic) → relay valve → servomotor → spear (Pelton) or wicket gates (Francis/Kaplan). Pelton uses a deflector to prevent water hammer during rapid load rejection; reaction turbines use a relief valve.



★ **PYQ-TREND:** Unit quantities and N ∝ √H, P ∝ H^1.5 (frequent); runaway speed; synchronous speed 120f/p; governing mechanisms (spear, deflector, wicket gates, servomotor); Muschel curves.

✖ **COMMON MISTAKE:** Specific speed of a turbine uses √P and H^(5/4); of a pump it uses √Q and H^(3/4). Using the wrong one is a common slip.

✎ **DESCRIPTIVE Q:** Explain the function of a draft tube and why it is used only in reaction turbines.

In a reaction turbine water leaves the runner with considerable velocity and the runner works full of water under pressure. A draft tube – a gradually expanding airtight pipe from the runner exit down into the tail race – does two things. First, the water column in it creates a pressure below atmospheric at the runner outlet, so the runner can be placed above the tail-race level (for easy maintenance) without losing the static head between the runner and tail race. Second, by gradually decelerating the flow it converts much of the exit kinetic energy into pressure energy, reducing the exit loss and increasing the efficiency. In an impulse turbine like the Pelton wheel the jet acts at atmospheric pressure in open air and the wheel must stand clear of the tail-race water; a draft tube cannot create suction there or recover energy, so it is not used.

CHAPTER 16

Pumps and Hydraulic Devices

16.1 Centrifugal Pumps – Construction and Working

- A centrifugal pump converts mechanical energy into pressure energy of the liquid by centrifugal action – a forced vortex in the impeller raises pressure. It is the reverse of an inward-flow reaction turbine.
- Main parts: impeller (vanes; closed/shrouded for clean liquids, semi-open, open for sewage/slurries), casing (volute – gradually increasing area converts velocity to pressure; vortex casing with a whirlpool chamber; casing with diffuser guide blades for multistage high-head pumps), suction pipe with foot valve (non-return) and strainer, delivery pipe with delivery valve.
- Vanes are backward-curved (outlet angle $\beta_2 < 90°$, typically 20°–25°): gives a falling H–Q curve, non-overloading power characteristic, stable operation and higher efficiency. Radial ($\beta_2 = 90°$) and forward-curved ($\beta_2 > 90°$) vanes give rising power curves – risk of motor overload.
- Priming: filling the suction pipe, casing and impeller with liquid before starting, to remove air. Needed because the head developed is independent of the fluid, so the pressure developed with air ($\rho_{air} \approx \rho_{water}/800$) is too small to lift water.
- Started with the delivery valve closed (zero discharge → minimum power for radial pumps); axial-flow pumps are started with the valve open (their power is maximum at shut-off).

★ **PYQ-TREND:** Priming and why it is needed (very frequent); backward-curved vanes; volute casing converts KE to pressure; foot valve function; starting against closed delivery valve; centrifugal pump = reverse of reaction turbine.

16.2 Heads, Work and Efficiencies

Static head $H_s = h_s$ (suction lift) + h_d (delivery head)

Manometric head $H_m = H_s + h_{fs} + h_{fd} + V_d^2/2g$ (head imparted to liquid, measured by gauges)

Euler head (work per unit weight, radial entry $V_{w1} = 0$): $H_e = V_{w2}\, u_2 / g$

Manometric efficiency $\eta_{mano} = g\, H_m / (V_{w2}\, u_2)$

Mechanical efficiency η_m = (power at impeller) / (shaft power)

Overall efficiency $\eta_o = \rho\, g\, Q\, H_m$ / shaft power = $\eta_{mano} \times \eta_m \times \eta_v$

Power required $P = \rho\, g\, Q\, H_m / \eta_o$

Minimum starting speed (centrifugal head ≥ manometric head): $u_2^2/2g - u_1^2/2g = H_m \rightarrow N_{min} = 120\, \eta_{mano}\, V_{w2}\, D_2 / (\pi(D_2^2 - D_1^2))$

Slip: actual V_{w2} < theoretical (finite number of vanes); slip factor $\sigma = V_{w2}'/V_{w2}$

★ **PYQ-TREND:** Manometric head definition; manometric efficiency formula (frequent); Euler head with radial entry; minimum starting speed; slip.

16.3 Similarity Laws, Specific Speed and Characteristics

Affinity laws (same pump, speed change): $Q \propto N$, $H \propto N^2$, $P \propto N^3$

Same speed, impeller diameter change (trimming): $Q \propto D$, $H \propto D^2$, $P \propto D^3$ (approx.)

Geometrically similar pumps: $Q/(ND^3) = \text{const}$, $H/(N^2D^2) = \text{const}$, $P/(N^3D^5) = \text{const}$

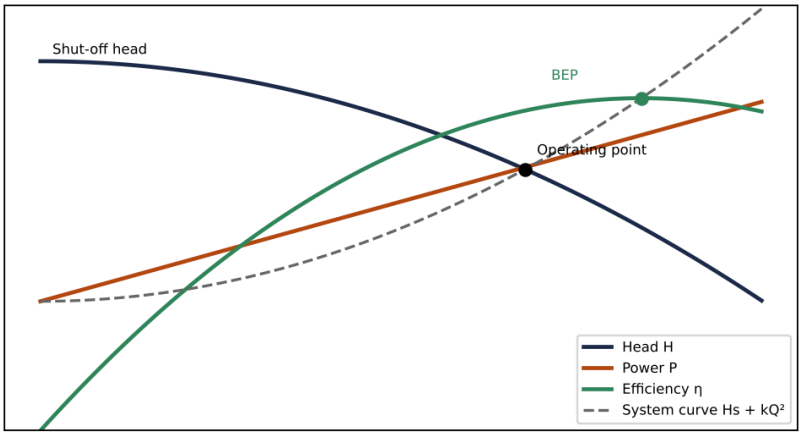
Specific speed of a pump: $N_s = N \sqrt{Q} / H^{(3/4)}$ (speed of a similar pump delivering 1 m^3/s against 1 m head)

Multistage: N_s uses head per stage; double-suction: use $Q/2$

Pump type by Ns (SI, rpm, m^3/s, m)	Ns range
Radial-flow (centrifugal)	10–80 (slow/medium/high)
Mixed-flow	80–160
Axial-flow (propeller)	160–500

- Characteristic curves: main (H, P, η vs Q at constant N), operating (at design speed), Muschel (iso-efficiency). The H–Q curve of a backward-curved pump falls with Q; η is maximum at the best efficiency point (BEP); shut-off head at Q = 0.
- Operating point: intersection of the pump H–Q curve and the system curve ($H_{static} + kQ^2$).
- Pumps in series: heads add at the same discharge – used for high head (multistage pumps, deep wells, boiler feed). Pumps in parallel: discharges add at the same head – used for large/variable discharge.
- Multistage pumps: several impellers in series on one shaft – high head (borewell submersible pumps, boiler feed pumps).

Centrifugal pump characteristics (constant speed)



★ **PYQ-TREND:** Affinity laws $P \propto N^3$ (very frequent); pump specific speed formula with √Q and H^(3/4); series for head, parallel for discharge; BEP; multistage pumps; operating point.

✔ **TOPPER TRICK:** If speed increases by 10%: Q +10%, H +21%, P +33% (1.1, 1.21, 1.331).

16.4 Cavitation and NPSH in Pumps

- Cavitation in centrifugal pumps occurs at the impeller eye/inlet (lowest pressure). Effects: pitting, noise, vibration, drop in head and efficiency ('break-off').

NPSH available = $p_{atm}/\rho g - h_s - h_{fs} - p_v/\rho g$ (suction lift case; add h if the liquid is above the pump)

Condition: $NPSH_A > NPSH_R$ (given by manufacturer)

Thoma's cavitation parameter for pumps $\sigma = NPSH / H_m$

Maximum suction lift (theoretical) = 10.3 m of water at sea level; practically 6–7 m

- Prevention: keep suction lift small (or flooded suction), short and large-diameter suction pipe with few bends, cool liquid (lower vapour pressure), avoid excessive speed.

★ **PYQ-TREND:** NPSH definition and NPSH_A > NPSH_R; cavitation at impeller eye; practical suction lift limit ≈ 6–7 m.

16.5 Reciprocating Pumps

- Positive-displacement pump: a piston/plunger moves in a cylinder with suction and delivery valves. Suitable for high head and low discharge (oil drilling, boiler feed in small plants, hydraulic testing, pressure washers). Self-priming in principle.

Theoretical discharge: single-acting $Q_{th} = ALN/60$; double-acting $Q_{th} = 2ALN/60$ (neglecting rod area; with rod: $(2A - a)LN/60$)

Slip = $Q_{th} - Q_{act}$; % slip = $(1 - C_d) \times 100$; $C_d = Q_{act}/Q_{th}$ (0.95–0.98)

Negative slip occurs when $Q_{act} > Q_{th}$ — long suction pipe, short delivery pipe, high speed (inertia of the suction column keeps delivery valve open)

Work done per second (single-acting) = $\rho g A L N (h_s + h_d)/60$; double-acting × 2

Acceleration head: $h_a = (L_{pipe}/g)(A/a)\ \omega^2 r \cos\theta$ (max at start and end of stroke: $(L/g)(A/a)\omega^2 r$)

Friction head (with SHM): $h_f = (4f\ L/(2gd))\ [(A/a)\ \omega\ r \sin\theta]^2$ — maximum at mid-stroke

Effect of acceleration head on work: none (positive and negative areas cancel); friction adds area to the indicator diagram

- Indicator diagram: plot of pressure head in the cylinder vs stroke; its area ∝ work done per cycle. Acceleration head tilts the suction and delivery lines; friction makes them parabolic.
- Separation: occurs if pressure in the suction pipe at the beginning of the suction stroke falls below about 2.5 m of water (absolute) – limits speed.
- Air vessels: closed chambers with compressed air fitted on suction and delivery sides close to the cylinder. Functions: obtain nearly uniform flow, reduce acceleration head (allowing higher speed and longer pipes), avoid separation, save work against friction. Saving ≈ 84.8% (single-acting), 39.2% (double-acting).

Feature	Centrifugal	Reciprocating
Discharge	Large, continuous, smooth	Small, pulsating
Head	Low to medium (multistage for high)	High
Priming	Needed	Not needed (self-priming)
Viscous liquids/slurries	Can handle	Difficult
Cost, size, maintenance	Low, compact, simple	High, bulky, valves need care
Efficiency	Good at BEP	Higher at high heads

★ **PYQ-TREND:** Negative slip conditions (very frequent); discharge formulas; air vessel functions and work saved; acceleration head max at stroke ends; indicator diagram; comparison with centrifugal pumps; separation.

16.6 Other Pumps and Hydraulic Devices

Device	Principle / use
Gear pump	Rotary positive displacement; lubrication systems, hydraulics

Device	Principle / use
Vane pump	Rotary positive displacement; power steering, hydraulics
Screw pump	Positive displacement; viscous liquids, Archimedean screw for lifting water
Lobe pump	Food processing
Jet pump (ejector)	Momentum transfer of a high-velocity jet; no moving parts; deep wells
Air-lift pump	Compressed air mixed with water reduces density in a well pipe
Submersible pump	Multistage centrifugal with motor below water; borewells
Hydraulic ram	Uses water hammer of a large flow from a small head to lift a small quantity to a large height; no external power; D'Aubuisson efficiency $\eta = qH/((Q + q)h)$; Rankine efficiency $\eta = q(H - h)/(Q\,h)$ (Q = waste flow, q = flow lifted to height H, h = supply head)
Hydraulic accumulator	Stores pressure energy (energy = pAL) – supplies peak demand for lifts, cranes, presses
Differential accumulator	Smaller volume, higher pressure
Hydraulic intensifier	Increases liquid pressure: $p_2 = p_1$ (A/a)
Hydraulic press	Pascal's law; W/F = A/a
Hydraulic crane, lift, jack	Ram-driven lifting (direct-acting or suspended lifts)
Fluid (hydraulic) coupling	Pump impeller + turbine runner in one casing with oil; torque out = torque in; slip 2–3%; efficiency = N_2/N_1
Torque converter	Coupling + fixed stator (guide vanes); output torque can exceed input (torque multiplication); automatic transmissions

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Hydraulic coupling: efficiency η = N_turbine / N_pump = 1 − slip
Torque converter: T_turbine = T_pump + T_stator
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★ **PYQ-TREND:** Hydraulic ram works on water hammer; accumulator stores energy; intensifier raises pressure; torque converter has a stator (torque multiplication) whereas fluid coupling does not; gear and vane pumps are positive displacement.

✎ **DESCRIPTIVE Q:** Why is priming necessary in a centrifugal pump? What happens if the pump runs without priming?

A centrifugal pump develops a head $H = V_{w2}\,u_2/g$ that depends only on impeller speed and geometry, not on the fluid. The pressure rise equals $\rho g H$, so it is proportional to the density of the fluid in the impeller. If the impeller contains air (density about 1/800 of water), the pressure developed is only about 1/800 of that with water – far too small to lift water up the suction pipe – and the pump only churns air. Priming fills the suction pipe, casing and impeller with water before starting, removing the air so that the full pressure can be developed. A foot valve (non-return valve) at the bottom of the suction pipe holds the water in after priming. Running dry also damages glands and wear rings through overheating. Self-priming pumps and reciprocating pumps do not need separate priming.

CHAPTER 17

Solved Numericals – Set 1 (Statics to Pipe Flow)

SSC-style problems solved step by step (no options). g = 9.81 m/s², ρ_water = 1000 kg/m³ unless stated.

17.1 Properties

Problem 1 – Viscosity conversion

An oil has dynamic viscosity 6 poise and specific gravity 0.9. Find μ in SI and ν in stokes.

μ = 6 × 0.1 = 0.6 Pa·s; ρ = 900 kg/m³; ν = 0.6/900 = 6.67 × 10⁻⁴ m²/s = 6.67 stokes

Problem 2 – Shear between plates

Two plates 2 mm apart; upper plate moves at 1 m/s with oil (μ = 0.5 Pa·s) between them. Find the shear stress and force on a 0.5 m² plate.

τ = μV/h = 0.5 × 1/0.002 = 250 N/m²; F = τA = 125 N

Problem 3 – Journal bearing power loss

Shaft D = 100 mm, bearing length 200 mm, clearance 0.5 mm, N = 600 rpm, μ = 0.1 Pa·s. Find the power lost.

u = πDN/60 = π × 0.1 × 600/60 = 3.142 m/s; τ = μu/t = 0.1 × 3.142/0.0005 = 628.3 N/m²

F = τ × πDL = 628.3 × π × 0.1 × 0.2 = 39.48 N; T = F × D/2 = 1.974 N·m; P = Tω = 1.974 × 62.83 = 124 W

Problem 4 – Bulk modulus

Pressure on 1 m³ of water increases from 1 bar to 101 bar; its volume decreases by 0.5%. Find K.

K = Δp/(ΔV/V) = 100 × 10⁵ / 0.005 = 2 × 10⁹ Pa = 2 GPa

Problem 5 – Soap bubble

Find the pressure inside a 40 mm soap bubble (σ = 0.04 N/m) above atmospheric.

Δp = 8σ/d = 8 × 0.04/0.04 = 8 N/m²

Problem 6 – Capillary rise

Find the rise of water in a 2 mm glass tube (σ = 0.075 N/m, θ = 0).

h = 4σ/(ρgd) = 4 × 0.075/(1000 × 9.81 × 0.002) = 0.0153 m = 15.3 mm

Problem 7 – Mercury depression

Find the depression of mercury in a 3 mm tube (σ = 0.52 N/m, θ = 130°, S = 13.6).

h = 4σ cosθ/(ρgd) = 4 × 0.52 × (−0.643)/(13600 × 9.81 × 0.003) = −0.0033 m → 3.3 mm depression

17.2 Statics and Hydrostatic Forces

Problem 8 – Pressure at depth

Find the gauge pressure 5 m below the surface of oil (S = 0.8).

$p = \rho g h = 800 \times 9.81 \times 5 = 39.24$ kPa; head of water = 4 m

Problem 9 – Hydraulic press

Plunger diameter 30 mm, ram diameter 300 mm, force on plunger 500 N. Find the load lifted.

$W = F\,(D/d)^2 = 500 \times 100 = 50$ kN

Problem 10 – U-tube manometer

A U-tube with mercury is connected to a water pipe. Pipe centre is 0.15 m above the mercury surface in the left limb; deflection 0.25 m. Find pipe pressure.

$p + 1000 \times 9.81 \times 0.15 = 13600 \times 9.81 \times 0.25 \rightarrow p = 33354 - 1471.5 = 31.88$ kPa (gauge)

Problem 11 – Differential manometer

A mercury differential manometer connected to two points of a horizontal water pipe reads 0.1 m. Find the pressure difference.

$\Delta p = g x (\rho_m - \rho) = 9.81 \times 0.1 \times 12600 = 12.36$ kPa; head = 1.26 m of water

Problem 12 – Absolute pressure

A vacuum gauge reads 30 kPa; atmospheric pressure is 101.3 kPa. Find absolute pressure.

$p_{abs} = 101.3 - 30 = 71.3$ kPa

Problem 13 – Force on vertical gate

A rectangular gate 3 m wide holds water 4 m deep. Find the total force and its location.

$F = \rho g b H^2/2 = 9810 \times 3 \times 16/2 = 235.4$ kN; acts at $H/3 = 1.33$ m above the base (2.67 m below surface)

Problem 14 – Circular plate

A circular plate of 2 m diameter is immersed vertically with its top 1 m below the water surface. Find F and centre of pressure.

$A = 3.142$ m²; $\bar{h} = 2$ m; $F = 9810 \times 3.142 \times 2 = 61.6$ kN

$I_G = \pi d^4/64 = 0.785$ m⁴; $h^* = 2 + 0.785/(3.142 \times 2) = 2.125$ m

Problem 15 – Inclined plate

A rectangular plate 1 m × 2 m (2 m along the slope) is inclined at 30° to the horizontal with its upper edge 1 m below the surface. Find F.

Depth of centroid $\bar{h} = 1 + 1 \times \sin 30° = 1.5$ m; $F = 9810 \times 2 \times 1.5 = 29.4$ kN

Problem 16 – Curved surface

A quarter-circle gate of radius 2 m, 1 m wide, holds water up to its top. Find F_H, F_V and resultant.

$F_H = \rho g \times (2 \times 1) \times 1 = 19.62$ kN; $F_V = \rho g \times (\pi \times 2^2/4) \times 1 = 30.82$ kN

$F = \sqrt{(19.62^2 + 30.82^2)} = 36.5$ kN at $\tan^{-1}(30.82/19.62) = 57.5°$ to horizontal

Problem 17 – Rotating vessel

An open cylinder of 0.6 m diameter rotates at 120 rpm about its axis. Find the height of the paraboloid.

$\omega = 12.57$ rad/s; $h = \omega^2 R^2/(2g) = 158 \times 0.09/19.62 = 0.725$ m; rise at wall = fall at centre = 0.362 m

Problem 18 – Accelerating tank

A tank accelerates horizontally at 2.45 m/s². Find the slope of the free surface.

$\tan\theta = a/g = 0.25 \rightarrow \theta = 14.04°$

17.3 Buoyancy and Stability

Problem 19 – Fraction submerged

A wooden log (S = 0.7) floats in water. What fraction is above water?

Submerged fraction = 0.7 → 30% above water

Problem 20 – Metacentric height of a block

A block 2 m wide, 4 m long, 1 m high (S = 0.6) floats with the 1 m side vertical. Find GM (rolling about the long axis).

Draft $d = 0.6$ m; $BM = b^2/(12d) = 4/(7.2) = 0.556$ m

OB = 0.3 m, OG = 0.5 m, BG = 0.2 m; GM = 0.556 − 0.2 = 0.356 m (stable)

Problem 21 – Experimental GM

A 50 kN weight moved 6 m across a ship of 20,000 kN displacement tilts it by 1.5°. Find GM.

$GM = wx/(W \tan\theta) = 50 \times 6/(20000 \times 0.02619) = 0.573$ m

17.4 Kinematics and Dynamics

Problem 22 – Continuity check

Is u = 2x, v = −2y a possible incompressible flow? Is it irrotational?

$\partial u/\partial x + \partial v/\partial y = 2 - 2 = 0 \rightarrow$ possible; $\omega_z = \frac{1}{2}(\partial v/\partial x - \partial u/\partial y) = 0 \rightarrow$ irrotational

Problem 23 – Acceleration

Velocity field u = 2x, v = −2y. Find acceleration at (1, 1).

$a_x = u\,\partial u/\partial x = 2x \times 2 = 4$ at $x = 1$; $a_y = v\,\partial v/\partial y = -2y \times (-2) = 4$; $|a| = 5.66$ m/s²

Problem 24 – Stream function discharge

Two streamlines have ψ = 2 and ψ = 5 m²/s. Find discharge per metre width between them.

$q = 5 - 2 = 3$ m³/s per m

Problem 25 – Bernoulli

Water flows through a pipe from 300 mm to 150 mm diameter at Q = 0.1 m³/s. Pressure at the larger section is 200 kPa. Find pressure at the smaller section (same level, no loss).

$V_1 = 0.1/0.0707 = 1.415$ m/s; $V_2 = 0.1/0.01767 = 5.66$ m/s

$p_2 = 200000 + 500(1.415^2 - 5.66^2) = 200000 - 15016 = 185$ kPa

Problem 26 – Pitot tube

A pitot tube in water shows a stagnation head 0.5 m above the static head. Find velocity (C_v = 0.98).

$V = 0.98\sqrt{(2 \times 9.81 \times 0.5)} = 0.98 \times 3.13 = 3.07$ m/s

Problem 27 – Force on a bend

A 90° horizontal bend carries 0.2 m³/s of water in a 300 mm pipe at 100 kPa (neglect loss). Find the resultant force.

$A = 0.0707\ m^2$; $V = 2.83\ m/s$; $\rho QV = 566\ N$; $pA = 7070\ N$

$F_x = pA + \rho QV = 7636\ N$; $F_y = 7636\ N$; $F = 7636\sqrt{2} = 10.8\ kN$ at 45°

17.5 Flow Measurement

Problem 28 – Venturimeter

A 300 × 150 mm venturimeter shows 0.2 m mercury deflection with water. $C_d = 0.97$. Find Q.

$h = 0.2 \times 12.6 = 2.52\ m$; $A_1 = 0.0707$, $A_2 = 0.01767$; $\sqrt{(A_1^2 - A_2^2)} = 0.06845$

$Q = 0.97 \times 0.0707 \times 0.01767 \times \sqrt{(2 \times 9.81 \times 2.52)}/0.06845 = 0.97 \times 0.01825 \times 7.03 = 0.1245\ m^3/s$

Problem 29 – Orifice coefficients

A 50 mm orifice under 4 m head discharges 9 L/s; jet diameter at vena contracta 40 mm. Find C_c, C_d, C_v.

$C_c = (40/50)^2 = 0.64$; $Q_{th} = 0.001963 \times \sqrt{(78.48)} = 0.01739\ m^3/s$; $C_d = 0.009/0.01739 = 0.518$; $C_v = 0.518/0.64 = 0.81$

Problem 30 – Tank emptying

A tank of area 2 m² with water 4 m deep empties through a 50 mm orifice ($C_d = 0.6$). Find time to empty.

$a = 0.001963\ m^2$; $T = 2A\sqrt{H}/(C_d\ a\ \sqrt{(2g)}) = 2 \times 2 \times 2/(0.6 \times 0.001963 \times 4.429) = 8/0.005217 = 1533\ s \approx 25.6\ min$

Problem 31 – Rectangular notch

Find Q over a rectangular notch 1.5 m long with head 0.3 m ($C_d = 0.62$).

$Q = (2/3) \times 0.62 \times 1.5 \times 4.429 \times 0.3^{1.5} = 2.746 \times 0.1643 = 0.451\ m^3/s$

Problem 32 – V-notch

Find Q over a 90° V-notch with head 0.2 m ($C_d = 0.6$).

$Q = (8/15) \times 0.6 \times 4.429 \times 1 \times 0.2^{2.5} = 1.417 \times 0.01789 = 0.0254\ m^3/s = 25.4\ L/s$

Problem 33 – Error in head

A 1% error in measuring head over a V-notch causes what error in discharge?

$dQ/Q = 2.5 \times dH/H = 2.5\%$

17.6 Laminar and Pipe Flow

Problem 34 – Laminar check and loss

Oil ($\nu = 1 \times 10^{-4}\ m^2/s$, S = 0.9) flows at 1.5 m/s in a 100 mm pipe, 200 m long. Find h_f.

$Re = 1.5 \times 0.1/10^{-4} = 1500$ (laminar); $f = 64/1500 = 0.0427$

$h_f = fLV^2/(2gD) = 0.0427 \times 200 \times 2.25/(1.962) = 9.78\ m$ of oil

Problem 35 – Maximum velocity in laminar flow

Average velocity in a pipe is 0.6 m/s (laminar). Find the maximum velocity and the radius where u = V.

$u_{max} = 2V = 1.2\ m/s$; $r = R/\sqrt{2} = 0.707R$

Problem 36 – Darcy loss

Water flows 0.05 m³/s through a 200 mm pipe, 1 km long, f = 0.02. Find h_f.

$h_f = fLQ^2/(12.1\,D^5) = 0.02 \times 1000 \times 0.0025/(12.1 \times 0.00032) = 0.05/0.003872 = 12.9$ m

Problem 37 – Sudden enlargement

Pipe diameter changes suddenly from 150 mm to 300 mm; Q = 0.05 m³/s. Find the loss.

$V_1 = 2.83$ m/s; $V_2 = 0.707$ m/s; $h = (2.12)^2/19.62 = 0.229$ m

Problem 38 – Power transmission

Water at 500 m head is transmitted through a pipe; friction loss is 100 m. Find efficiency of transmission and check maximum-power condition.

$\eta = (500 - 100)/500 = 80\%$; max power would need $h_f = 500/3 = 166.7$ m ($\eta = 66.7\%$)

Problem 39 – Water hammer

Water flowing at 1.5 m/s in a rigid pipe is stopped suddenly (K = 2.1 GPa). Find the pressure rise.

$c = \sqrt{(2.1 \times 10^9/1000)} = 1449$ m/s; $\Delta p = \rho Vc = 1000 \times 1.5 \times 1449 = 2.17$ MPa

Problem 40 – Equivalent pipe

Pipes 600 m of 0.3 m and 400 m of 0.2 m are in series. Find the equivalent length of a 0.3 m pipe.

$L_e = 600 + 400 \times (0.3/0.2)^5 = 600 + 400 \times 7.594 = 3638$ m

CHAPTER 17B

Solved Numericals – Set 2 (BL, Models, Channels, Machines) and Ratio Twisters

17B.1 Boundary Layer, Drag, Similitude, Compressible

Problem 41 – Boundary layer thickness

Air ($\nu = 1.5 \times 10^{-5}$ m²/s) flows at 3 m/s over a plate. Find δ at x = 1 m (laminar check).

$Re_x = 3 \times 1/1.5 \times 10^{-5} = 2 \times 10^5$ (< 5×10^5, laminar); $\delta = 5x/\sqrt{Re_x} = 5/447 = 0.0112$ m = 11.2 mm

Problem 42 – Drag force

A car (frontal area 2 m², $C_D = 0.3$) moves at 30 m/s in air (ρ = 1.2). Find drag and the power to overcome it.

$F_D = 0.3 \times 0.5 \times 1.2 \times 900 \times 2 = 324$ N; $P = 324 \times 30 = 9.72$ kW

Problem 43 – Stokes' terminal velocity

A 0.1 mm sand particle (ρ = 2650) settles in water ($\mu = 10^{-3}$ Pa·s). Find terminal velocity.

$V = D^2(\rho s - \rho)g/(18\mu) = (10^{-4})^2 \times 1650 \times 9.81/(0.018) = 8.99 \times 10^{-3}$ m/s ≈ 9 mm/s; Re ≈ 0.9 (Stokes valid)

Problem 44 – Froude model

A 1:36 spillway model discharges 0.05 m³/s. Find the prototype discharge and time scale.

$Q_p = Q_m \times 36^{2.5} = 0.05 \times 7776 = 388.8$ m³/s; $T_r = \sqrt{36} = 6$

Problem 45 – Reynolds model

Water flows at 2 m/s in a 1 m pipe. A 1:10 model uses the same water. Find model velocity.

$V_m = V_p \times L_p/L_m = 2 \times 10 = 20$ m/s

Problem 46 – Mach number

An aircraft flies at 900 km/h where air temperature is −50 °C. Find M (γ = 1.4, R = 287).

$c = \sqrt{(1.4 \times 287 \times 223)} = 299.3$ m/s; V = 250 m/s; M = 0.835 (subsonic, near transonic)

Problem 47 – Stagnation temperature

Air at 300 K moves at M = 2. Find T_0.

$T_0 = T(1 + 0.2M^2) = 300 \times 1.8 = 540$ K

Problem 48 – π-terms

A problem has 7 variables in M, L, T. How many π-terms?

$n - m = 7 - 3 = 4$

17B.2 Open Channels

Problem 49 – Economical rectangle

A rectangular channel must carry 2 m³/s at V = 1 m/s. Find the most economical dimensions.

$A = 2\ m^2 = by = 2y^2 \rightarrow y = 1\ m,\ b = 2\ m;\ R = 0.5\ m$

Problem 50 – Manning

A channel has R = 0.8 m, S = 1/1600, n = 0.015. Find velocity.

$V = (1/0.015) \times 0.8^{(2/3)} \times (1/40) = 66.7 \times 0.862 \times 0.025 = 1.44\ m/s$

Problem 51 – Critical depth

q = 4 m³/s per metre width. Find y_c, E_min and V_c.

$y_c = (16/9.81)^{(1/3)} = 1.177\ m;\ E_{min} = 1.765\ m;\ V_c = q/y_c = 3.40\ m/s\ (= \sqrt{g\ y_c})$

Problem 52 – Hydraulic jump

Before a jump y_1 = 0.5 m and V_1 = 8 m/s. Find y_2 and energy loss.

$Fr_1 = 8/\sqrt{4.905} = 3.61;\ y_2 = 0.25(-1 + \sqrt{1 + 8 \times 13.05}) = 0.25 \times 9.27 = 2.32\ m$

$\Delta E = (2.32 - 0.5)^3/(4 \times 0.5 \times 2.32) = 6.03/4.64 = 1.30\ m$

17B.3 Jets and Turbines

Problem 53 – Jet on fixed plate

A 50 mm jet at 25 m/s strikes a fixed flat plate normally. Find the force.

$a = 1.963 \times 10^{-3}\ m^2;\ F = \rho a V^2 = 1000 \times 1.963 \times 10^{-3} \times 625 = 1227\ N$

Problem 54 – Series of plates

Same jet strikes a series of flat plates moving at 10 m/s. Find force, power and efficiency.

$F = \rho a V(V - u) = 1000 \times 1.963 \times 10^{-3} \times 25 \times 15 = 736\ N;\ P = 736 \times 10 = 7.36\ kW$

$\eta = 2u(V - u)/V^2 = 2 \times 10 \times 15/625 = 48\%$

Problem 55 – Pelton wheel

Net head 400 m, Q = 0.3 m³/s, C_v = 0.98, speed ratio 0.46, η_o = 0.85. Find jet velocity, bucket speed and power.

$V = 0.98\sqrt{2 \times 9.81 \times 400} = 0.98 \times 88.6 = 86.8\ m/s;\ u = 0.46 \times 88.6 = 40.8\ m/s$

$P = 0.85 \times 9810 \times 0.3 \times 400 = 1.0\ MW$

Problem 56 – Pelton dimensions

Pitch diameter 1.8 m, jet diameter 0.15 m. Find jet ratio and number of buckets.

$m = 12;\ z = 15 + 12/2 = 21\ buckets$

Problem 57 – Turbine specific speed

A turbine gives 5,000 kW under 64 m head at 300 rpm. Find N_s and the turbine type.

$H^{1.25} = 64^{1.25} = 181;\ N_s = 300 \times 70.7/181 = 117 \rightarrow Francis$

Problem 58 – Unit quantities

A turbine gives 1,000 kW at 100 m head and 500 rpm. Find speed and power at 64 m head.

$N \propto \sqrt{H} \rightarrow 500 \times 0.8 = 400\ rpm;\ P \propto H^{1.5} \rightarrow 1000 \times 0.512 = 512\ kW$

Problem 59 – Synchronous speed

A generator with 24 poles runs on 50 Hz. Find its speed.

```
N = 120f/p = 6000/24 = 250 rpm
```

Problem 60 – Kaplan discharge

Outer diameter 4 m, hub 1.6 m, flow velocity 8 m/s. Find Q.

```
Q = (π/4)(16 − 2.56) × 8 = 10.56 × 8 = 84.4 m³/s
```

17B.4 Pumps

Problem 61 – Centrifugal pump head

Impeller D_2 = 0.5 m at 1450 rpm, V_w2 = 30 m/s, radial entry. Manometric head 60 m. Find Euler head and manometric efficiency.

```
u₂ = π × 0.5 × 1450/60 = 37.96 m/s; H_e = 30 × 37.96/9.81 = 116 m; η_mano = 60/116 = 51.7%
```

Problem 62 – Pump power

A pump delivers 40 L/s against 25 m head with η_o = 0.8. Find shaft power.

```
P = 9810 × 0.04 × 25/0.8 = 12.26 kW
```

Problem 63 – Affinity laws

A pump gives 50 L/s, 20 m head, 15 kW at 1200 rpm. Find values at 1800 rpm.

```
Q = 75 L/s; H = 20 × 2.25 = 45 m; P = 15 × 3.375 = 50.6 kW
```

Problem 64 – Pump specific speed

Pump: 1450 rpm, Q = 0.1 m³/s, H = 25 m. Find N_s and type.

```
N_s = 1450 × 0.316/25^0.75 = 458.5/11.18 = 41 → radial-flow (centrifugal)
```

Problem 65 – Reciprocating pump

Single-acting: bore 200 mm, stroke 300 mm, 60 rpm, actual Q = 9 L/s. Find Q_th, C_d and slip.

```
Q_th = 0.03142 × 0.3 × 60/60 = 9.42 L/s; C_d = 0.955; slip = 4.5%
```

Problem 66 – NPSH

Atmospheric head 10.3 m, suction lift 4 m, suction loss 0.8 m, vapour head 0.24 m. Find NPSH available.

```
NPSH_A = 10.3 − 4 − 0.8 − 0.24 = 5.26 m
```

Problem 67 – Hydraulic coupling

Pump runs at 1500 rpm, turbine at 1455 rpm. Find slip and efficiency.

```
slip = 45/1500 = 3%; η = 1455/1500 = 97%
```

Problem 68 – Intensifier

Low-pressure water at 2 MPa acts on a 200 mm ram; the high-pressure plunger is 50 mm. Find output pressure.

```
p₂ = 2 × (200/50)² = 32 MPa
```

17B.5 Ratio Twisters (Most Time-Saving)

If this changes…	Effect
Tube diameter halved (capillary)	Rise doubles (h ∝ 1/d)

If this changes…	Effect
Droplet diameter doubled	Excess pressure halves ($\propto 1/d$)
Depth of water doubled on a vertical gate	Total force × 4 ($\propto H^2$)
Pipe diameter halved (same Q)	Velocity × 4
Laminar, same Q, D doubled	$\Delta p \div 16$ ($\propto 1/D^4$)
Turbulent, same Q and f, D doubled	$h_f \div 32$ ($\propto 1/D^5$)
Turbulent, Q doubled	$h_f \times 4$
Head over rectangular weir doubled	Q × 2.83
Head over V-notch doubled	Q × 5.66
Tank head quartered (time to empty)	Time halves ($\propto \sqrt{H}$)
Jet velocity doubled (fixed plate)	Force × 4
Pump speed doubled	Q × 2, H × 4, P × 8
Turbine head doubled (same turbine)	N × 1.41, Q × 1.41, P × 2.83
Froude model 1:25	V ÷ 5, Q ÷ 3125, T ÷ 5, F ÷ 15625
Speed doubled (drag $\propto V^2$)	Drag × 4, drag power × 8
Sphere diameter doubled (Stokes)	Terminal velocity × 4

★ **PYQ-TREND:** SSC repeatedly asks these one-step ratio questions; know the power of each variable.

CHAPTER 17C

Concept Recall Bank

Short question → crisp answer. Cover the right column and test yourself (active recall, not MCQs).

R1. Properties and Statics

Question	Answer
What defines a fluid?	Continuous deformation under any shear stress
SI unit of dynamic viscosity?	N·s/m² (Pa·s); 1 poise = 0.1 Pa·s
SI unit of kinematic viscosity?	m²/s; 1 stoke = 10^{-4} m²/s
Dimensions of μ?	$ML^{-1}T^{-1}$
Viscosity of liquids with temperature?	Decreases
Viscosity of gases with temperature?	Increases
Fluid with $\tau \propto du/dy$ through origin?	Newtonian
Blood is which type of fluid?	Pseudoplastic (shear-thinning)
Toothpaste is which type?	Bingham plastic
Printer's ink is which type?	Thixotropic
Quicksand / starch solution?	Dilatant
Ideal fluid has?	Zero viscosity, incompressible
Bulk modulus of water?	≈ 2.1 GPa
Isothermal bulk modulus of a gas?	Equal to its pressure p
Adiabatic bulk modulus of a gas?	γp
Pressure inside a droplet?	4σ/d
Pressure inside a soap bubble?	8σ/d
Pressure inside a liquid jet?	2σ/d
Capillary rise formula?	$4\sigma \cos\theta/(\rho g d)$
Mercury in glass tube shows?	Capillary depression
Cavitation occurs when?	Local pressure < vapour pressure
Pascal's law?	Pressure at a point in a static fluid is equal in all directions
Hydraulic press works on?	Pascal's law
1 atm in m of water?	10.33 m
Absolute pressure?	Atmospheric + gauge
Manometer for small pressures?	Inclined manometer
Manometer for small differences in liquids?	Inverted U-tube
Piezometer cannot measure?	Negative pressure, gas pressure, high pressure
Instrument to calibrate pressure gauges?	Dead-weight tester
Free surface of a rotating liquid?	Paraboloid
Rise at wall of rotating vessel?	$\omega^2R^2/4g$ above original level (half of $\omega^2R^2/2g$)

Question	Answer
Slope of liquid surface in horizontally accelerating tank?	$\tan\theta = a/g$

R2. Forces, Buoyancy, Kinematics

Question	Answer
Total pressure on a plane surface?	$\rho g A \bar{h}$
Centre of pressure relative to centroid?	Below (except horizontal surfaces)
CP of vertical rectangle with top at surface?	2d/3 depth
CP of circle touching surface (diameter d)?	5d/8
Horizontal force on curved surface?	Force on its vertical projection
Vertical force on curved surface?	Weight of liquid above it
Archimedes' principle?	Buoyant force = weight of fluid displaced
Centre of buoyancy?	Centroid of displaced volume
Stability of a floating body?	M above G
Stability of a submerged body?	B above G
Metacentric height formula?	$GM = I/V - BG$
Effect of large GM on rolling?	Shorter period (stiff, uncomfortable)
Streamline, pathline, streakline coincide in?	Steady flow
Dye injected continuously gives?	Streakline
Continuity for incompressible flow?	$\partial u/\partial x + \partial v/\partial y + \partial w/\partial z = 0$
Acceleration in steady non-uniform flow?	Convective only
Condition for irrotational flow?	$\partial v/\partial x = \partial u/\partial y$
Vorticity equals?	Twice the angular velocity
Velocity potential exists for?	Irrotational flow
Stream function exists for?	2-D incompressible flow
φ and ψ lines intersect at?	Right angles
Discharge between streamlines?	$\psi_2 - \psi_1$
Free vortex law?	$V \times r = \text{constant}$
Forced vortex law?	$V = \omega r$
Flow in a whirlpool / sink?	Free vortex
Flow in a rotating cylinder?	Forced vortex

R3. Dynamics and Measurement

Question	Answer
Euler's equation considers?	Gravity and pressure forces
Navier–Stokes adds?	Viscous forces
Bernoulli's equation is based on?	Conservation of energy
Bernoulli assumption on viscosity?	Frictionless (inviscid)
TEL vs HGL gap?	Velocity head $V^2/2g$
Line that can rise along flow?	HGL
α for laminar pipe flow?	2

Question	Answer
β for laminar pipe flow?	4/3
Force on a pipe bend found by?	Momentum equation
Venturimeter divergent angle?	5–7°
C_d of venturimeter?	0.95–0.98
C_d of orificemeter?	≈ 0.62
Meter with least head loss?	Venturimeter
Pitot tube measures?	Stagnation pressure (point velocity)
Variable-area flowmeter?	Rotameter
C_d = ?	C_c × C_v
Vena contracta position?	About d/2 from orifice
C_d of external cylindrical mouthpiece?	0.855
Borda's mouthpiece running free?	0.5
Discharge over rectangular notch ∝?	H^(3/2)
Discharge over V-notch ∝?	H^(5/2)
Cipolletti weir side slope?	1 H : 4 V
Notch best for small discharge?	V-notch
Time to empty a tank ∝?	√H

R4. Laminar, Turbulent, Pipe Flow

Question	Answer
Reynolds number = ratio of?	Inertia to viscous force
Laminar limit in pipes?	Re < 2000
Laminar velocity profile?	Parabolic
Average/max velocity (laminar pipe)?	1/2
Average/max velocity (fixed plates)?	2/3
Shear stress in laminar pipe flow max at?	Wall
Hagen–Poiseuille head loss?	$32\mu VL/(\rho g D^2)$
Darcy f for laminar flow?	64/Re
Stokes' drag?	$3\pi\mu DV$
Falling-sphere viscometer uses?	Stokes' law
Turbulent velocity profile?	1/7 power law, flatter
Darcy–Weisbach formula?	$h_f = fLV^2/(2gD)$
h_f in terms of Q?	$fLQ^2/(12.1D^5)$
Hydraulic mean depth of full pipe?	D/4
Loss at sudden enlargement?	$(V_1 - V_2)^2/2g$
Exit loss?	$V^2/2g$
Entrance loss (sharp)?	$0.5\ V^2/2g$
Pipes in series – what is same?	Discharge
Pipes in parallel – what is same?	Head loss
Equivalent pipe equation?	Dupuit: $L/D^5 = \Sigma L_i/D_i^5$
Max power transmission condition?	h_f = H/3

Question	Answer
Max transmission efficiency?	66.67%
Water hammer (sudden closure)?	ρVc
Critical time of valve closure?	2L/c
Device to protect penstock?	Surge tank
Network analysis method?	Hardy Cross

R5. BL, Models, Channels, Machines

Question	Answer
BL thickness defined at u = ?	0.99U
Laminar BL grows as?	√x
Blasius δ?	5x/√Re_x
Separation occurs with?	Adverse pressure gradient
At separation point?	(∂u/∂y)wall = 0
Drag on bluff bodies is mainly?	Pressure (form) drag
Lift on a rotating cylinder?	Magnus effect
Number of π-terms?	n − m
Model law for ships/spillways?	Froude
Model law for pipes/submarines?	Reynolds
Discharge ratio in Froude model?	L_r^2.5
Mach number significance?	Compressibility
Weber number significance?	Surface tension
Mach angle?	$\sin^{-1}(1/M)$
Across a normal shock, constant?	Stagnation temperature
Most economical rectangle?	b = 2y
Critical depth (rectangular)?	(q²/g)^(1/3)
Minimum specific energy?	1.5 y_c
Hydraulic jump converts?	Supercritical to subcritical
Max efficiency, series of flat plates?	50% at u = V/2
Turbine for high head?	Pelton
Turbine for low head?	Kaplan
Turbine with adjustable runner blades?	Kaplan
Draft tube used in?	Reaction turbines
Pelton jet ratio?	≈ 12
Number of buckets?	15 + m/2
Turbine specific speed?	N√P/H^(5/4)
Pump specific speed?	N√Q/H^(3/4)
Synchronous speed?	120f/p
Priming needed in?	Centrifugal pumps
Pump vane shape used?	Backward-curved
Pump power vs speed?	P ∝ N³
Pumps for high head arranged in?	Series

Question	Answer
Negative slip occurs when?	Long suction, short delivery, high speed
Air vessel saves work (single-acting)?	≈ 84.8%
Device using water hammer to lift water?	Hydraulic ram
Device that stores hydraulic energy?	Accumulator
Device that multiplies torque?	Torque converter

CHAPTER 17D

Paper-II Model Answers

P1. Statics and Kinematics

✎ **DESCRIPTIVE Q:** Explain the different types of manometers with their uses and limitations.

A manometer measures pressure by balancing it against a column of liquid. (1) Piezometer – a vertical tube open to air; simplest, measures moderate positive gauge pressure of liquids only; cannot measure vacuum, gas pressure or high pressures (column too tall). (2) Simple U-tube – one limb connected to the pipe, the other open, containing a heavier immiscible liquid (mercury); measures positive or negative gauge pressure of liquids and gases. (3) Single-column (well-type) manometer – one limb replaced by a large reservoir, so only one level change needs reading; the inclined version magnifies small readings (L = h/sinθ) for low pressures such as air ducts. (4) Differential U-tube – both limbs connected to two points; gives the pressure difference, e.g., across a venturimeter: p_A − p_B = gx(ρm − ρ). (5) Inverted U-tube – a lighter fluid (air/oil) at the top; for small pressure differences between two points in a liquid. (6) Micromanometers for very small differences. Limitations: slow response, unsuitable for fluctuating pressures, need careful reading, capillary error in narrow tubes, mercury toxicity.

✎ **DESCRIPTIVE Q:** Derive the continuity equation in Cartesian coordinates.

Consider a fluid element of sides dx, dy, dz. Mass flux entering the face normal to x = ρu dy dz; leaving = [ρu + ∂(ρu)/∂x dx] dy dz; net outflow in x = ∂(ρu)/∂x dx dy dz. Similarly for y and z. The rate of decrease of mass inside = −∂ρ/∂t dx dy dz. By conservation of mass, net outflow = rate of decrease of mass: ∂ρ/∂t + ∂(ρu)/∂x + ∂(ρv)/∂y + ∂(ρw)/∂z = 0. For steady flow ∂ρ/∂t = 0; for incompressible flow ρ is constant and the equation reduces to ∂u/∂x + ∂v/∂y + ∂w/∂z = 0. In one-dimensional pipe flow it gives $\rho_1 A_1 V_1 = \rho_2 A_2 V_2$ (or $A_1 V_1 = A_2 V_2$ for liquids).

✎ **DESCRIPTIVE Q:** Explain free and forced vortex flows with examples, and derive the equation of the free surface in a forced vortex.

In a vortex the fluid moves along curved paths. In a forced vortex the fluid rotates like a rigid body under an external torque, with V = ωr; the flow is rotational and energy is supplied continuously – e.g., liquid in a rotating vessel, flow inside a centrifugal pump impeller or turbine runner. In a free vortex no external torque acts, angular momentum is conserved and Vr = constant; the flow is irrotational – e.g., whirlpool, flow into a sink drain, flow in a pump casing outside the impeller. For a forced vortex, consider pressure variation: $\partial p/\partial r = \rho V^2/r = \rho\omega^2 r$ and $\partial p/\partial z = -\rho g$. Integrating, $p = \rho\omega^2 r^2/2 - \rho g z + C$. On the free surface p = atmospheric (constant), so $z = \omega^2 r^2/(2g)$ + constant: the free surface is a paraboloid of revolution. The height of the paraboloid for radius R is $h = \omega^2 R^2/(2g)$; because a paraboloid's volume is half its circumscribing cylinder, the liquid rises h/2 at the wall and falls h/2 at the centre from the original level.

P2. Dynamics and Pipe Flow

✎ **DESCRIPTIVE Q:** Derive the expression for the force exerted by a flowing fluid on a pipe bend.

Take a control volume of fluid in the bend between sections 1 (area A_1, pressure p_1, velocity V_1 along x) and 2 (A_2, p_2, V_2 inclined at θ to x). Forces on the fluid: pressure forces p_1A_1 (in +x) and p_2A_2 (opposite to V_2), and the reaction of the bend ($-F_x$, $-F_y$). Applying the momentum equation in x: $p_1A_1 - p_2A_2\cos\theta - F_x = \rho Q(V_2\cos\theta - V_1)$, so $F_x = \rho Q(V_1 - V_2\cos\theta) + p_1A_1 - p_2A_2\cos\theta$. In y: $-p_2A_2\sin\theta - F_y = \rho Q(V_2\sin\theta - 0)$, so $F_y = -\rho QV_2\sin\theta - p_2A_2\sin\theta$ (i.e., magnitude $\rho QV_2\sin\theta + p_2A_2\sin\theta$ in the −y direction). The resultant $F = \sqrt{(F_x^2 + F_y^2)}$ at $\tan^{-1}(F_y/F_x)$. Anchor blocks are designed to resist this force.

✎ **DESCRIPTIVE Q:** What is the Moody diagram? Explain its zones.

The Moody diagram is a log–log chart of the Darcy friction factor f against Reynolds number for different relative roughness ε/D, based on the Colebrook–White equation. Laminar zone (Re < 2000): f = 64/Re, a single straight line independent of roughness. Critical zone (2000–4000): flow is unstable and f uncertain. Transition (turbulent) zone: f depends on both Re and ε/D; the smooth-pipe curve is the lower limit (Blasius f = 0.316/Re^0.25 for $Re < 10^5$). Fully rough zone (high Re, to the right of a dashed boundary): curves become horizontal and f depends only on ε/D, because roughness elements project through the laminar sub-layer and losses are due to form drag on them; head loss then varies as V^2. The diagram is used to find f for Darcy–Weisbach calculations and to solve pipe-sizing problems.

✎ **DESCRIPTIVE Q:** Derive the condition for maximum power transmission through a pipe.

Water from a reservoir of head H flows through a pipe of length L and diameter D. Head at the outlet = H − h_f, with $h_f = fLV^2/(2gD)$. Power transmitted $P = \rho gQ(H - h_f) = \rho g(\pi D^2/4)V(H - fLV^2/(2gD))$. For a given pipe, P is maximum when $dP/dV = 0$: $H - 3fLV^2/(2gD) = 0$, i.e., $h_f = H/3$. The efficiency of transmission $\eta = (H - h_f)/H$ then equals 2/3 or 66.67%. In hydro power plants the penstock is designed for lower losses than this (higher efficiency), but the result explains why friction must be limited.

P3. Machines

✎ **DESCRIPTIVE Q:** Explain the working of a Kaplan turbine and state why it is suitable for low heads.

A Kaplan turbine is an axial-flow reaction turbine. Water from the scroll casing passes through guide vanes (wicket gates), turns through 90° in the whirl space and flows axially through a runner shaped like a ship's propeller with 3–8 blades on a large hub. The runner blades are twisted and can be rotated about their own axes by a servomotor in the hub, together with the guide vanes ('double regulation'), so the blade angles match the flow at every load. Water then leaves through an elbow draft tube. At low heads (2–70 m) a large discharge is needed for appreciable power; the axial runner passes large flows through a small diameter at high rotational speed, giving a high specific speed (255–860), which allows a compact, high-speed generator. Adjustable blades keep efficiency high (≈ 90%) over a wide range of load, unlike a fixed-blade propeller turbine.

✎ **DESCRIPTIVE Q:** Draw and explain the characteristic curves of a centrifugal pump.

Main characteristic curves are drawn at constant speed with discharge Q on the x-axis. Head H: maximum at zero flow (shut-off head) and falling with Q for backward-curved vanes, because of increasing losses and slip. Shaft power P: rises gradually with Q; for backward-curved vanes it levels off (non-overloading), so the motor cannot be overloaded at high flows. Efficiency η: zero at Q = 0, rising to a maximum at the design (best efficiency) point, then falling. The intersection of the H–Q curve with the system curve $H = H_{static} + kQ^2$ gives the operating

point. Operating characteristic curves are the main curves at the rated speed; iso-efficiency (Muschel) curves join points of equal efficiency on H–Q plots at different speeds and help choose the best speed. Pumps are selected so that the operating point lies near the BEP.

✎ **DESCRIPTIVE Q:** Explain the working of a reciprocating pump with an indicator diagram.

A single-acting reciprocating pump has a piston moving in a cylinder with a suction valve and a delivery valve, driven by a crank. During the suction stroke the piston moves away, pressure in the cylinder falls below atmospheric, the suction valve opens and liquid enters. During the delivery stroke the piston pushes the liquid out through the delivery valve. The ideal indicator diagram (pressure head vs stroke) is a rectangle: the suction line lies $H_{atm} - h_s$ and the delivery line $H_{atm} + h_d$, so its area $\propto \rho g A L N(h_s + h_d)/60$, the work done. Acceleration of the water columns ($h_a = (L/g)(A/a)\omega^2 r \cos\theta$) tilts these lines – pressure is lowest at the start of suction and highest at the start of delivery – but does not change the area. Pipe friction (maximum at mid-stroke) bulges the lines parabolically and increases the area (more work). Air vessels near the cylinder reduce both effects, smooth the flow and allow higher speeds without separation.

✎ **DESCRIPTIVE Q:** Compare centrifugal and reciprocating pumps.

Centrifugal pumps give a continuous, smooth discharge, handle large flows at low to medium heads, run at high speeds with direct motor coupling, are compact, cheap and simple to maintain, can handle dirty liquids, slurries and viscous liquids (with suitable impellers), but need priming and lose efficiency away from the design point. Reciprocating pumps give a small, pulsating discharge at high heads, are self-priming, maintain efficiency over a range of heads, but are bulky, costly, have many wearing parts (valves, packing), run at low speeds, need air vessels to smooth flow and are unsuitable for liquids with suspended solids. Hence centrifugal pumps dominate water supply, irrigation and process industries, while reciprocating pumps are used for high-pressure duties such as hydraulic testing, oil-well and boiler-feed service in small plants, and metering.

CHAPTER 17E

Solved Numericals – Set 3 (Velocity Triangles & Mixed Practice)

17E.1 Velocity Triangles

Problem 69 – Francis turbine work

A Francis turbine: inlet whirl velocity V_{w1} = 20 m/s, inlet blade speed u_1 = 25 m/s, radial discharge, net head 60 m. Find work per unit weight and hydraulic efficiency.

Work/weight = $V_{w1}\,u_1/g$ = 20 × 25/9.81 = 51.0 m; η_h = 51/60 = 85%

Problem 70 – Francis discharge

Runner D_1 = 1 m, width B_1 = 0.1 m, flow velocity V_{f1} = 4 m/s (neglect vane thickness). Find Q.

$Q = \pi D_1 B_1 V_{f1} = \pi \times 1 \times 0.1 \times 4 = 1.257\ m^3/s$

Problem 71 – Pelton hydraulic efficiency

V_1 = 80 m/s, u = 38 m/s, bucket deflection 165° (φ = 15°), k = 0.9. Find η_h.

$\eta_h = 2(V_1 - u)(1 + k\cos\varphi)u/V_1^2 = 2 \times 42 \times (1 + 0.869) \times 38/6400 = 5965.7/6400 = 93.2\%$

Problem 72 – Jet on moving curved vane

A 40 mm jet at 30 m/s strikes a series of semicircular vanes moving at 15 m/s. Find power and efficiency (no friction).

$a = 1.257 \times 10^{-3}\ m^2$; ρaV = 37.7 kg/s; work/s = $\rho aV \times 2(V - u) \times u$ = 37.7 × 30 × 15 = 16.97 kW

Jet KE/s = $\frac{1}{2} \times 37.7 \times 900$ = 16.97 kW → η = 100% (theoretical at u = V/2 with full reversal)

Problem 73 – Centrifugal pump outlet triangle

D_2 = 0.4 m, N = 1450 rpm, blade outlet angle β_2 = 30°, V_{f2} = 2.5 m/s, radial entry. Find V_{w2} and Euler head.

$u_2 = \pi \times 0.4 \times 1450/60 = 30.37$ m/s; $V_{w2} = u_2 - V_{f2}/\tan\beta_2 = 30.37 - 2.5/0.577 = 26.04$ m/s

$H_e = V_{w2}\,u_2/g = 26.04 \times 30.37/9.81 = 80.6$ m

Problem 74 – Minimum starting speed

D_2 = 0.5 m, D_1 = 0.25 m, manometric head 20 m. Find the minimum starting speed (centrifugal head = manometric head).

$(u_2^2 - u_1^2)/2g = 20 \rightarrow (\pi N/60)^2\,(0.25 - 0.0625) = 392.4 \rightarrow (\pi N/60)^2 = 2092.8 \rightarrow \pi N/60 = 45.75 \rightarrow N = 874$ rpm

17E.2 Mixed Practice

Problem 75 – Head of mercury to water

Convert 380 mm of mercury into kPa and metres of water.

$p = 13600 \times 9.81 \times 0.38 = 50.7$ kPa; water head $= 0.38 \times 13.6 = 5.17$ m

Problem 76 – Pressure in a lift

A tank with water 2 m deep is in a lift accelerating upward at 4.9 m/s². Find pressure at the bottom.

$p = \rho h(g + a) = 1000 \times 2 \times 14.71 = 29.4$ kPa

Problem 77 – Centre of pressure of triangle

A triangle of base 2 m and height 3 m is immersed vertically with its apex at the free surface (base horizontal at the bottom). Find $\bar{h}$, F and h*.

$\bar{h} = 2$ m (2h/3 from apex); $A = 3$ m²; $F = 9810 \times 3 \times 2 = 58.86$ kN

$I_G = bh^3/36 = 2 \times 27/36 = 1.5$ m⁴; $h^* = 2 + 1.5/(3 \times 2) = 2.25$ m (= 3h/4)

Problem 78 – Stagnation pressure

Water flows at 5 m/s with static pressure 150 kPa. Find stagnation pressure.

$p_0 = p + \rho V^2/2 = 150 + 12.5 = 162.5$ kPa

Problem 79 – Jet range

A water jet leaves a nozzle at 20 m/s at 45°. Find the horizontal range and maximum height.

$R = V^2/g = 400/9.81 = 40.8$ m; $H = V^2 \sin^2 45°/2g = 10.2$ m

Problem 80 – Mouthpiece discharge

A 50 mm external cylindrical mouthpiece works under 4 m head. Find Q ($C_d = 0.855$).

$Q = 0.855 \times 1.963 \times 10^{-3} \times 8.86 = 0.01487$ m³/s $= 14.9$ L/s

Problem 81 – Viscous flow between plates

Oil ($\mu = 0.2$ Pa·s) flows between fixed plates 10 mm apart with a pressure gradient of −2,400 Pa/m. Find u_{max} and mean velocity.

$u_{max} = (1/8\mu)(-dp/dx)\, t^2 = (1/1.6) \times 2400 \times 10^{-4} = 0.15$ m/s; $V = (2/3) \times 0.15 = 0.10$ m/s

Problem 82 – Pipe diameter for a given loss

Water at 0.1 m³/s must flow 2 km with $h_f \le 20$ m, $f = 0.02$. Find minimum D.

$D^5 = fLQ^2/(12.1\, h_f) = 0.02 \times 2000 \times 0.01/(242) = 1.653 \times 10^{-3} \rightarrow D = 0.278$ m (adopt 300 mm)

Problem 83 – Siphon summit

A siphon's summit is 3 m above the upstream water level; velocity 2 m/s; loss up to summit 0.5 m. Find summit pressure head (atmospheric 10.3 m).

$0 = p/\rho g + 3 + 0.204 + 0.5 \rightarrow p/\rho g = -3.70$ m (gauge) → absolute 6.6 m of water (above vapour limit)

Problem 84 – Parallel pipes

Two parallel pipes of equal length and f have diameters 0.2 m and 0.3 m. Find the discharge ratio Q_2/Q_1.

Same $h_f \rightarrow Q \propto D^{(5/2)} \rightarrow Q_2/Q_1 = (1.5)^{2.5} = 2.76$

Problem 85 – Model drag

A 1:20 ship model is tested at 2 m/s. Find the prototype speed and the ratio of wave-resistance forces.

$V_p = 2 \times \sqrt{20} = 8.94$ m/s; $F_p/F_m = 20^3 = 8000$

Problem 86 – Turbine speed change

A Pelton wheel runs at 500 rpm under 200 m. At what speed should it run under 50 m for similar conditions?

$N \propto \sqrt{H} \rightarrow 500 \times 0.5 = 250$ rpm

Problem 87 – Pumps in series

Two identical pumps each give 30 m head at 0.05 m³/s. Find total head in series and discharge in parallel (ideal).

Series: 60 m at 0.05 m³/s; parallel: 0.10 m³/s at 30 m (actual depends on system curve)

Problem 88 – Double-acting pump

Bore 150 mm, stroke 300 mm, 50 rpm, rod diameter neglected. Find Q_{th}.

$Q_{th} = 2ALN/60 = 2 \times 0.01767 \times 0.3 \times 50/60 = 8.84 \times 10^{-3}$ m³/s = 8.84 L/s

Problem 89 – Accumulator energy

An accumulator ram 200 mm diameter, lift 5 m, pressure 6 MPa. Find stored energy.

Energy = $pAL = 6 \times 10^6 \times 0.03142 \times 5 = 942.5$ kJ

Problem 90 – Hydraulic ram efficiency

Waste flow 100 L/min under 3 m supply head lifts 20 L/min to 15 m. Find D'Aubuisson and Rankine efficiencies.

D'Aubuisson $\eta = qH/((Q + q)h) = (20 \times 15)/(120 \times 3) = 83.3\%$; Rankine $\eta = q(H - h)/(Qh) = 20 \times 12/(100 \times 3) = 80\%$

CHAPTER 18

Last-Minute Revision Sheet

18.1 Must-Know Values

Quantity	Value
ρ water / mercury / sea water	1000 / 13,600 / 1025 kg/m³
γ water	9810 N/m³
μ water (20 °C) / ν water	1×10^{-3} Pa·s / 1×10^{-6} m²/s
1 poise / 1 stoke	0.1 Pa·s / 10^{-4} m²/s
K water	≈ 2.1 GPa
σ water–air	≈ 0.073 N/m
1 atm	101.325 kPa = 10.33 m water = 760 mm Hg
Speed of sound: water / air	≈ 1450 / 343 m/s
Critical Re (pipe)	2000 (laminar), 4000 (turbulent)
Critical Re (flat plate)	5×10^5
f laminar (Darcy)	64/Re
α, β laminar pipe	2, 4/3
V/u_max laminar pipe / plates	1/2 / 2/3
C_d venturi / orifice / nozzle	0.95–0.98 / 0.62 / 0.95
C_c, C_v orifice	0.64, 0.98
C_d external mouthpiece; Borda free/full	0.855; 0.5 / 0.707
Max power transmission	h_f = H/3, η = 66.7%
Jet on series of plates	η_max 50% at u = V/2
Pelton speed ratio; jet ratio	0.46; ≈ 12
Critical pressure ratio (air)	0.528
Practical siphon / suction lift	≈ 7.5 m / 6–7 m
Most economical rectangle	b = 2y, R = y/2
Critical depth; E_min	(q²/g)^(1/3); 1.5 y_c
Turbine Ns ranges (SI)	Pelton 8.5–51, Francis 50–340, Kaplan 255–860
Pump Ns ranges (SI)	Radial 10–80, mixed 80–160, axial 160–500

18.2 Commonly Confused Pairs

A	B
Liquid viscosity ↓ with temperature	Gas viscosity ↑ with temperature
Droplet Δp = 4σ/d	Soap bubble Δp = 8σ/d
Water rises (θ ≈ 0°)	Mercury falls (θ ≈ 130°)
Piezometer – no vacuum/gas	U-tube – vacuum and gases ok
Floating: M above G for stability	Submerged: B above G

A	B
Free vortex Vr = const (irrotational)	Forced vortex V = ωr (rotational)
φ – irrotational only	ψ – any 2-D incompressible
Steady: streamline = pathline = streakline	Unsteady: they differ
TEL always falls	HGL may rise
Euler eq – no viscosity	Navier–Stokes – with viscosity
Venturi divergent 5–7° (long)	Convergent ~20° (short)
Rectangular notch Q ∝ H^1.5	V-notch Q ∝ H^2.5
Laminar f = 64/Re (no roughness effect)	Fully rough: f = function of ε/D only
Sudden enlargement loss $(V_1-V_2)^2/2g$	Sudden contraction ≈ $0.5V_2^2/2g$
Water hammer sudden Δp = ρVc	Gradual Δp = ρLV/t
Separation – adverse gradient	No separation – favourable gradient
Reynolds model – pipes, submarines	Froude model – channels, ships, spillways
Impulse turbine – no draft tube	Reaction turbine – draft tube
Kaplan – adjustable blades	Propeller – fixed blades
Turbine Ns = N√P/H^(5/4)	Pump Ns = N√Q/H^(3/4)
Pumps in series – head adds	Pumps in parallel – discharge adds
Centrifugal – priming needed	Reciprocating – self-priming
Fluid coupling – T_out = T_in	Torque converter – torque multiplied (stator)
Cavitation in turbine – runner exit	Cavitation in pump – impeller eye

18.3 Paper-II Quick Answers (5-mark style)

✎ **DESCRIPTIVE Q:** Distinguish between laminar and turbulent flow.

Laminar flow: fluid moves in smooth parallel layers with no mixing across them; occurs at low Re (< 2000 in pipes); shear stress is purely viscous (τ = μ du/dy); velocity profile in a pipe is parabolic with V = u_max/2; head loss ∝ V ($h_f = 32\mu VL/\rho gD^2$), f = 64/Re, independent of roughness. Examples: oil in narrow tubes, blood in capillaries, groundwater flow. Turbulent flow: fluid particles move randomly with eddies and strong mixing; occurs at high Re (> 4000); shear stress includes large turbulent (Reynolds) stresses; velocity profile is flatter (≈ 1/7 power law, V ≈ 0.82 u_max); head loss ∝ $V^{1.75}$–V^2; f depends on Re and relative roughness (Moody chart). Examples: water in town mains, rivers, air around vehicles.

✎ **DESCRIPTIVE Q:** What is cavitation? State its effects on hydraulic machines and how it is prevented.

Cavitation is the formation of vapour bubbles in a flowing liquid wherever the local absolute pressure falls to the vapour pressure, followed by the violent collapse of these bubbles when they are carried into regions of higher pressure. The collapse produces very high local pressures (hundreds of MPa) and micro-jets. Effects: pitting and erosion of runner blades, impeller vanes and draft tubes; noise and vibration; sudden drop in head, discharge and efficiency. In turbines it occurs at the runner outlet/draft-tube inlet, in pumps at the impeller eye. Prevention: keep pressure above vapour pressure – limit the suction height of pumps and the setting height of turbines (Thoma's σ above critical; NPSH_A > NPSH_R), avoid high speeds, use cavitation-resistant materials (stainless steel, aluminium bronze), polish surfaces, and avoid operation far from design conditions.

✎ **DESCRIPTIVE Q:** Compare impulse and reaction turbines.

Impulse turbine (Pelton): all available head is converted to kinetic energy in the nozzle before the runner; water acts at atmospheric pressure throughout; the casing has no hydraulic function; partial admission (water strikes only a few buckets at a time); can be installed above tail race and needs no draft tube; flow is regulated by a spear valve; suited to high heads and small discharges; low specific speed. Reaction turbine (Francis, Kaplan): part of the head is pressure energy at the runner inlet, and pressure falls as water passes through the runner; the runner is full of water and enclosed in an airtight casing; full admission all round; a draft tube is essential; flow is regulated by guide vanes (and runner blades in Kaplan); suited to medium/low heads and large discharges; higher specific speeds.

✎ **DESCRIPTIVE Q:** Explain the working of a hydraulic ram.

A hydraulic ram lifts a small quantity of water to a large height using the energy of a larger quantity of water falling through a small height – without any external power. Water from a supply tank flows down a drive pipe to a valve chamber and escapes through the waste valve. As the velocity rises, the dynamic force closes the waste valve suddenly. The resulting water hammer raises the pressure in the chamber, opening the delivery valve and forcing water into an air vessel and up the delivery pipe. When the pressure falls, the delivery valve closes and the waste valve reopens under its own weight, and the cycle repeats 50–150 times a minute. Efficiency (D'Aubuisson) = $q(H - h)/(Qh)$, where q is the water lifted to height H, Q is the waste discharge and h the supply head. Used to supply water to hilly villages and farms.

✔ **TOPPER TRICK:** In the last 24 hours, read only the formula boxes (Appendix A), the PYQ-trend boxes (Appendix B), the ratio-twister table and this chapter. Don't open new books.

APPENDIX

Appendix A – One-Place Formula Sheet

Every formula box from the theory chapters, collected chapter by chapter for the final revision. Read it with a pen and rewrite any formula you hesitate on.

Properties of Fluids

Mass density ρ = m/V (kg/m³); water 1000 at 4 °C, mercury 13,600, air ≈ 1.225 at 15 °C sea level

Specific weight w = γ = ρg (N/m³); water 9810 N/m³ = 9.81 kN/m³

Specific volume v = 1/ρ (m³/kg)

Specific gravity (relative density) S = ρ_fluid/ρ_water (gases: relative to air or hydrogen)

Newton's law of viscosity: τ = μ (du/dy)

Dynamic viscosity μ — unit N·s/m² = Pa·s; CGS: poise (1 P = 1 dyne·s/cm² = 0.1 Pa·s); 1 centipoise = 10^{-3} Pa·s

Kinematic viscosity ν = μ/ρ — unit m²/s; CGS: stoke (1 St = 1 cm²/s = 10^{-4} m²/s)

Dimensions: μ = $[ML^{-1}T^{-1}]$; ν = $[L^2T^{-1}]$

Power law (Ostwald–de Waele): τ = K $(du/dy)^n$ (n = 1 Newtonian; n < 1 pseudoplastic; n > 1 dilatant)

Bingham plastic: τ = τ_y + μ_p (du/dy)

Bulk modulus K = −dp/(dV/V) = dp/(dρ/ρ) (N/m²); compressibility β = 1/K

For water K ≈ 2.1–2.2 × 10^9 Pa (2.1 GPa); steel ≈ 160 GPa; air at 1 atm (isothermal) ≈ 1.01 × 10^5 Pa

Gas: isothermal K = p; adiabatic (isentropic) K = γp

Speed of sound c = √(K/ρ); in ideal gas c = √(γRT)

Pressure excess inside a liquid droplet: Δp = 4σ/d = 2σ/R

Inside a soap bubble (two free surfaces): Δp = 8σ/d = 4σ/R

Inside a liquid jet (cylinder): Δp = 2σ/d = σ/R

Capillary rise/fall: h = 4σ cosθ / (ρgd) (θ = contact angle)

Rise between two parallel plates (gap t): h = 2σ cosθ / (ρgt)

Perfect gas: p = ρRT; R_air = 287 J/kg·K

Isothermal: p/ρ = constant; adiabatic: p/ρ^γ = constant

Shear force between plates: F = τA = μ A V / h (linear velocity profile)

Torque on a rotating shaft in a journal bearing (thin oil film t): T = μ (πDN/60)/t × (πDL) × D/2

Power lost P = T ω = 2πNT/60

Fluid Statics and Pressure Measurement

Hydraulic press: F/a = W/A → mechanical advantage W/F = A/a = $(D/d)^2$

dp/dz = −ρg (z upward) → p = ρgh (h measured downward from the free surface)

Pressure head h = p/(ρg) = p/w; converting heads: $h_1S_1 = h_2S_2$

Atmospheric pressure variation (linear lapse): p = p_0 (1 − Lz/T_0)^(g/(LR))

Absolute pressure = atmospheric pressure + gauge pressure

Vacuum (negative gauge) pressure = atmospheric − absolute

U-tube (mercury), pipe centre h_1 above mercury in left limb, deflection h_2: $p_A + \rho_1 g h_1 = \rho_2 g h_2 \rightarrow p_A = \rho_2 g h_2 - \rho_1 g h_1$

Vacuum with U-tube: $p_A + \rho_1 g h_1 + \rho_2 g h_2 = 0$ (gauge) → $p_A = -(\rho_1 g h_1 + \rho_2 g h_2)$

Differential U-tube, same fluid in both pipes at same level: $p_A - p_B = gx(\rho_m - \rho)$

Inverted U-tube: $p_A - p_B = gx(\rho - \rho_{light})$

Single column: $p_A = \rho_2 g h_2 (1 + a/A) - \rho_1 g h_1 \approx \rho_2 g h_2 - \rho_1 g h_1$ when A □ a

Inclined manometer: $h_2 = L \sin\theta$

Tank accelerating horizontally with a: free surface tilts, $\tan\theta = a/g$

Vertical acceleration a (upwards): $p = \rho h(g + a)$; downwards: $p = \rho h(g - a)$; free fall: $p = 0$ (gauge)

General: $\tan\theta = a_x/(g + a_y)$

Free surface is a paraboloid: $z = \omega^2 r^2/(2g)$; height rise at rim above vertex $h = \omega^2 R^2/(2g)$

Volume of paraboloid = ½ × volume of circumscribing cylinder → rise at wall = fall at centre = h/2 (from original level)

Pressure: $p = \rho\omega^2 r^2/2 - \rho g z + C$

Hydrostatic Forces on Surfaces

Total pressure on any plane surface: $F = \rho\, g\, A\, \bar{h}$ ($\bar{h}$ = vertical depth of centroid below free surface)

Vertical plane surface: $h^* = \bar{h} + I_G/(A\,\bar{h})$

Inclined plane surface (angle θ with free surface): $h^* = \bar{h} + I_G \sin^2\theta / (A\,\bar{h})$

Horizontal surface: $F = \rho g A h$, acts at the centroid ($h^* = \bar{h}$)

Horizontal component F_H = force on the vertical projection of the curved surface ($\rho\, g\, A_{proj}\, \bar{h}_{proj}$), acting at its centre of pressure

Vertical component F_V = weight of the liquid (real or imaginary) directly above the curved surface up to the free surface, acting through its centroid

Resultant $F = \sqrt{(F_H^2 + F_V^2)}$; inclination $\tan\theta = F_V/F_H$

Force on vertical gate (width b, water depth H): $F = \rho g b H^2/2$ acting at H/3 from the bottom

Lock gates: reaction between gates R = force on each gate P, with P = R when gates make 120° (angle with the normal 30°): $P = F/(2\sin\alpha)$

Gravity dam stability: no tension at heel if the resultant falls within the middle third of the base ($e \le b/6$)

Buoyancy, Floatation and Stability

$F_B = \rho_{fluid}\, g\, V_{displaced}$

Floating body: weight W = F_B → fraction submerged = ρ_{body}/ρ_{fluid}

Apparent weight of a submerged body = $W - \rho_{fluid}\, g\, V$

BM = I/V (I = least second moment of area of the water-plane section about the tilt axis; V = displaced volume)

GM = BM − BG = I/V − BG (BG positive when G above B)

Experimental: $GM = w\, x\, /\, (W \tan\theta)$ (moving a load w through distance x across the deck tilts by θ)

Time period of rolling (oscillation): $T = 2\pi \sqrt{(k^2 / (g \cdot GM))}$ (k = radius of gyration about the roll axis)

Fluid Kinematics

Streamline equation: $dx/u = dy/v = dz/w$

General (3-D, compressible): $\partial\rho/\partial t + \partial(\rho u)/\partial x + \partial(\rho v)/\partial y + \partial(\rho w)/\partial z = 0$

Steady compressible: $\nabla\cdot(\rho V) = 0$; 1-D: $\rho_1 A_1 V_1 = \rho_2 A_2 V_2$ (mass flow constant)

Incompressible (steady or unsteady): $\partial u/\partial x + \partial v/\partial y + \partial w/\partial z = 0$; 1-D: $A_1 V_1 = A_2 V_2 = Q$

Cylindrical: $\partial(r\, v_r)/(r\partial r) + \partial v_\theta/(r\partial\theta) + \partial v_z/\partial z = 0$

$a_x = \partial u/\partial t + u\, \partial u/\partial x + v\, \partial u/\partial y + w\, \partial u/\partial z$ (local + convective)

Along a streamline: $a_s = \partial V/\partial t + V\, \partial V/\partial s$; normal (centripetal): $a_n = V^2/R + \partial V_n/\partial t$

Rotation about z: $\omega_z = \tfrac{1}{2}(\partial v/\partial x - \partial u/\partial y)$; similarly ω_x, ω_y

Vorticity $\Omega = 2\omega$ (curl of velocity)

Circulation $\Gamma = \oint V\cdot ds$ = vorticity × enclosed area (Stokes' theorem)

Angular deformation (shear strain rate) $\dot{\gamma}_{xy} = \partial v/\partial x + \partial u/\partial y$

Velocity potential ϕ: $u = -\partial\phi/\partial x$, $v = -\partial\phi/\partial y$ (some texts use + sign)

Stream function ψ: $u = -\partial\psi/\partial y$, $v = \partial\psi/\partial x$ (or $u = \partial\psi/\partial y$, $v = -\partial\psi/\partial x$ – sign convention varies)

Existence: ϕ exists only for irrotational flow; ϕ satisfies Laplace's equation $\nabla^2\phi = 0$ if the flow is also incompressible

ψ exists for any 2-D incompressible flow (rotational or irrotational); $\nabla^2\psi = 0$ only if irrotational

Discharge per unit width between two streamlines $q = \psi_2 - \psi_1$

Lines of constant ϕ (equipotentials) and constant ψ (streamlines) intersect at right angles: slope product = −1

Free vortex (irrotational): $V\cdot r$ = constant; pressure rises outward: $p + \rho V^2/2$ = const across streamlines

Forced vortex (rotational, rigid-body rotation): $V = \omega r$; free surface paraboloid, $h = \omega^2 R^2/(2g)$

Rankine combined vortex: forced core ($r < R$) + free vortex outside

Fluid Dynamics – Energy and Momentum

Euler's equation along a streamline: $dp/\rho + g\, dz + V\, dV = 0$

$p/(\rho g) + V^2/(2g) + z$ = constant (pressure head + velocity (kinetic) head + datum (potential) head = total head)

Energy form per unit mass: $p/\rho + V^2/2 + gz$ = const; per unit volume: $p + \rho V^2/2 + \rho g z$ = const

Real fluid between sections 1 and 2: $p_1/\rho g + V_1^2/2g + z_1 = p_2/\rho g + V_2^2/2g + z_2 + h_L$

With pump (h_p) and turbine (h_t): $H_1 + h_p = H_2 + h_t + h_L$

Kinetic energy correction factor $\alpha = (1/(AV^3))\int u^3\, dA$

Momentum correction factor $\beta = (1/(AV^2))\int u^2\, dA$

Momentum equation (steady): $\Sigma F = \rho Q\,(V_2 - V_1)$ (vector, per direction)

Force on a pipe bend (x-direction): $F_x = \rho Q(V_1 - V_2\cos\theta) + p_1 A_1 - p_2 A_2\cos\theta$

y-direction: $F_y = \rho Q(-V_2\sin\theta) - p_2 A_2\sin\theta$ (sign depends on the coordinate choice); resultant $F = \sqrt{(F_x^2 + F_y^2)}$

Force on a nozzle (water exerts on nozzle): $F = p_1 A_1 - \rho Q(V_2 - V_1)$

Moment of momentum (torque): $T = \rho Q(V_2 r_2 - V_1 r_1)$

Propulsive (Froude) efficiency of a jet: $\eta = 2u/(u + V_j)$ (u = vehicle speed, V_j = jet velocity relative to the vehicle) – maximum as $V_j \to u$

Jet issuing at angle θ with velocity V: max height = $V^2\sin^2\theta/(2g)$; horizontal range = $V^2\sin 2\theta/g$ (max at 45°)

Equation of jet path: $y = x\tan\theta - g x^2/(2V^2\cos^2\theta)$

Horizontal jet from an orifice at height y above ground: horizontal distance $x = V\sqrt{(2y/g)}$; $Cv = x/(2\sqrt{(yH)})$

Flow Measurement – Meters, Orifices, Mouthpieces, Notches

Theoretical discharge $Q_{th} = A_1A_2\sqrt{(2gh)} / \sqrt{(A_1^2 - A_2^2)}$

Actual discharge $Q = C_d Q_{th}$; $C_d \approx 0.95–0.98$

Differential head (mercury–water): $h = x (S_m/S - 1) = 12.6 x$ (for water)

Inclined venturimeter: $h = (p_1/\rho g + z_1) - (p_2/\rho g + z_2)$ – same formula; the manometer reading is unaffected by inclination

Orificemeter: $Q = C_d a_0a_1\sqrt{(2gh)}/\sqrt{(a_1^2 - a_0^2)}$; $C_d \approx 0.6–0.65$

Pitot tube: $V = \sqrt{(2gh)}$; with differential manometer: $h = x(S_m/S - 1)$

Coefficient of contraction $C_c = a_c/a \approx 0.61–0.69$ (≈ 0.64)

Coefficient of velocity $C_v = V_{actual}/\sqrt{(2gH)} \approx 0.95–0.99$

Coefficient of discharge $C_d = C_c \times C_v \approx 0.62$

Coefficient of resistance $C_r = (1/C_v^2) - 1$ (loss of head = $C_r \times$ actual velocity head)

Discharge through a small orifice: $Q = C_d a\sqrt{(2gH)}$

Large rectangular orifice (breadth b, top H_1, bottom H_2): $Q = (2/3) C_d b\sqrt{(2g)} (H_2^{(3/2)} - H_1^{(3/2)})$

Fully drowned orifice: $Q = C_d a\sqrt{(2gH)}$, H = difference of water levels

Time to empty a tank (area A) from H_1 to H_2 through orifice a: $T = 2A(\sqrt{H_1} - \sqrt{H_2})/(C_d a\sqrt{(2g)})$

Time to empty completely: $T = 2A\sqrt{H}/(C_d a\sqrt{(2g)})$

Hemispherical tank of radius R emptying from full: $T = (14\pi R^{(5/2)})/(15 C_d a\sqrt{(2g)})$

Time to empty a circular horizontal cylinder (length L, radius R) from full: $T = (4L/(3C_d a\sqrt{(2g)})) (2R)^{(3/2)}$

Loss at entrance of external mouthpiece $\approx 0.375 V^2/2g$; pressure at vena contracta ≈ 2.5 m below atmospheric (absolute 7.8 m of water)

Maximum head for a convergent–divergent mouthpiece to avoid separation: $H \approx 2.5–8$ m depending on area ratio

Rectangular notch/weir: $Q = (2/3) C_d L\sqrt{(2g)} H^{(3/2)}$

Triangular (V) notch: $Q = (8/15) C_d\sqrt{(2g)}\tan(\theta/2) H^{(5/2)}$; for 90° V-notch with $C_d = 0.6$: $Q \approx 1.417 H^{(5/2)}$ (m^3/s)

Trapezoidal notch = rectangular + triangular parts

Cipolletti weir (trapezoidal, side slope 1 horizontal : 4 vertical, i.e., $\approx 14°$): $Q = 1.86 L H^{(3/2)}$ (m^3/s) – compensates end contractions

Stepped notch = sum of rectangular notches

Francis' formula with end contractions: $Q = 1.84 (L - 0.1 n H) H^{(3/2)}$ (n = number of end contractions)

Bazin's formula (velocity of approach included): $Q = m\sqrt{(2g)} L H^{(3/2)}$, $m = 0.405 + 0.003/H$

Error in discharge due to error in head: rectangular $dQ/Q = (3/2) dH/H$; triangular $dQ/Q = (5/2) dH/H$

Time to lower water level from H_1 to H_2 over a rectangular weir: $T = (3A/(C_d L\sqrt{(2g)})) (1/\sqrt{H_2} - 1/\sqrt{H_1})$

Viscous (Laminar) Flow

Reynolds number $Re = \rho VD/\mu = VD/\nu$ (ratio of inertia force to viscous force)

Shear stress distribution: $\tau = (-\partial p/\partial x)(r/2)$ → zero at centre, maximum at wall $\tau_0 = (-\partial p/\partial x)(R/2)$

Velocity profile: $u = (1/4\mu)(-\partial p/\partial x)(R^2 - r^2) = u_{max}\,(1 - r^2/R^2)$ (parabolic)

Maximum velocity $u_{max} = (1/4\mu)(-\partial p/\partial x)\,R^2$; average velocity $V = u_{max}/2$

Discharge $Q = (\pi/8\mu)(-\partial p/\partial x)\,R^4 = (\pi D^4 \Delta p)/(128\,\mu L)$

Pressure drop (Hagen–Poiseuille): $\Delta p = 32\,\mu V L / D^2$; head loss $h_f = 32\,\mu V L/(\rho g D^2)$

Darcy friction factor (laminar): $f = 64/Re$; Fanning friction factor $f' = 16/Re$

Wall shear stress $\tau_0 = 8\mu V/D$; power required $P = \Delta p \times Q$

Radius where local velocity equals the average velocity: $r = R/\sqrt{2} = 0.707R$

Both plates fixed (gap t): $u = (1/2\mu)(-\partial p/\partial x)(t\,y - y^2)$; $u_{max} = (1/8\mu)(-\partial p/\partial x)\,t^2$; $V_{avg} = (2/3)\,u_{max}$

Pressure drop: $\Delta p = 12\,\mu V L / t^2$

Shear stress $\tau = (-\partial p/\partial x)(t/2 - y)$ (zero at mid-plane)

One plate moving with velocity U, no pressure gradient (Couette flow): $u = U\,y/t$ (linear); $\tau = \mu U/t$

Couette flow with pressure gradient: $u = U\,y/t - (1/2\mu)(\partial p/\partial x)(t\,y - y^2)$

Journal bearing viscous torque: $T = \mu \pi^2 N D^3 L / (120\,t)$; power $P = 2\pi NT/60 = \mu \pi^3 N^2 D^3 L/(3600\,t)$

Footstep bearing: $T = \mu \pi^2 N R^4/(60\,t)$; collar bearing: $T = \mu \pi^2 N (R_2^4 - R_1^4)/(60\,t)$

Dashpot (piston of diameter D, length L, clearance t, velocity V): $\mu = 4\,W\,t^3/(3\pi L D^3 V)$

Stokes' law (drag on a sphere in creeping flow, $Re < 1$): $F_D = 3\pi\mu DV$; $C_D = 24/Re$

Terminal velocity of a falling sphere: $V_t = D^2(\rho_s - \rho_f)\,g / (18\mu)$

Turbulent Flow and Flow Through Pipes

Turbulent shear stress: $\tau = \tau_{laminar} + \tau_{turbulent} = \mu\,d\bar{u}/dy - \rho\,\overline{u'v'}$ (Reynolds stress)

Boussinesq eddy viscosity: $\tau_t = \eta\,d\bar{u}/dy$; Prandtl mixing length: $\tau_t = \rho\,l^2\,(d\bar{u}/dy)^2$

Power-law profile (smooth pipe, $Re \approx 10^5$): $u/u_{max} = (y/R)^{(1/7)}$; $V \approx 0.817\,u_{max}$ (often taken ≈ 0.8–0.85 u_{max})

Shear (friction) velocity $u^* = \sqrt{(\tau_0/\rho)}$

Laminar sub-layer thickness $\delta' = 11.6\,\nu/u^*$

Darcy–Weisbach: $h_f = f\,L\,V^2 / (2gD)$ (f = Darcy friction factor)

In terms of discharge: $h_f = f\,L\,Q^2 / (12.1\,D^5) = 8fLQ^2/(\pi^2 g D^5)$

Fanning friction factor $f' = f/4$ → $h_f = 4f'\,L\,V^2/(2gD)$

Wall shear stress $\tau_0 = (f/8)\,\rho\,V^2$

Blasius (smooth pipes, $4000 < Re < 10^5$): $f = 0.316/Re^{0.25}$

Chezy's formula: $V = C\sqrt{(m\,i)}$; m = hydraulic mean depth = A/P (= D/4 for full circular pipe); i = h_f/L; $C = \sqrt{(8g/f)}$

Manning's formula: $V = (1/n)\,R^{(2/3)}\,S^{(1/2)}$

Hazen–Williams (water supply): $V = 0.849\,C_{HW}\,R^{0.63}\,S^{0.54}$

Colebrook–White: $1/\sqrt{f} = -2\log_{10}(\varepsilon/(3.7D) + 2.51/(Re\sqrt{f}))$ (basis of the Moody diagram)

Sudden enlargement loss as fraction of initial KE: $h_L = (1 - A_1/A_2)^2\,V_1^2/2g$

Series: Q same; H = Σh_f + minor losses; equivalent pipe (Dupuit): $L/D^5 = L_1/D_1^5 + L_2/D_2^5 + \ldots$

Parallel: h_f same in each branch; $Q = Q_1 + Q_2 + \ldots$

Three-reservoir problem: continuity at the junction + equal piezometric head at the junction for all branches

Power available at outlet: $P = \rho\,g\,Q\,(H - h_f)$

Efficiency of transmission $\eta = (H - h_f)/H$

Condition for maximum power transmitted: $h_f = H/3 \rightarrow \eta_{max} = 66.67\%$

Nozzle at end of pipe: maximum power when the velocity head of the jet = 2H/3, i.e., $h_f = H/3$

Diameter of nozzle for maximum power: $d = (D^5/(8fL))^{1/4}$ (Darcy f)

Pressure wave velocity: $c = \sqrt{(K/\rho)}$ (rigid pipe); elastic pipe: $c = \sqrt{((K/\rho)/(1 + KD/(Et)))}$

Critical time of closure $t_c = 2L/c$

Sudden (instantaneous, $t < 2L/c$) closure: $\Delta p = \rho V c$

Gradual closure ($t > 2L/c$): $\Delta p = \rho L V / t$

Boundary Layer, Drag and Lift

Boundary layer thickness δ: distance where $u = 0.99 U$

Displacement thickness $\delta^* = \int_0^{\delta} (1 - u/U)\, dy$ (distance by which the outer flow is displaced)

Momentum thickness $\theta = \int_0^{\delta} (u/U)(1 - u/U)\, dy$ (loss of momentum flux)

Energy thickness $\delta^{**} = \int_0^{\delta} (u/U)(1 - u^2/U^2)\, dy$

Shape factor $H = \delta^*/\theta$ (Blasius laminar ≈ 2.59; turbulent ≈ 1.3–1.4)

Order (Blasius profile): $\delta > \delta^* > \delta^{**} > \theta$ ($\delta^* = 1.72$, $\delta^{**} \approx 1.04$, $\theta = 0.664$ in units of $x/\sqrt{Re_x}$)

Linear profile $u/U = y/\delta$: $\delta^* = \delta/2$, $\theta = \delta/6$, $H = 3$

Parabolic profile $u/U = 2(y/\delta) - (y/\delta)^2$: $\delta^* = \delta/3$, $\theta = 2\delta/15$, $H = 2.5$

Laminar (Blasius): $\delta = 5x/\sqrt{Re_x}$ (exact ≈ 4.91); $\delta^* = 1.72x/\sqrt{Re_x}$; $\theta = 0.664x/\sqrt{Re_x}$

Local skin-friction coefficient (laminar): $C_f = 0.664/\sqrt{Re_x}$; average $C_D = 1.328/\sqrt{Re_L}$

Turbulent (1/7 power law): $\delta = 0.37x/Re_x^{1/5}$; $C_D = 0.074/Re_L^{1/5}$ ($5\times10^5 < Re < 10^7$)

Von Kármán momentum integral equation: $\tau_0/(\rho U^2) = d\theta/dx$

Drag $F_D = C_D (\frac{1}{2} \rho U^2) A$; Lift $F_L = C_L (\frac{1}{2} \rho U^2) A$ (A = projected/frontal area for drag, plan area for aerofoils)

Kutta–Joukowski: Lift per unit span $L = \rho U \Gamma$

Magnus effect – rotating cylinder/ball in a stream experiences lift: swing of a cricket ball, spin of a tennis ball, Flettner rotor ships

Dimensional Analysis and Model Similitude

Number of π-terms = n − m

Froude law (same g): power ratio $P_r = L_r^{7/2}$; pressure ratio $p_r = L_r$; acceleration ratio = 1

Reynolds law: $V_r L_r / \nu_r = 1$

Compressible Flow (Basics)

Speed of sound: $c = \sqrt{(dp/d\rho)} = \sqrt{(K/\rho)}$; ideal gas $c = \sqrt{(\gamma RT)}$; air at 15 °C ≈ 340 m/s, 20 °C ≈ 343 m/s

Mach number $M = V/c$

Mach angle: $\sin \alpha = 1/M$ (Mach cone exists only for $M \geq 1$)

Energy equation (adiabatic): $h + V^2/2 = h_0 \rightarrow c_pT + V^2/2 = c_pT_0$

Stagnation temperature: $T_0/T = 1 + ((\gamma - 1)/2) M^2$

Stagnation pressure: $p_0/p = [1 + ((\gamma - 1)/2) M^2]^{(\gamma/(\gamma-1))}$

Stagnation density: $\rho_0/\rho = [1 + ((\gamma - 1)/2) M^2]^{(1/(\gamma-1))}$

Critical (sonic, M = 1) ratios for air (γ = 1.4): $T^*/T_0 = 0.833$, $p^*/p_0 = 0.528$, $\rho^*/\rho_0 = 0.634$

Area–velocity relation: $dA/A = (M^2 - 1)\, dV/V$

Maximum mass flow (choked): $\dot{m}_{max} = 0.0404\, A^*\, p_0/\sqrt{T_0}$ (air, SI)

$M_2^2 = (M_1^2 + 2/(\gamma - 1)) / ((2\gamma/(\gamma - 1))\, M_1^2 - 1)$

Open Channel Flow

Hydraulic radius (hydraulic mean depth) R = A/P; hydraulic depth $D_h = A/T$ (T = top width)

Froude number $Fr = V/\sqrt{(g\, D_h)}$ (rectangular: $V/\sqrt{(gy)}$)

Chezy: $V = C\sqrt{(R\, S)}$; $Q = A\, C\sqrt{(R\, S)}$

Manning: V = (1/n) R^(2/3) S^(1/2); Chezy C related: C = R^(1/6)/n

Bazin: C = 87/(1 + K/√R); Kutter's formula (older)

Specific energy $E = y + V^2/(2g) = y + q^2/(2gy^2)$ (q = discharge per unit width)

Critical depth (rectangular): $y_c = (q^2/g)^{(1/3)}$; $E_{min} = (3/2)\, y_c$; $V_c = \sqrt{(g\, y_c)}$

At critical flow: Fr = 1, specific energy minimum for given q, discharge maximum for given E

General critical condition: $Q^2\, T / (g\, A^3) = 1$

Sequent (conjugate) depths: $y_2/y_1 = \frac{1}{2}\,[-1 + \sqrt{(1 + 8\, Fr_1^2)}]$

Energy loss: $\Delta E = (y_2 - y_1)^3 / (4\, y_1\, y_2)$

Length of jump ≈ 5–7 $(y_2 - y_1)$ (≈ 6.1 y_2 for many designs)

Impact of Jets

Fixed flat plate normal to jet: $F = \rho\, a\, V^2$

Fixed flat plate inclined at θ to the jet (normal force): $F_n = \rho\, a\, V^2 \sin\theta$; component along jet $F_x = \rho a V^2 \sin^2\theta$; across $F_y = \rho a V^2 \sin\theta \cos\theta$

Fixed curved vane, jet striking at centre and deflected through angle (180° − φ): $F_x = \rho\, a\, V^2\, (1 + \cos\varphi)$

Semicircular vane (jet reversed, φ = 0): $F = 2\rho a V^2$

Curved plate, jet at one tip (unsymmetrical): $F_x = \rho a V(V_1\cos\theta + V_2\cos\varphi)$

Single flat plate moving in the jet direction with u: $F = \rho\, a\, (V - u)^2$; work done per second $= \rho a (V - u)^2\, u$

Efficiency (single moving plate): $\eta = 2u(V - u)^2/V^3$; maximum at u = V/3, $\eta_{max} = 8/27 = 29.6\%$

Series of flat plates (wheel): $F = \rho\, a\, V\, (V - u)$; $\eta = 2u(V - u)/V^2$; maximum at u = V/2, $\eta_{max} = 50\%$

Series of radial curved vanes, jet deflected by (180° − φ): $\eta_{max} = (1 + \cos\varphi)/2$ at u = V/2

Semicircular buckets (φ = 0): $\eta_{max} = 100\%$ (theoretical)

Curved moving vane, velocity triangles: work done per second $= \rho\, a\, V_1\, (V_{w1} \pm V_{w2})\, u$; per unit weight $= (V_{w1} \pm V_{w2})\, u\, /\, g$

Plate hinged at the top, jet striking at the centre of gravity: $\sin\theta = \rho a V^2/W$ (W = weight of plate)

Hydraulic Turbines

Gross head H_g = head race level − tail race level

Net (effective) head H = H_g − h_f (penstock losses); Pelton: H measured at the nozzle base; reaction: H = (p₁/ρg + V₁²/2g + z₁) − (V₂²/2g at draft tube exit)

Water power (input) = ρ g Q H

Hydraulic efficiency η_h = power developed by runner / water power = (V_w1 u₁ ± V_w2 u₂)/(gH)

Mechanical efficiency η_m = shaft power / runner power

Volumetric efficiency η_v = Q actually striking runner / Q supplied

Overall efficiency η_o = shaft power / (ρgQH) = η_h × η_m (× η_v)

Euler's turbine equation: work per unit mass = V_w1 u₁ ± V_w2 u₂

Specific speed of turbine: N_s = N √P / H^(5/4) (P in kW, H in m, N in rpm)

Jet velocity V₁ = C_v √(2gH), C_v ≈ 0.97–0.99

Bucket speed u = φ √(2gH), speed ratio φ ≈ 0.43–0.48

Work per second = ρ a V₁ (V₁ − u)(1 + k cosφ) u (k = bucket friction factor, φ = outlet angle)

Hydraulic efficiency η_h = 2(V₁ − u)(1 + k cosφ) u / V₁²; maximum at u = V₁/2: η_h,max = (1 + k cosφ)/2

Jet ratio m = D/d (pitch circle dia / jet dia) ≈ 12 (range 11–14; minimum ≈ 6)

Number of buckets z = 15 + m/2 = 15 + D/(2d) (Taygun's formula)

Bucket width ≈ 5d; depth ≈ 1.2d

Number of jets n: Q = n × (π/4)d² V₁

Flow ratio ψ = V_f1/√(2gH) ≈ 0.15–0.30

Speed ratio K_u = u₁/√(2gH) ≈ 0.6–0.9

Discharge Q = π D₁ B₁ V_f1 (× blade thickness factor); width ratio n = B₁/D₁ ≈ 0.10–0.40

Hydraulic efficiency (radial discharge) η_h = V_w1 u₁/(gH)

Discharge Q = (π/4)(D_o² − D_b²) V_f1

Peripheral velocity at outer diameter u = π D_o N/60

Draft tube efficiency η_d = [(V₂² − V₃²)/2g − h_f] / (V₂²/2g)

Pressure head at draft-tube inlet: p₂/ρg = H_atm − H_s − (V₂² − V₃²)/2g + h_f (H_s = height of runner exit above tail race)

Thoma's cavitation factor σ = (H_atm − H_v − H_s)/H; cavitation avoided if σ > σ_critical

Critical σ increases with specific speed (Kaplan needs higher σ → must be set lower relative to tail race)

Unit speed N_u = N/√H; unit discharge Q_u = Q/√H; unit power P_u = P/H^(3/2)

Same turbine at different heads: N ∝ √H, Q ∝ √H, P ∝ H^(3/2)

Model–prototype (geometrically similar): N D/√H = const, Q/(D²√H) = const, P/(D²H^(3/2)) = const

Synchronous speed of generator: N = 120 f / p (p = number of poles; f = 50 Hz in India)

Pumps and Hydraulic Devices

Static head H_s = h_s (suction lift) + h_d (delivery head)

Manometric head H_m = H_s + h_fs + h_fd + V_d²/2g (head imparted to liquid, measured by gauges)

Euler head (work per unit weight, radial entry V_w1 = 0): H_e = V_w2 u₂ / g

Manometric efficiency η_mano = g H_m / (V_w2 u₂)

Mechanical efficiency η_m = (power at impeller) / (shaft power)

Overall efficiency η_o = ρ g Q H_m / shaft power = η_mano × η_m × η_v

Power required P = ρ g Q H_m / η_o

Minimum starting speed (centrifugal head ≥ manometric head): $u_2^2/2g - u_1^2/2g = H_m \rightarrow N_{min} = 120\ \eta_{mano}\ V_{w2}\ D_2 / (\pi(D_2^2 - D_1^2))$

Slip: actual V_{w2} < theoretical (finite number of vanes); slip factor $\sigma = V_{w2}'/V_{w2}$

Affinity laws (same pump, speed change): $Q \propto N$, $H \propto N^2$, $P \propto N^3$

Same speed, impeller diameter change (trimming): $Q \propto D$, $H \propto D^2$, $P \propto D^3$ (approx.)

Geometrically similar pumps: $Q/(ND^3) = \text{const}$, $H/(N^2D^2) = \text{const}$, $P/(N^3D^5) = \text{const}$

Specific speed of a pump: $N_s = N \sqrt{Q} / H^{(3/4)}$ (speed of a similar pump delivering 1 m^3/s against 1 m head)

Multistage: N_s uses head per stage; double-suction: use $Q/2$

NPSH available = $p_{atm}/\rho g - h_s - h_{fs} - p_v/\rho g$ (suction lift case; add h if the liquid is above the pump)

Condition: $NPSH_A > NPSH_R$ (given by manufacturer)

Thoma's cavitation parameter for pumps $\sigma = NPSH / H_m$

Maximum suction lift (theoretical) = 10.3 m of water at sea level; practically 6–7 m

Theoretical discharge: single-acting $Q_{th} = ALN/60$; double-acting $Q_{th} = 2ALN/60$ (neglecting rod area; with rod: $(2A - a)LN/60$)

Slip = $Q_{th} - Q_{act}$; % slip = $(1 - C_d) \times 100$; $C_d = Q_{act}/Q_{th}$ (0.95–0.98)

Negative slip occurs when $Q_{act} > Q_{th}$ — long suction pipe, short delivery pipe, high speed (inertia of the suction column keeps delivery valve open)

Work done per second (single-acting) = $\rho\ g\ A\ L\ N\ (h_s + h_d)/60$; double-acting × 2

Acceleration head: $h_a = (L_{pipe}/g)(A/a)\ \omega^2\ r \cos\theta$ (max at start and end of stroke: $(L/g)(A/a)\omega^2 r$)

Friction head (with SHM): $h_f = (4f\ L/(2gd))\ [(A/a)\ \omega\ r \sin\theta]^2$ — maximum at mid-stroke

Effect of acceleration head on work: none (positive and negative areas cancel); friction adds area to the indicator diagram

Hydraulic coupling: efficiency $\eta = N_{turbine} / N_{pump} = 1 - \text{slip}$

Torque converter: $T_{turbine} = T_{pump} + T_{stator}$

APPENDIX

Appendix B – PYQ-Trend Concept Map

All PYQ-Trend boxes in one place, chapter-wise. These are the concepts SSC JE has tested repeatedly; revise this list in the last week.

Properties of Fluids

★ Definition of fluid based on continuous deformation under shear. Continuum assumption and Knudsen number.

★ Specific gravity of mercury (13.6). Unit of specific weight (N/m³). Specific volume is the reciprocal of density. Water has maximum density at 4 °C.

★ Viscosity variation with temperature (liquids ↓, gases ↑) is asked every year. Units/dimensions of μ and ν; poise and stoke conversions. Examples of non-Newtonian fluids (blood – pseudoplastic, toothpaste – Bingham, printer's ink – thixotropic, quicksand – dilatant). Newton's law of viscosity.

★ K of water; K = p (isothermal) and γp (adiabatic) for gases; compressibility as the reciprocal of bulk modulus; speed of sound √(K/ρ).

★ Pressure inside a soap bubble 8σ/d vs droplet 4σ/d (very frequent). Capillary rise formula and h ∝ 1/d. Mercury shows capillary depression. Unit of surface tension (N/m). Effect of temperature on surface tension.

★ Cavitation occurs when pressure falls below vapour pressure. Why mercury is used in barometers (high density, low vapour pressure, does not wet glass). Boiling point lower at higher altitude.

★ Ideal vs real fluid. Shear stress between two plates with oil film (τ = μV/h). Power lost in a journal bearing due to viscous friction.

Fluid Statics and Pressure Measurement

★ Statement and applications of Pascal's law; mechanical advantage of hydraulic press; unit conversions (1 atm = 10.33 m water = 760 mm Hg).

★ p = ρgh and its independence from container shape; head conversions between fluids (e.g., 10 m of water = 0.735 m of mercury); thick dam base.

★ Relation among absolute, gauge, vacuum pressure; which instrument measures which (barometer – atmospheric; Bourdon – gauge).

★ Limitations of piezometer; differential manometer reading $p_A - p_B = gx(\rho m - \rho)$ (very frequent for venturimeter); inclined manometer for small pressures; inverted U-tube for small differences between liquids.

★ Bourdon gauge measures gauge pressure; dead-weight tester for calibration; McLeod for vacuum; aneroid barometer has no liquid.

★ Inclination of free surface $\tan\theta = a/g$; pressure in an accelerating lift; paraboloid free surface in rotating vessel and $h = \omega^2R^2/2g$; equal rise and fall about the original level.

Hydrostatic Forces on Surfaces

★ Centre of pressure is below the centroid; $h^* = 2d/3$ for a rectangle with top at the surface; $F = \rho g A\bar{h}$ independent of inclination; formula $h^* = \bar{h} + I_G/(A\bar{h})$; CP approaches centroid with depth.

★ Horizontal component = force on vertical projection; vertical component = weight of liquid above; resultant on a circular arc passes through the centre.

★ Force on a vertical gate $\rho g b H^2/2$ at H/3 above base; middle-third rule for dams; resultant on Tainter gate through the hinge.

Buoyancy, Floatation and Stability

★ Archimedes' principle; centre of buoyancy = centroid of displaced volume; fraction submerged = ρ_body/ρ_liquid; ship rises when entering sea; hydrometer principle.

★ Stability conditions (M above G for floating; B above G for submerged) – very frequent. Metacentric height formula $I/V - BG$. Effect of GM on time period of rolling. Experimental determination of GM.

Fluid Kinematics

★ Examples of steady/unsteady and uniform/non-uniform flows (very frequent). Lagrangian vs Eulerian.

★ Definitions and coincidence of streamline/pathline/streakline in steady flow; smoke from a chimney shows a streakline; no flow across a streamline.

★ Checking whether a given velocity field satisfies continuity; $A_1V_1 = A_2V_2$; velocity $\propto 1/D^2$ in pipes.

★ Local vs convective acceleration; which is zero in steady/uniform flow; computing acceleration from a velocity field.

★ Condition of irrotational flow ($\partial v/\partial x = \partial u/\partial y$); vorticity = twice the rotation; circulation definition.

★ Existence of φ (irrotational only) vs ψ (2-D incompressible); orthogonality of φ and ψ lines; discharge $= \psi_2 - \psi_1$; flow nets and their limitations.

★ Free vortex Vr = const vs forced vortex $V = \omega r$; examples of each; which is rotational (forced) and irrotational (free); Rankine vortex; potential-flow combinations (Magnus effect).

Fluid Dynamics – Energy and Momentum

★ Which forces are included in Euler's vs Navier–Stokes vs Reynolds equations.

★ Assumptions of Bernoulli's equation (very frequent). Bernoulli is energy conservation. Terms as heads (m) and their names. Everyday applications (carburettor, aerofoil, atomiser).

★ TEL always slopes down in flow direction; HGL can rise; vertical distance between TEL and HGL = velocity head; HGL below pipe → negative pressure.

★ $\alpha = 2$ and $\beta = 4/3$ for laminar pipe flow (frequent). $\alpha > \beta > 1$ for non-uniform profiles.

★ Impulse-momentum equation; force on pipe bends; momentum equation used for sudden enlargement loss and hydraulic jump; lawn sprinkler uses moment of momentum.

★ Range of a liquid jet max at 45°; Cv from jet trajectory method.

Flow Measurement – Meters, Orifices, Mouthpieces, Notches

★ Divergent cone angle 5–7° to avoid separation; C_d 0.95–0.98; differential manometer reading independent of inclination; why the divergent cone is longer than the convergent cone.

★ Order of C_d (venturi > nozzle > orifice) and head loss (reverse). Pitot tube measures point velocity/stagnation pressure. Rotameter is a variable-area meter. Electromagnetic flowmeter needs conductive liquid.

★ $C_d = C_c \times C_v$; typical values (0.64, 0.98, 0.62); vena contracta location; time to empty a tank ∝ √H; drowned orifice uses difference of levels.

★ C_d values of mouthpieces (external 0.855; Borda's running free 0.5, running full 0.707). Why a mouthpiece discharges more than an orifice.

★ Q ∝ H^(3/2) (rectangular) and H^(5/2) (triangular) – very frequent. V-notch preferred for low flows. Cipolletti slope 1:4. Error ratios 3/2 and 5/2. Francis' end-contraction correction. Ventilation of weirs.

Viscous (Laminar) Flow

★ Reynolds number definition and critical values (2000/4000); Re = inertia/viscous forces; dye experiment.

★ Parabolic velocity profile; $V_{avg} = u_{max}/2$; shear stress linear (zero at centre, max at wall); $h_f = 32\mu VL/\rho gD^2$; $f = 64/Re$ (most frequent); $Q \propto D^4$ at constant pressure gradient; $\alpha = 2$, $\beta = 4/3$.

★ $V_{avg} = (2/3)u_{max}$ between fixed plates; $\Delta p = 12\mu VL/t^2$; Couette flow linear profile; comparison with pipe flow (1/2 vs 2/3).

★ Stokes' drag $3\pi\mu DV$ and $C_D = 24/Re$; terminal velocity ∝ D^2; viscous torque on journal bearing ∝ $\mu N D^3 L/t$.

★ Falling-sphere viscometer uses Stokes' law; Redwood/Saybolt are orifice-type; rotating cylinder method.

Turbulent Flow and Flow Through Pipes

★ Velocity profile flatter in turbulent flow; 1/7th power law; laminar sub-layer and smooth/rough classification; Reynolds stress; Prandtl mixing length.

★ Darcy–Weisbach formula; $h_f \propto Q^2/D^5$; hydraulic mean depth D/4; Chezy and Manning formulas; Moody diagram zones; Blasius equation.

★ Sudden enlargement loss $(V_1-V_2)^2/2g$ (very frequent); exit loss = $V^2/2g$; entrance $0.5V^2/2g$; equivalent length; momentum used to derive enlargement loss.

★ Series vs parallel rules; Dupuit's equation; siphon summit pressure negative and practical height limit ≈ 7.6–8 m.

★ Maximum power transmission when $h_f = H/3$ and η = 66.7% (very frequent); nozzle diameter for maximum power.

★ Water hammer pressure ρVc for sudden closure; critical time 2L/c; surge tank function; $c = \sqrt{(K/\rho)}$.

★ Hardy Cross method for networks; effect of ageing on roughness.

Boundary Layer, Drag and Lift

★ Definitions of δ, δ*, θ; δ ∝ √x for laminar; Blasius result $5x/\sqrt{Re_x}$; displacement thickness meaning; critical Re for flat plate 5×10^5; shape factor; momentum integral equation.

★ Separation occurs with adverse pressure gradient and at $(\partial u/\partial y)_{wall} = 0$ (very frequent). Methods to prevent separation. Why golf balls have dimples. Stall of aerofoils.

★ Form drag dominates bluff bodies and skin friction dominates streamlined bodies; Magnus effect and Kutta–Joukowski theorem; drag crisis; Karman vortex street and Strouhal number; drag formula with $\frac{1}{2}\rho U^2 A$.

Dimensional Analysis and Model Similitude

★ Dimensions of viscosity, surface tension, power, kinematic viscosity (very frequent); principle of dimensional homogeneity.

★ Buckingham π-theorem statement and n − m; choice of repeating variables; Rayleigh method.

★ Meaning and application of Re, Fr, We, M, Eu (very frequent). Froude for open channels and ships; Reynolds for pipes and submarines; Mach for compressible flow; Weber for surface tension.

★ Froude model ratios ($V \propto \sqrt{L}$, $Q \propto L^{2.5}$, $T \propto \sqrt{L}$, $F \propto L^3$, $P \propto L^{3.5}$) – very frequent. Reynolds model $V \propto 1/L$ for the same fluid. Distorted models for rivers. Types of similarity.

Compressible Flow (Basics)

★ Speed of sound $\sqrt{(\gamma RT)}$; Mach angle $\sin \alpha = 1/M$; flow regimes by Mach number; zone of silence.

★ Stagnation temperature relation; critical pressure ratio 0.528 for air; area–velocity relation and behaviour of nozzles/diffusers in subsonic vs supersonic flow; choking at M = 1; de Laval nozzle.

★ Across a normal shock stagnation temperature is constant and stagnation pressure falls; shock converts supersonic to subsonic; entropy increases; Fanno vs Rayleigh lines.

Open Channel Flow

★ Hydraulic radius; Froude number classification; uniform flow – all slopes equal; examples of GVF and RVF.

★ Chezy and Manning formulas; relationship $C = R^{(1/6)}/n$; typical n values.

★ Most economical rectangular section b = 2y and R = y/2 (very frequent); trapezoidal as half hexagon; triangular 90° vertex; circular channel depths for max velocity (0.81D) and max discharge (0.95D).

★ Critical depth $(q^2/g)^{(1/3)}$ and $E_{min} = 1.5\, y_c$ (very frequent); minimum specific energy at critical depth; alternate depths; Parshall flume.

★ Jump converts supercritical to subcritical; sequent depth formula; energy loss $(y_2 - y_1)^3/4y_1y_2$; momentum (not energy) is conserved; uses in stilling basins.

Impact of Jets

★ $F = \rho a V^2$ for fixed flat plate; semicircular vane doubles the force ($2\rho a V^2$); inclined plate normal force $\rho a V^2 \sin\theta$.

★ $\eta_{max} = 50\%$ at u = V/2 for a series of flat plates (very frequent); single moving plate u = V/3, 8/27; curved vanes give higher efficiency; series vs single plate mass flow.

★ Angle of deflection of a hinged plate struck by a jet.

Hydraulic Turbines

★ Gross vs net head; definitions of hydraulic, mechanical, overall efficiency; Euler's equation; radial discharge for max efficiency.

★ Turbine selection by head and specific speed (most frequent hydraulic-machines question); Pelton – tangential impulse; Kaplan – axial reaction; Francis – mixed/radial reaction; Ns formula with $H^{(5/4)}$.

★ Spear valve regulates discharge; splitter divides the jet; casing has no hydraulic function; deflector avoids water hammer; jet ratio ≈ 12; number of buckets 15 + m/2; max efficiency at u = V/2; no draft tube for Pelton.

★ Guide vanes (wicket gates) regulate flow; scroll casing gives uniform velocity; flow ratio range; Francis is inward/mixed-flow reaction turbine; runner full of water.

★ Kaplan has adjustable runner blades (double regulation); propeller has fixed blades; axial flow; flat part-load efficiency curve.

★ Draft tube used only in reaction turbines (not Pelton); functions (suction head + KE recovery); conical type most efficient; angle < 8° to avoid separation.

★ Location of cavitation in reaction turbines; Thoma's factor; σ_critical increases with Ns.

★ Unit quantities and N ∝ √H, P ∝ H^1.5 (frequent); runaway speed; synchronous speed 120f/p; governing mechanisms (spear, deflector, wicket gates, servomotor); Muschel curves.

Pumps and Hydraulic Devices

★ Priming and why it is needed (very frequent); backward-curved vanes; volute casing converts KE to pressure; foot valve function; starting against closed delivery valve; centrifugal pump = reverse of reaction turbine.

★ Manometric head definition; manometric efficiency formula (frequent); Euler head with radial entry; minimum starting speed; slip.

★ Affinity laws P ∝ N³ (very frequent); pump specific speed formula with √Q and H^(3/4); series for head, parallel for discharge; BEP; multistage pumps; operating point.

★ NPSH definition and NPSH_A > NPSH_R; cavitation at impeller eye; practical suction lift limit ≈ 6–7 m.

★ Negative slip conditions (very frequent); discharge formulas; air vessel functions and work saved; acceleration head max at stroke ends; indicator diagram; comparison with centrifugal pumps; separation.

★ Hydraulic ram works on water hammer; accumulator stores energy; intensifier raises pressure; torque converter has a stator (torque multiplication) whereas fluid coupling does not; gear and vane pumps are positive displacement.